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Concept of Operations (CONOPS 2019)

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PJ19-02

ATM OPERATIONS

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Abstract

The Concept Of Operations (CONOPS 2019) document targets the future Air Traffic Management (ATM) operation envisaged in Europe, with a focus on the detailed concepts developed under the overall SESAR 2020 programme. It represents the end state of the SESAR 2020 R&D. The objective being that Civil and Military Airspace Users, Air Navigation Service Providers, Airports, Aviation & ATM industries and SESAR 2020 Programme tasks, gain common understanding of the operational characteristics of ATM in the future European airspace coming from the development of the SESAR Programme, and the main significant changes they imply in operating practices along with the capabilities they require.

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Executive summary

Objective and Scope

The Concept of Operations (CONOPS) acts as the primary point of entry to the concept for organisations and individuals both external and internal to the SESAR Programme. The context of the CONOPS sees only those concept elements that are relevant to SESAR 2020. This captures development in a timeframe to 2035 and describes areas where operational improvements, supported by technical enablers, bring the expected performance gains to deliver the overall performance benefits in the Single European Sky High-Level Goals [6] set out in the European ATM Master Plan [5].

Trajectory-based operations (TBO) will enable Airspace Users to operate as close as feasible to their preferred trajectories, thereby satisfying their business and operational needs. TBO enables effective dynamic adjustment of capacity (via airspace characteristics) in order to meet demand, making full use of developed civil/military collaboration.

Optimised ATM Network Services aim to meet SESAR goals of increasing ATM capacity and providing greater opportunities for stakeholders to optimise their operations - in terms of operational efficiency and environmental gains. This sees full integration of Dynamic Airspace Configurations (DAC) with Demand Capacity Balancing (DCB) and dynamic DCB with ATC planning, alongside Queue Management. This new operation is able to support optimised trajectories in an environment that includes Free Route, Flight and/or Flow-Centric operations, and use of Collaborative Control.

Improved airport performance and access utilise solutions to allow airports to operate efficiently during periods of high traffic density, extending capability to operate at maximum capacity, even during periods of adverse weather conditions. This is achieved by implementing enhanced runway throughput capabilities, safety nets and more accurate navigation and routing tools. New approach procedures improve operational flexibility and accessibility to airports - even with limited ground navigation infrastructure. Solutions for remote tower services that enable operational coverage to be extended at low and medium-traffic airports also provide safety and operational efficiency benefits to the tower operations.

The main Operational Improvements are based on advances in **digitisation and automation**. This sees use of artificial intelligence where appropriate, and is accompanied by robust cyber-security, which reflects the growing need for secure data exchange in the aviation community. Continuity of operations and resilience are essential to maintain the trust in the aviation system.

The role of the human, and its continued importance within the operational concept, is recognised in the future system, which takes into account the need to balance efficiency created by automation with human capabilities. The principle remains whereby the human decision always needs to take primacy over a system - independent of the level of automation.

Safety is the overarching aim of the entire Single European Sky (SES) II initiative and is the central element of any technical solution developed by SESAR.

1 Introduction

1.1 Purpose of the document

The Concept Of Operations (CONOPS) document in SESAR is a document describing the operational concept, answering the question “what?” and also describing on a high level the way this concept shall be applied, answering the question “how?” thereby providing the overall concept level guidance to ensure consistent development of the operational details.

SESAR operational concept, fitting the European ATM Master Plan [12], formulates and describes the evolution of the European ATM System.

The European ATM Master Plan outlines the essential operational and technological changes that are foreseen to provide SESAR contributions to achieving the European SES strategic performance objectives. It is the agreed European “roadmap” connecting research and development with deployment scenarios and is the key tool for SESAR deployments providing the basis for timely, coordinated and efficient deployment of new technologies and procedures.

1.2 Current scope and evolution of the CONOPS

SESAR 2020 CONOPS describes on a high level the ATM operation envisaged in Europe. It allows civil and military Airspace Users, Service Providers, Airports, Aviation and ATM industries, Aviation Regulation bodies and SESAR Programme tasks to gain a common understanding of the operational characteristics of ATM and the main changes they imply in operating practices along with the capabilities they require.

The CONOPS covers the complete air traffic management process from early planning through flight execution to post flight activities.

It should be noted that the concepts described may only apply to operational environments where they provide clear benefits to ATM and are practicable. There may in some instances exist optional solutions that can be applied to gain the same operational benefit. These options shall be interoperable with the solutions presented in the CONOPS.

Remark:

This document is limited to SESAR 2020 wave 1 and 2

Note:

SESAR JOINT UNDERTAKING wave 2 call, Technical Specifications [10] forms the baseline for this CONOPS 2019 as far as wave 2 is concerned.

As far as wave 1 is concerned ATM Master Plan Level 2, (DS19) forms the reference [4].

In wave 2 there might be a discrepancy as some OI steps in DS 19 are not allocated to solutions. Furthermore some OI steps mention wave 2 call which is not part of DS 19.

1.3 Document Structure

The document addresses the three main operational key features of SESAR and links the concept elements to the expected performance. High Level Operational Requirements address Operational Solutions and ensure that detailed operational concept descriptions (e.g. in OSEDs) are consistent with the global concept as well as with other detailed concept elements.

Chapter 2 – SESAR 2020 Operational Concept addresses the new operating methods brought by SESAR 2020. It is structured around the three operational key features, Optimised ATM Network Management, Advanced Air Traffic Services and High Performing Airport Operations.

Chapter 3 – High Level Operational Requirements describes the requirements expressed by Operational Projects and Operational Solutions, considered as concept parts (parts of Key Areas).

Chapter 4 – References

Appendix A – Glossary

1.4 Intended readership

The target audience of this document is twofold:

- Primarily SESAR Joint Undertaking partners and stakeholders involved in the development and validation of the SESAR concept, Solution and Enabling projects and Very Large Scale demonstration projects. The CONOPS serves as the common high-level reference document. It sets the framework for development of further detailed operational concept descriptions in Operational Service and Environment Definition (OSED) documents developed in SESAR 2020 Solution projects.
- Secondly, this CONOPS is also aiming at providing the global ATM community with information on how ECAC intends to develop and implement the ATM System to meet the overarching goals of the Global Air Transport.

1.5 CONOPS and the bigger picture

In the context of SESAR Research Programme, two high-level concept documents are developed, the **Operational Concept Document (OCD)** [14] and the **Concept of Operations (CONOPS)**. The OCD complements the CONOPS and explains why we make the changes of ATM in Europe and what we change. The CONOPS goes further into detail describing more how we will operate in this new ATM environment.

1.5.1 Operational Key Features

The Concept of Operations Document (CONOPS) describes areas identified by **The SESAR Vision** documented in the European ATM Master Plan [5] where operational improvements supported by technical enablers will bring performance gains that yield the overall performance expected in Single European Sky High-Level Goals as set out in the European ATM Master Plan. The description follows the **three operational key features** of SESAR:



- Optimised ATM Network Management



- Advanced ATS



- High Performing Airport Operations

The 4th SESAR key feature “Enabling Infrastructure” is not elaborated in this document, see further information below.

An overview section provides a high-level description of the main solutions developed in the SESAR 2020 programme. It allows civil and military Airspace Users, Service Providers, Airports, Aviation and ATM industries, Aviation Regulation bodies to gain a high level view of the operational characteristics of the likely future ATM environment and the main changes implied in operating practices along with the capabilities required.

1.5.2 Enabling Aviation Infrastructure

The enhancements described in chapter 2 (three Key Features) will be underpinned by an integrated and rationalised aviation infrastructure, providing the required technical capabilities in a resource-efficient manner.

This feature will rely on enhanced integration and interfacing between aircraft and ground systems, including ATC and other stakeholder systems, such as flight operations centres and military wing operations centres. Communications, navigation and surveillance (CNS) systems, SWIM, trajectory management, Common Support Services and the evolving role of the human will be considered in a

coordinated way for application across the ATM system in a globally interoperable and harmonised manner.

A performance-based CNS approach allows evolving from system/technology-based operations, where systems/technologies are prescribed, towards the delivery of Performances-Based services that specify which performance is required within a specific environment based on the operational requirements and considering CNS as a whole and integrated system.

The introduction of U-space services [11] to support access to airspace for a large number of Unmanned Aircraft Systems (UAS) is a major development in connection to this feature.

The development and deployment of several technologies (VDL Mode 2, ADSB-out, etc.) is running in parallel whilst there should be convergence and integration via the performance based approach. The ATM needs that UAS/RPAS/drones are equipped with Detect and Avoid Capabilities that are interoperable with the manned aviation ones (e.g., ACAS or its evolutions), compliant with local Air/Ground procedures and compatible with ground safety nets.

Automation and Virtualisation enabled by Digitalisation

The main operational Improvements described in this CONOPS are based on step-change advances in automation and virtualisation enabled by digitalisation- with use of artificial intelligence where appropriate, is accompanied by the growing need in the aviation world for robust cyber-security, and is addressed with the securing of data exchange. Human-system integration will evolve across the domain into highly integrated human-machine work systems through co-active design that shapes and does not devalue the human in order to justify the machine.

Service oriented approach

The service-oriented architecture concept helps generating new business models with a positive impact on the Air Navigation Service Providers costs. Thanks to the service-oriented approach applied to CNS provision it is possible to support several ATM changes, like data link services in support of trajectory-based operations.

These various technological solutions will have to be packaged or merged in a way, which guarantees to end users the availability, integrity, safety, security and other performance requirements that could be mandated by the relevant authority and facilitates the transparent application of the user-pays principle for the delivered services.

Developments within ATM are marked by virtualisation, which first appeared in support of ATS for remote towers and will be further developed to En-route and TMA sectors by decoupling the CWP from the ATM data service provision with the implementation of virtual centres supported by data centre services.

1.5.3 Digitalisation in ATM

Digital technologies are identified as fundamental enablers to support many ATM operational processes and their underlying systems. The resulting digitalisation process is expected to unlock the performance expectations envisaged in the CONOPS, thus creating the Digital European Sky, advocated by the whole European aviation industry and depicted in European ATM Master Plan, the Airspace Architecture Study [15] and the Wise Persons Group Report [16].

The process of digitalisation is however not new in ATM and significant effort has been spent in the last 10 years in modernising the tools used by ANSPs, NM, AUs, Airport operators and AUs to automate their procedures and to efficiently interact among them.

Digital solutions have been developed to transpose in a digital format the aeronautical information to be exchanged at network level (e.g. the EAD), thus allowing higher accessibility and the application of uniform quality standards for the creation and manipulation of data. However, this just constitutes a first digitisation step within the complete digital transformation journey affecting ATM.

Digitalisation, in fact, is much more than exchanging and storing data in digital format. A myriad of new applications are enabled by the availability of these digital data to users, in particular providing additional capabilities for prediction (4D traffic demand, traffic complexity, etc.), based on the observation and processing of past data, permitting the extraction of meaningful knowledge on the complex patterns and interactions behind the ATM system dynamics.

It is obvious that such a profound change needs to be accompanied since its inception by an effective change management process, ensuring that economic and social concerns are factored in the equation and adequately reflected by the regulatory environment.

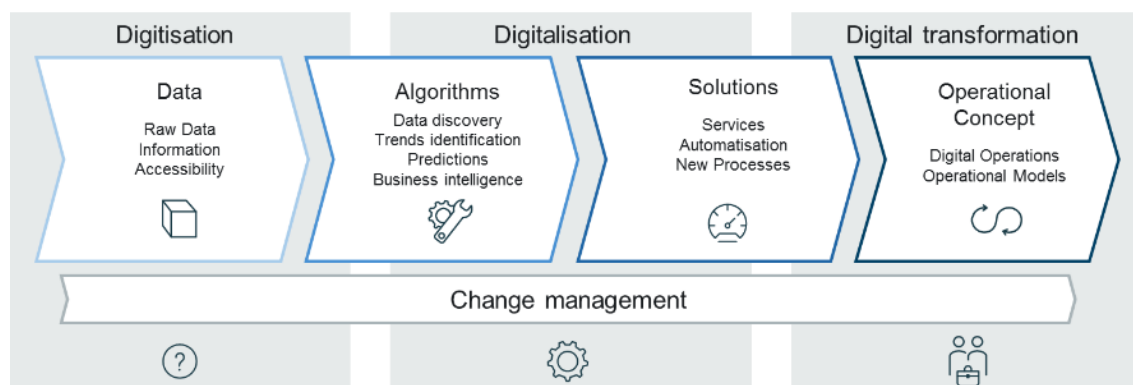


Figure 1: Typical phases supporting digital transformation

Key digital technologies

A number of key digital technologies and techniques enhancing ATM performance that make it possible generating new knowledge and operational insights out of the data collected are identified.

BIG DATA

Big Data technology is used in a context where stringent requirements for data management are necessary. These requirements are defined by the 5 Vs of Big Data: Volume, Variety, Velocity, Veracity and Value. Big data will be key to capture data from a variety of sources at network level and exploit them to improve ATM efficiency (e.g. DCB process, post-ops analysis, etc.).

DATA SCIENCE AND ANALYTICS

Data science and analytics include all those algorithms that focus on the generation of valuable information from the data, thus allowing inferring intelligence and knowledge out of it. It is important that data is prepared and follows strict standards to enable and facilitate the analytics tasks. Pre-processing the data to unify formats and perform quality checks is usually required prior the analysis. Data science models will be particularly relevant to detect complex patterns among different key performance areas and to enhance predictions accordingly, thus enabling performance-based operations.

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Artificial intelligence (AI) broadly groups all those computer programs that exhibit human-like intelligence such as logical reasoning, problem solving and learning. Expert systems are designed to

solve complex problems by reasoning through bodies of knowledge, represented mainly as if-then rules rather than through conventional procedural code.

Artificial intelligence is expected to play a fundamental role in air traffic complexity management and airspace design, supporting a more effective management of ATC resources based on enhanced predictions of demand.

CLOUD

Cloud is a computer data architecture according to which the collected information is located in multiple servers, usually in different physical locations. Cloud storage and computing is based on a highly virtualised infrastructure, which allows effective and consistent access to the same information from multiple locations (or by multiple entities) in a synchronous or asynchronous manner, while at the same time providing elasticity and scalability of the infrastructure based on the real needs of users. The interoperability among ground systems and the virtualisation of service provision in ATM will be highly reliant on cloud technologies.

INTERNET OF THINGS

The Internet of Things (IoT) concept considers the integration of multiple sensors in almost all the devices/systems. The future of IoT in air transport could drastically change once IP-based A/G communications are realised. Airbus and Boeing explore aircraft Internet Protocols to support future air/ground services communications.

VIRTUAL REALITY AND AUGMENTED REALITY

Virtual reality refers to the technologies used to immerse users in a fully digital environment, i.e. created by a computer. This may notably improve the training of controllers and pilots, allowing the recreation of complex operational scenarios to enhance the competencies of operators to deal with similar situations in the real world.

On the other hand, augmented reality refers to the technologies allowing the overlay of virtual objects on the real-world environment. This technology will have applications both in the cockpit in head-up displays and on the ground, particularly at airport towers in low visibility conditions, increasing the situational awareness of operators.

CYBERSECURITY

The introduction of digital resources in safety-critical applications and management systems poses severe risks against cyber-attacks. Cybersecurity is of extreme importance and shall be developed along the implementation of digital systems. Embedded cybersecurity software is just a small part of a global security system, where physical credentials, personnel protocols, etc. are equally relevant. As ATM operations rely on the effective exchange of digital data among relevant stakeholders, a resilient and robust data sharing network, including relevant cyber protection and cyber resilience will be essential in order to guarantee the availability and quality of information, according to its features and use.

BLOCKCHAIN

Blockchain consists of a transactions database that is shared and synchronised in multiple computers and locations without a centralised control through a peer-to-peer network. A particularly interesting feature of blockchain is its immutability, which enables the creation of smart contracts. It may therefore provide, as an important supporting technology, guarantee of secure exchange of data among different parties in ATM.

1.5.4 Empowering Human Performance in ATM Operations

With the aim to enhance the ATM processes, there is a perceived need to go beyond the state-of-the-art concerning the efficiency of air- and ground tools, and appropriate complementarity with both human roles and definition of human tasks.

There is clearly an increasing opportunity to enhance the global Air/Ground ATM processes via the right understanding of global ATM automation, supported by enhanced communication between respective Air and Ground Automation & Assistance functionalities.

As regards automation, progressively more accurate and relevant information will be available to ATM processes: e.g. wide implementation of Extended Projected Profile (EPP) and/or accurate 4D Trajectory sent by Flight Operations and Wing Operations Centres as well as more up to date and accurate airspace status and constraints. Such will be used to enhance e.g. trajectory calculations, traffic and sectors load predictions and will provide a Trajectory Based Operating environment supporting advanced tools.

As a result, there is an absolute need to take into account the necessary balance between the efficiency created by automation and the human capability (when assisted by more basic functions). This is particularly valid for recovery from non-nominal and/or degraded modes of operations. The identification of most sizing failures, fallback and degradation (i.e. most frequent and/or impacting ones) requires a correct understanding of the needed robustness of the tools and a detailed description of the related human roles. The human decision always needs to take primacy over a system, independent of the automation level¹. During the design of automated systems, safety analysis will have to be performed in order to implement appropriate level of redundancy so that the human takeover is avoided due to the lack of situational awareness. This needs to be addressed in the certification process at the latest.

As an example, the airborne domain has experienced significant enhancements to safety, flight efficiency and regularity, cost-efficiency and comfort of operations. Contributing to those enhancements, automation of cockpit functions was enabled thanks to the combination of three pillars:

- Advanced functionalities supported by powerful hardware and software
- Data of appropriate quality (accuracy; reliability and integrity; availability etc.)
- Operational procedures to efficiently implement the above evolutions, with the support of the needed training

As a conclusion, attention must be drawn to the importance of the capabilities offered by sensible automation developments as a key contribution to ATM enhancements. The exploration of such capabilities in support to modern operations shall be performed with the right experts, dedicating

¹ For a more complete material on Human Performance Guidance, see ref. [8]

the appropriate effort and time for this, and prior to decisions for prototyping and validation at V3 level or beyond (e.g. Very Large Scale Demonstrations).

1.5.5 Performance Driven Approach in SESAR 2020

OVERVIEW OF THE SESAR 2020 HIGH-LEVEL PROCESS

The ATM Master Plan describes the Performance Ambitions that SESAR may enable through the full implementation of its vision within the 2035 timeframe. The Performance Ambitions are outlined according to several Key Performance Areas (KPA), including those captured in the Single European Sky (SES) High-Level Goals and by the SES Performance Scheme. The principle behind the programme is that these Performance Ambitions can be achieved by defining and validating a Concept Of Operations (CONOPS) made up of a set of discrete operational improvements or solutions that solve identified operational problems.

The performance information cascading from the Master Plan should be linked to the operational solutions defined by the SESAR Solution projects and integrated in the SESAR 2020 Concept of Operations.

The SESAR Performance Management Process, as presented in Figure 2, is thus structured to steer the overall R&D work in respect of these SESAR Performance Ambitions. The SESAR 2020 CONOPS reconciles them by defining Performance Expectations². Those are mapped with the set of operational improvements described as SESAR Solutions.

² Performance Expectations were defined in cooperation with PJ.19-04 to set the validation targets [9] and the military dimension for performance partnership

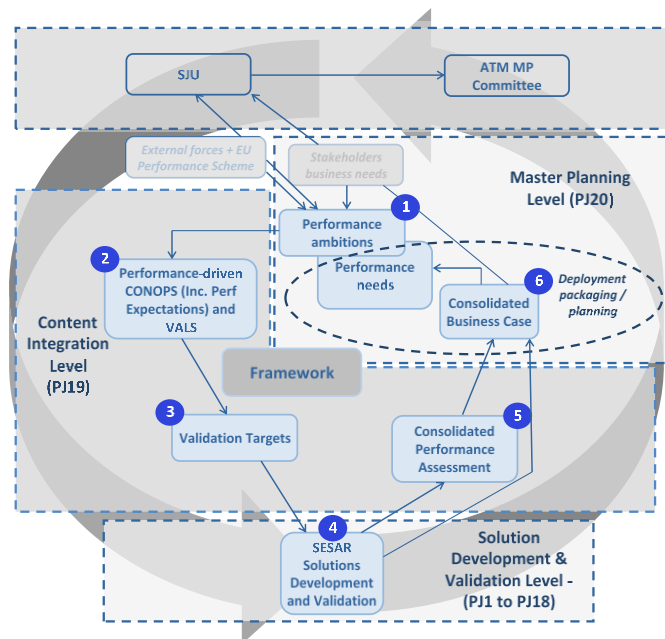


Figure 2: SESAR 2020 Performance Management Process

The iterative nature of this process means that it is also a cycle. This includes the analysis of performance assessments from validation exercises and CBAs including gaps. The findings are fed back into CONOPS and VALS [13]. To ensure the quality of this activity evolution of the SESAR Solution validation should demonstrate increasing confidence in the performance assessment result.

Through validation exercises and dedicated analysis, the SESAR projects assess the performance benefits and costs of the operational improvements brought by SESAR Solutions. The results of the validation are then extrapolated at an ECAC level to provide an overall performance assessment of the solution. All the results at Solution level are then collected and consolidated at Content Integration level. This is necessary to ensure that the Deployment Planning process executed at Master Plan level is adequately de-risked against operational, technological and economic/financial issues.

This process includes analysis of performance assessments from validation exercises and Cost Benefit Analysis (CBAs) including gaps. The results are fed back into CONOPS and Validation Strategy (VALS).

The following elements of the Performance Management Process are described in more detail in the SESAR 2020 Performance Framework (2019) [9]:

- SESAR Performance Ambitions
- Performance Expectations
- SESAR Validation Targets
- Performance validation objectives, assessment, results and CBA
- Consolidated Performance Assessment
- Consolidated Business Case

Performance Expectations in the CONOPS

The SESAR 2020 CONOPS developed at the Content Integration-level (PJ19-02), provides a comprehensive and performance driven operational concept for all ATM phases of operation, whilst ensuring a stepwise evolution of the entire ATM system.

The operational concept is internally consistent and meets the overall Performance Ambitions of SESAR. This requires that the portfolio of SESAR Solutions under development, together with their validations are presented through a coherent description of their operations, and are associated to specific expectations in terms of performance improvement, based on available validation results, evidences and expert judgment.

Performance expectations are defined as qualitative estimations of a potential performance outcome or impact that could be achieved as a result of the implementation of an envisaged operational improvement. Nevertheless, it needs to be validated and confirmed. Performance expectations should be seen as initial expert judgement of the magnitude of potential benefits for each KPI (prior to validation).

Performance Expectations in the CONOPS therefore represent the main initial qualitative source of performance that concept designers expect to be achievable by relevant SESAR concept components in relevant operating environments. They should be seen as a very initial expert judgement of the magnitude of potential benefits for each KPI. The gap between these very initial qualitative estimations and more precise quantitative Performance Assessment from the solutions and consolidated results is continuously filled by expert judgment and more validation results from the solutions.

This will help with the allocation of Validation Targets among the different solutions and will provide a realistic bottom-up picture of potential performance improvements, particularly when compared to the aspirational, top-down Performance Ambitions.

In addition, and at overall concept level, SESAR Solutions that could typically be expected to interact or have dependencies on others (and how that interaction operates) are identified.

The expectations are not to be interpreted as universally applicable expectations, i.e. each implementation location has specific local circumstances that dictate achievability and benefit priority.

It is also important to stress the following:

- The expectations are subject to validation by the programme. Even if confirmed by validation, in practice it is unlikely that an actual deployment will deliver all the identified potential benefits.
- Specific assumptions, judgements and the analyses carried out to derive the judgements are not described. This background information will be provided separately.

Goals applicable in the Enabling Aviation Infrastructure Projects are outside the scope of the CONOPS.

Dependencies and trade-offs

Interacting solutions, whereby the ATM process developed in one solution may influence the process of another solution, may create trade-offs. It should be considered that dependencies and exclusivities are best addressed through the validation programme.

It is also noted that SESAR Solution dependency should initially be addressed at Concept design level so that the performance assessment task is able to understand the likely combination of interacting concept elements. In particular, additional integrated validation exercises with improved exercise controls would likely provide results that would a) better identify the individual contribution of a solution, and b) better identify the total benefit when solutions are combined in representative and realistic ways.

1.5.6 Operational environment

The R&D solutions under SESAR 2020, will contribute to the improvements and benefits to be realised through the gradual implementation and deployment of the SESAR CONOPS. The achievement of the above described improvements with the maximum performance gains will require a change in the way in which solutions are deployed and services are provided.

Following aspects of the operational environment need to be considered for SESAR 2020:

- According to the forecast, by 2040 traffic in Europe is expected to grow over 16.2 million flights in the Regulation and Growth scenario (most-likely), 53% more than the 2017 volume. Higher growth is expected in the Global Growth scenario, with around 20 million flights [6].
 - Airport Capacity presented in Annex1 Flight Forecast to 2040
 - Environmental Impact presented in Annex 2 Adapting Aviation to a Changing Climate
 - Other factors³: e.g. Increase of aircraft size, High-speed train, passenger forecast

³ chapter 4.4 of Challenges of Growth 2013 Task 4: European Air Traffic in 2035

2 SESAR 2020 Operational Concept

Concept Overview

SESAR Operational concept and objective

The **SESAR 2020 Concept** of Operations has the objective to meet the overall SESAR Performance Ambitions whilst being coherent and consistent within the research programme. It anticipates that the portfolio of SESAR Solutions under development, together with their validations, are presented through a coherent description of their operations, and are associated to specific expectations in terms of performance improvement.

The main Operational Improvements detailed below are based on advances in digitalisation and automation - with use of artificial intelligence where appropriate, accompanied by the growing need in the aviation world for robust cyber-security, which is addressed with the securing of data exchange. Continuity of operations and resilience will be essential to maintain the trust in the aviation system.

Trajectory Based Operations (TBO)

A significant and fundamental shift takes place with the adoption of 4D Trajectory Management Principles to manage flights, which facilitates the performance-friendly transition from tactical intervention towards a more strategic focus on planning and intervention by exception. This enables airspace users to operate as close as feasible to their preferred trajectories and to perform continuous descent and climb, generating environmental benefits in terms of both emissions and noise. TBO enables effective dynamic adjustment of airspace characteristics to meet predicted demand and makes full use of developed civil/military collaboration whilst aiming to keep any distortions to the Business/Mission Trajectories to the absolute minimum without compromising the flexibility required for optimisation purposes.

RPAS are integrated into the ATM network. This CONOPS addresses their initial integration into non-segregated controlled airspace with ICAO classification A, B and C, and between dedicated airfields. They are managed alongside all other trajectories to meet the required safety and performance objectives. RPAS trajectory planning captures any specific performance/communication/coordination needs that are of relevance to a planned and optimised network. Their subsequent execution is enabled by technical capabilities and procedures to enable compliance with ATC instructions. These capabilities and procedures also allow for loss of command & control data link and other emergencies. For the Military, RPAS Integration is of utmost importance, notably for the large ones.

The Network

Network optimisation is supported by improved flight trajectories provided through integration of a number of key processes. This sees Dynamic Airspace Configurations (DAC) being fully integrated with DCB, and dynamic DCB being fully integrated with ATC planning. These, together with Queue Management, are able to optimise new operations, supporting optimised trajectories in an environment that includes e.g. Free Route, flight and/or flow-centric and use of Collaborative Control.

The dynamic adjustment of airspace through DAC delivers an optimised and coordinated organisation of airspace reservations in the context of advanced airspace management and ATC sectorisation to match capacity to demand as much as feasible. This delivers performance gains by responding flexibly to the Airspace Users' expectations.

Airspace Users are able to indicate preference and recommend priority for flights for use in Demand/Capacity balancing. Access to DCB processes provides a means for their business/mission needs to be taken into account.

Managing the transition between planning and execution, the Integrated Network Management and (Extended) ATC Planning (INAP) function is introduced. This enables the seamless management of traffic and reduction in the number of potentially conflicting trajectories to be handled by tactical ATC. AUs are able to fly their preferred routes, satisfying their business needs, and to perform continuous descent and climb, generating environmental benefits both on emissions and on noise, whilst at the same time offering the best possible performance at network level.

For military aircraft operators, increased operational flexibility is provided to better integrate flights into the ATM Network operations, catering for better conditions to achieve mission effectiveness with greater freedom to access training and exercise areas.

Airport performance and access

Airport performance and access is improved through increased capacity and predictability at high traffic airports, implementation of enhanced runway throughput technologies, more accurate navigation and routing tools and technical solutions such as vision-based system and operational procedures, to maintain performance in all weather conditions and to separate the flow of the different airports.

In the context of integrating the Airport into the network, the operational information affecting individual aircraft is reflected in the Airport Transit View (ATV), ensuring that airport performance capabilities and ground operations are known to all actors – allowing them to act accordingly and anticipate potential problems. Capability to take into account diversity of aircraft equipage and diversity of performances, will improve flexibility of operations and enable accessibility to airports even with no ground navigation infrastructure, allowing better flexibility and enhanced environmentally-friendly operations for all aircraft. The introduction of new approach and landing procedures such as Second Runway Aiming Points, Increased Glide Slope, Double Slope Approach and Dual Threshold will allow a reduction in the noise footprint on final approach and with management of ROT, collaborates to deliver a more efficient use of the runway.

New airborne and ground-based systems are introduced – able to detect collision risk with other traffic and fixed obstacles during runway and taxi operations, non-compliance with airport configuration as well as non-conformance to procedure or clearances. This includes vehicles that operate on runways and taxiways of the airport. Flight Crews receive dynamic traffic context information superimposed onto the airport layout in the cockpit, including status of runways and taxiways, obstacles, route to runway or stand.

The introduction of solutions such as remote tower services for single and multiple airports will enable operational coverage to be extended at low and medium-traffic airports. This will bring benefit to isolated communities by providing the opportunity to preserve currently uneconomic airport operations, and by stimulating local economies through the opening up of new markets. The

co-location of remote tower modules within a centre permits workforce efficiency gains, and the use of similar technology solutions bring safety and/or efficiency benefits as well.

Regional airports are developed to address notable increase in demand, due in part to growth limitations at large airports and new AU business models. Developments considered include independent access to Hub aerodromes TMAs; adapted TMAs able to share traffic with Hub TMA; precision approaches to all runway (supported by PBN IR); Low visibility operations (thanks to vision-based system); ecological procedures; and precision approaches to all runways (supported by PBN IR). Airport performance includes improved access to small airports and also addresses non-instrument runway.

Digital arrival management sees new processes cover a series of improvements to the extended arrival area, airspace design serving several airports; potential interaction with dynamic DCB; improved information exchange (i.e. SWIM); and improved accuracy of trajectory predictions through the use of airborne predictions and of adapted sequencing algorithms. Information from multiple arrival management systems operating out to extended range is integrated with local traffic information within the En Route sectors, allowing balancing of the workload and needs of the TMA and En Route. This may include the use of artificial intelligence, in particular to build more robust arrival sequences supporting airspace users' preferred trajectories. With the implementation of advanced separation techniques, the spacing task may be delegated to the Flight Deck, optimising the flow of aircraft in the TMA and reducing controller workload.

Rotorcraft traffic has improved access both to airspace in high-density areas flying an optimised Low Level IFR route and to airports with the possibility to operate under IFR using Simultaneous Non-Interfering approach and departure procedures.⁴

Digitalisation, technological evolution with new tools and automation

Increased digitalisation and implementation of virtualisation technologies enable a cost-effective performance-oriented service provision that allows an increase of the global ATM system performance. Performance monitoring becomes an integral part of the ATM system, which allows both ANSPs and AUs to choose which key performance areas they want to prioritise to serve their customers' needs. The technological evolution of civil and military airborne and ground systems coincides with the harmonisation of performance requirements. In this context, cyber security is of particular importance, securing the confidentiality, integrity and availability of ATM operations. Cyber security will be an inherent and collaborative element in the development of ATM operational solutions.

New decision-making support tools and automation provide coupling of Arrival Manager/Advanced Surface Movement Guidance and Control System/Departure Manager (AMAN/A-SMGCS/DMAN). These play a major role in Surface Planning and Routing and now link to DCB, CDM and UDPP.

⁴ In this framework the implementation of PBN and the PBN related SESAR solutions (such as those for RC), and PBCNS in the future, plays an important role.

New ATC tools provide new capabilities for trajectory and traffic complexity prediction. The introduction of standardised and interoperable systems together with complexity tools, enable controllers to manage both flows of traffic and individual flights. Controllers use enhanced knowledge of the predicted vertical behaviour of the aircraft to support Continuous Climb and Descent Operations [1] [2], bridging the transition between the TMA and both Free Route and fixed route airspace, facilitating a broad range of climb and descent profiles, which are fuel-efficient for the specific flight.

The role of the human

The role of the human and its continued importance within the concept is recognised in the future system, which takes into account the necessary balance between the efficiency created by automation and human capabilities [8].

The improvement and integration of automation is supported by digitalisation technology, enhancing conflict detection and resolution tools and conformance monitoring tools as well as agreed protocols for delegation of spacing to the aircraft in order to reduce the need for ad hoc interventions. Automation of routine tasks and wide introduction of data communication also allows controllers to concentrate on added value tasks, handling more traffic in a safer way, improving ATCO productivity and enhancing flight efficiency.

Safety

Safety is the overarching aim of the entire Single European Sky (SES) II initiative and is the central element of any technical solution developed by SESAR.

The concepts are supported by new tools designed to:

- prevent or reduce the risk of loss of separation
- reduce the risk of runway excursion
- improve runway condition awareness
- alert for taxi non-conformance and for collision avoidance in surface operation.
- access to small airport, allowing separation and collision avoidance with all the users (VFR, light IFR)

Safety nets are updated to ensure operation within the new SESAR environment and with the operations of RPAS in non-segregated areas.

Reduction in environmental impact

Management of airborne and surface routing and associated procedures is an important element in reducing the effect on climate, noise and health for direct stakeholders and the wider community.

2.1 Transversal Topics

The SESAR 2020 Operational Concept describes the practical solutions achieving the objectives and overall performance expected in Single European Sky High-Level Goals [6] as set out in the European ATM Master Plan [5]. It is the main common reference for all operationally related SESAR tasks.

The underlying SESAR 2020 Operational Concept has the objective to meet the overall SESAR Performance Ambitions whilst being coherent and consistent within the research programme. It anticipates that the portfolio of SESAR Solutions under development, together with their validations, are presented through a coherent description of their operations and related performance improvement. It aims to illustrate how they are integrated and orchestrated into the ATM system to ensure its optimum performance while allowing Airspace Users (AUs) to fly their optimum trajectories as far as possible, satisfying their business and/or operational needs.

The main Operational Improvements detailed below are based on advances in digitalisation and automation - with the use of artificial intelligence where appropriate. They are accompanied by the growing need in the aviation world for robust cyber-security, which is addressed with the securing of data exchange. Continuity of operations and resilience will be essential to maintain the trust in the aviation system.

2.1.1 Trajectory Based Operations (TBO)

ATM accuracy, predictability and reliability are significantly improved through the sharing of common reference trajectory information via ground-ground and air-ground System wide information Management (SWIM) throughout the Business/Mission Trajectory lifecycle. This is supported by sharing of robust wind and temperature forecasts. New cutting-edge ground-ground trajectory exchange mechanisms see high quality 4D trajectories data with all relevant information automatically made available to all relevant stakeholders, improving common situation awareness, automation and global performance. Increased collaboration and operational predictability is also enabled by collaborative decision-making, allowing prioritisation processes for operations and increased predictability.

A description of the Trajectory Management process is contained in the section Advanced ATS (for the execution phase) and in the section Optimised ATM Network (for the planning phase), which taken together represents the seamless lifecycle of the trajectory.

Remotely Piloted Aircraft System (RPAS) are integrated into the ATM network, at the airport, and into non-segregated controlled airspace. They are managed alongside all other trajectories to meet the required safety and performance objectives.

RPAS trajectory planning captures any specific performance/communication/coordination needs that are of relevance to a planned and optimised network. Their subsequent execution is enabled by technical capabilities and procedures to enable compliance with ATC instructions. These capabilities and procedures also allow for loss of command & control data link and other emergencies.

MET Aspects

MET information distribution and consistency between all ATM actors are enhanced thanks to a unified access to MET Information Services for ATM called the 4DWxCube.

The implementation of enhanced meteorological services allows improving predictability, flight efficiency and network performance by supporting the design of optimum trajectories.

The 4DWxCube and the new MET Information Services it supports, is the major⁵ source of MET information for all applications within ATM, ensuring a common situational awareness among all ATM actors. It ensures that the planned trajectory and the agreed trajectory are based on the best available MET information and consistent with what is available to all other actors. MET information is used for various applications during both the planning and execution phases. The planned trajectory can be initiated much earlier with MET forecasts available many days in advance.

4DWxCube information sources are enriched by MET data from on-board sensors (e.g. icing, hail, turbulence, wind/temperature) provided via air-ground SWIM and/or data link.

The 4DWxCube provides the Flight Crews during the execution phase with relevant and reliable MET observations and updates relevant for each trajectory (e.g. wind, temperature, icing, convection, turbulence and when required, satellite/radar imagery showing how hazardous clouds evolved). Uplinked forecasts allow for visualisation of MET information by on-board systems.

Flight Crew continuously receives the latest airport information, including runway in use and MET information via data link and/or via air-ground SWIM.

Military specific aspects

For military AUs, trajectory management, together with performance-based navigation (PBN) and advanced surveillance, will be fundamental features of future TBO. A key enabler of trajectory management will be high capacity air-ground data link communications and the ability of military airborne functionalities to rely on flight guidance to process trajectory parameters at the level of flight management systems/military mission systems (FMS/MMS). The compliance approach will vary with aircraft types and mission.

⁵ The NOP is a consumer of MET Data that is available from the 4DWxCube through SWIM directly and/or by other communication means than SWIM. The NOP should hold and make available the MET data that NM is using in their processes.

2.1.2 Free Route operations

Free Route operations are extended in all complexity airspaces, including very high complexity and further enabled by the introduction of a high automation level in support of different processes:

- The result is an airspace managed as a continuum, which Wide implementation of automated tools for conflict detection and resolution, flight monitoring, electronic coordination
- Enhanced ATFCM with INAP offering a more optimised granularity level for complexity management and dynamic airspace configurations management where required
- Advanced processes for airspace management are facilitating more flexible use of ARES
- Dynamic airspace configurations, with a high level of modularity and flexibility
- Flight and/or flow-centric operations, where applicable

The result is an airspace managed as a continuum, which enables an optimum use of capacity, supports a better demand and capacity balancing and offers much more flexibility for flight profiles optimisation with the use of user defined trajectories. FRA must be interoperable with the solutions presented in the CONOPS as well as their associated technical supporting systems.

Cross-border Free Route operations are planned at sub-regional level, implemented H24 and extended to high complexity areas in all flight phases (i.e. climbing, cruising and descending), including lower airspace where possible (from TMA exit to TMA entry in best cases).

The main objective of the extension of Free Route airspace is to provide Airspace Users with significant opportunities to optimise their associated flights in line with their individual operator business needs or military requirements.

Free Route airspace characteristics must be defined in order to ensure safety and best performance balance in all performance areas (i.e. fuel efficiency, predictability, capacity).

In high and very high complexity environments, the minimisation of constraints and optimisation of flight profiles should be further enabled by the deployment of advanced automated support tools. However, in case of observed or anticipated significant airspace complexity increase and potential detrimental effects on capacity or safety harming airspace users' needs and global Network performance, some constraints may still be applied to manage the containment of agreed trajectories.

In order to guarantee access to airspace to all Airspace Users (and additionally support airspace capacity performance objective), but also to facilitate air-ground communication, some published waypoints have to be maintained in FRA, and can be used at Airspace Users' convenience and ATCOs' needs.

The extension of FRA to make available free routing for Airspace Users in lower En Route airspace will especially allow for improved time and fuel efficiencies in short haul flights.

This will modify the planning principles by allowing a greater number of user defined trajectories that allow AUs to plan the most flight cost efficient trajectory from TMA exit to TMA entry. Taking the highest degree of benefit from those planning principles will require the use of advanced planning systems that are able to plan the most flight cost efficient trajectory under consideration of the ATM

constraints, the respective aircraft specifics (e.g. RNP) as well as all ambient and business requirements having effect on the flight operations.

SESAR 2020 Significant Changes

Traffic management to enable Free Route operations in high and very high complexity environments (Solution PJ.06-01) – Free Routing for Flights both in cruise and vertically evolving within high and very high complexity environments in Upper En Route airspace (AOM-0505) Management of Performance Based Free Route in Lower Airspace (Solution PJ.06-02). – Free route for flights both in cruise and vertically evolving within high complexity environments in Lower En Route airspace (AOM-0506)
Expected Benefits The main fuel efficiency benefits are expected from the extension of Free Route operations, allowing Airspace Users to plan and fly more fuel optimum tracks. Fuel efficiency benefits are also expected from Continuous Descent / Continuous Climb Operations in high-density operations.

Free Routing solutions as such are not part of wave 2 R&D. Indeed, FRA is already widely implemented in Europe and its implementation is mandatory as from the 1st of January 2022 (as per the PCP) at and above FL310. FRA will thus be the default operating environment for all the Solutions addressing the En Route OEs. Performance of the Free Route operations will be progressed further through these Solutions.

2.2 Optimised ATM Network Services

Optimised ATM Network Services and Trajectory Management

Optimised ATM Network Services aim to meet SESAR goals of increasing ATM capacity and providing greater opportunities for stakeholders to optimise their operations - in terms of operational efficiency and environmental gains. Such capacity and efficiency increases, both for the network and for individual trajectories, have to be balanced against the SESAR goal to increase ANS productivity through a reduction in ATM cost per flight.

The activities are conducted in the new environment of Trajectory Based Operations (TBO), which rely on accurate and predictable trajectories shared with all stakeholders involved in the planning process and in the execution of the flight. This allows stakeholders to improve ATM resource planning, to optimise airspace configurations, to better exploit available capacity and to reduce the impact of DCB measures.

Trajectory Management will improve the coordination between the different processes including Airspace Management (ASM), DCB, Airport Sequencing and User Prioritisation. A detailed description of the Trajectory Management process relevant to the Execution Phase is contained in the beginning of Section 2.3 Advanced ATS, whilst the elements relevant to the Planning Phase are taken for description here.

The overall Planning Phase encompasses:

- The Scheduling Phase
- The Trajectory Planning Phase.

The scheduling phase is of particular interest to this chapter, as it covers those strategic planning processes able to deliver the best capacity plan in preparation for commencement of the trajectory planning phase. Notably, little actual trajectory data is available in this phase, with evaluations based on scheduled operations and/or historical data that is extracted from data repositories such as the NOP. The scheduling phase starts with the comparatively basic information of flight intention that is first available several months before day of operation. Note that the reference to several months represents the nominal case - the overall Trajectory Management concept is sufficiently flexible to cater for a wide range of timelines. The information is progressively updated with more detailed information affecting the flight as time moves towards the Execution Phase. The scheduling phase transitions into the trajectory planning phase once the AU has provided/shared its desired trajectory (i.e. the Airspace User's optimum trajectory according to its business model including published ATM constraints).

Therefore, calculation of demand is improved significantly as a consequence of using such iterative trajectory development. This sees traditional historic/forecast demand calculations enriched through use of probabilistic demand algorithms, incorporation of planned DCB measures and accurate MET data. Furthermore, consideration is given to the level of certainty of the forecast, with an associated computation of confidence index. Future application of Big Data and machine learning (and inevitably AI) will be accompanied by the necessary automation to take advantage of a more complex and nuanced demand forecast. The resulting integrated forecast traffic, composed of forecast trajectories, is used by all stakeholders involved in holistic network planning management.

Such planning management allows precise identification of capacity needed to meet this accurately quantified demand, enabling a more efficient utilisation of ATM resource - realised as optimised planned and executed airspace configurations.

Furthermore, TBO provides the basis for management of the network through collaborative agreement. To this end, the Collaborative NOP is developed to support the network operations planning as a continuum from planning through to execution, taking accurate and reliable SWIM-shared data. Capability is expanded to support airspace configuration planning, collaborative flight planning, and Network performance monitoring. The latter providing capability for stakeholders to assess the impact of their intentions and decisions on the wider network, and hence provides a trigger for CDM to mitigate negative impact.

The Collaborative NOP is developed to enable network operations planning activity to take place irrespective of network conditions, encompassing nominal, disruption, and crisis conditions. It provides a common awareness of key elements including demand/capacity imbalances; hotspots; DCB measures; MET information - from day-to-day weather to severe events.

Lastly, TBO provides the core for management of the planned operation as it moves into execution, with integration and modification of changes requested by stakeholders to optimise their operations. Such change occurs at granular level, and may be driven by new opportunity to maximise individual trajectory efficiency, or to fine-tune traffic complexity and controller workload at local level. Such complexity and workload management is identified at an earlier stage, thanks to automated tools being able to take advantage of the improved and more accurate trajectory data. Any resultant hotspot, shared with the network, is therefore more credible.

RPAS are integrated into the ATM network to meet the required performance objectives. This integration takes into account any additional procedures and requirements associated with nominal operations and their contingency to accommodate RPAS IFR flights in non-segregated controlled airspace. Such procedures and requirements may impose new controller workload and/or complexity, leading to capacity revision and new considerations for sector/airspace configuration management.

RPAS trajectories will be planned in a way that is fully compatible with the ATM network processes, capturing any specific performance/communication/coordination needs that are of relevance to a planned and optimised network. Provided the implications of such needs are well understood, quantified, and incorporated into the relevant tools and processes for network planning and execution, RPAS operations will be successfully integrated into the ATM network alongside all other trajectories. The overview of integration of RPAS in execution is further detailed in section 2.3 Advanced ATS.

For the Military AUs new generations of military aircraft and RPAS, with innovative technologies and new tactical capabilities, require more flexibility and adaptability from the ATM network in order to accommodate military mission-specific requirements. Consequently, trajectory management becomes more dynamic and complex with enriched data content managed in real time, providing opportunities for the optimisation of entire ATM network operations.

2.2.1 Optimised AU operations

The principle whereby AUs become more involved in DCB processes to allow their business/mission needs to be taken into account wherever possible is further developed:

- In order to provide greater flexibility, AUs are able to refine business/mission trajectories in planning and revise them in execution within trajectory management processes;
- AUs are able to provide the ATM network with accurate input to better reflect demand during the planning phase. During this process they receive an improved and harmonised picture of ATM constraints (which includes MET and AIS); AUs are able to indicate their preferences (used when choosing between different trajectory adjustment alternatives for DCB purposes, e.g. time based or lateral/vertical constraint) and priorities (allowing AUs to identify which flights shall be given priority over others to best reflect their own business needs). These preferences and priorities are used in any subsequent DCB process, including those managed at airport level.

User Driven Prioritisation Process (UDPP) primarily exists to provide AUs with operational flexibility to respond to delay allocated to manage capacity constrained situations at departure, En Route, or arrival. AUs are able to recommend a new priority order for their flights within the scope of dDCB or sequencing tools (i.e. in execution) that can be driven by AU business interest. Such reprioritisation requests are not restricted to situations of demand/capacity imbalance. Integration takes place with both airport and network processes to ensure consistency with the operational needs of affected ATSUs. UDPP is framed by high-level principles and rules for equity that include transparency, open access, and guard against negative impact on other stakeholders.

For military AUs, optimised operations offer increase in operational flexibility with better access to airspace, and are enabled through the use of the 4D trajectory - specifically the Mission Trajectory (MT) concept. Importantly, this concept does not challenge those military operations, which constitute national defence and security – these are out of scope of SESAR R&D.

The MT concept facilitates the integration of military IFR flights and military mission-specific requirements into ATM network operations, but this should not restrict other airspace users exploiting the concept in circumstances where their operational requirements cannot be fulfilled by applying the concept of business trajectory.

For military aircraft operators, the utilisation of the MT concept as a way to integrate flight intent and profile containing Airspace Reservation (ARES), allocated via ASM, represents a significant operational improvement. It reflects how complex processes managed by different actors and composed of several activities in the planning phase can be expressed using 4D trajectory data - hence sharing respective MT data with all ATM actors involved

Military AU operations are strongly supported by the Wing Operations Centre functionalities (WOC) with the system support that are performed at different levels based on the architecture of the military national ATM systems. This is a key capability for the military AU that facilitates the trajectory development, planning, execution, and management and interfaces with ATM network.

2.2.2 Advanced Airspace Management

Trajectory Based Operations provide the opportunity to better understand demand, and through improvements made to calculating confidence of predictions using forecast business trajectories, resources are optimised to meet forecast traffic demand and its associated workload. Optimisation of airspace configurations forms part of this process.

In order to best accommodate optimised trajectories, the Dynamic Airspace Configurations (DAC) concept is used to deliver an optimised and coordinated organisation of:

- airspace reservations (ARES) i.e. DMA (Dynamic Mobile Areas) of different types, and
- the ATC sector configuration,

This is achieved through planning, designing, and dynamically managing these capacity elements in a flexible way to meet required performance objectives and/or requirements.

The concept sees full integration of ASM/DCB/ATS processes, utilising CDM, and supported by automated tools. Configurations are fully dynamic, with a high level of modularity and flexibility up to the execution phase. The concept takes in new aspects as an input into the DAC process, such as the Hazard Zone, encompassing dynamic airspace volumes containing significant meteorological phenomena. Airspace/sector building block architecture is used as the basis for sector configuration planning, and includes dynamic definition and delineation in addition to predetermined boundaries. This provides a significantly greater number of sector configuration possibilities. The building blocks principle provides the possibility to assemble the airspace, with ARES activations, sector and route organisations managed throughout Europe in a seamless continuum.

Offering new opportunities in the context of the Advanced Airspace Management concept, ARES types - based on VPA and DMA design principles - are agreed upon in a CDM process, and may form part of the holistic trajectory description. The trajectory data, shared via ATM network systems for DCB and DAC processes, aims to fulfil the military mission requirements whilst minimising the potential impact on the wider ATM network performance. The complexity of the task involving dynamic management and dynamic updates, together with the interaction between processes, makes automation and new modelling capability for decision-making a necessity.

The percentage of non-equipped civil and military aircraft either not or partially capable for operations in TBO environment will require special treatment. In order to facilitate operations of such aircraft in TBO environment an aircraft-centric DMA (type 3) may become a solution and of utmost importance and benefit for military AU and ATM network to accommodate such flights in the ATM network.

2.2.3 Advanced DCB

Enhanced and fully integrated processes ensure that all levels of the network – regional, sub-regional, and local – across all ATM phases, are able to manage their operation to meet performance targets in a seamless and fully collaborative basis.

As noted earlier, the core of effective DCB across the planning and execution phases is trajectory management, making full use of iterative development of the 4D trajectory. The increase in information quality, improved accuracy of demand forecasting, together with the shared view and interface between DCB actors, provides new opportunities for Airspace Users to become involved (via Holistic Planning Management) and to better manage their business needs within the overall DCB process.

The integration and necessary coordination of constraints to address demand/capacity imbalance into 4D trajectories is required to ensure stability and performance of the network. The integration of these constraints from a number of sources – e.g. airports (including multi-APOC), ANSPs, AUs and NM – is achieved with new rules and mechanisms replacing legacy first-planned/first-served slot allocation. Holistic Planning Management allows each stakeholder, in particular the Airspace User, to get a comprehensive view of all constraints. This view provides the opportunity to find the global optimum among the various stakeholders' requirements. It includes Airspace Users' preferred trajectories and costs associated with deviation, as well as ATM Planning constraints issued for Airport optimisation, ASM, DCB, ATC sequencing and flow optimisation. Holistic Planning Management can be seen as a toolbox evolving along the timeline from planning to execution, where the different actors, can use the tools and what-if capabilities as and when needed, but always able to consider the network view.

Network Performance KPA/KPI allows the network to be monitored continuously, enabling targeted action to address inefficiency. Stakeholders are able to assess and evaluate the effect of their intentions and decisions using the Network Impact Assessment functionality, with the support of what-if capability. This capability is expanded to provide problem resolution, using multiple scenarios across the ATM layered planning spectrum. Such scenarios and associated tools (utilising what-if or automated resolutions) have different levels of granularity appropriate to the task. The availability of query mechanisms and what-if functionalities provide stakeholders with the information to support their needs e.g. 4D trajectory planning, DCB decision-making and approval processes. The consequence is an enhanced CDM and ultimately greater stability, with a reduced coordination need for such strategic tasks to be factored into the execution phase.

The integration of DAC with 4D targets takes place to provide the complete DCB solution. This solution sees the optimised adaptation of capacity to meet a minimally adjusted demand. Iterative and integrated CDM processes integrate ASM and DAC into advanced DCB and ATC planning, taking place from regional to sub-regional and local levels in the planning phase. The result is an improvement in ATM resource management, with increased efficiency for both airspace and ground resources.

The principle of holistic management continues into execution, where new constraints such as arrival sequencing are – where appropriate – reconciled with other ATM constraints, taking into account AU preference and priority wherever possible.

Complexity and its management form a key capability within Advanced DCB. Quantitative and qualitative assessments of complexity take into account the uncertainty of the trajectory prediction over time, aiming to capture the relationship with workload. Advanced airspace environments (including modular airspace and Free Route airspace) necessitate either new or evolved complexity algorithms to indicate instances of excessive workload.

Much is made of the integration of processes to provide an effective planned operation up to execution. However, without a similar degree of integration with tactical ATC, the expected performance benefits are at risk. To address this, the concept of Integrated Network Management and (extended) ATC Planning Function (INAP) is developed, the scope of which is to bridge the traditional gap between network activities, managing DCB, and ATC planning activities, managing and fine-tuning the tactical picture. This period within which INAP is applicable sees overlapping capabilities in order to deliver successful integration of network measures, local complexity management, traffic synchronisation and airspace & ATC sector configuration management within the context of a more localised area of responsibility. INAP actors are provided with the capability to identify and resolve local hotspots in a fully integrated way. They are able to assess potential actions to resolve these hotspots within the context of an evolving traffic picture close to execution, and evaluate such actions in terms of performance.

SESAR 2020 Significant Changes

Wave 1

PJ.07 - Optimised AU Operations

PJ.07-01 AU processes for trajectory definition:

FF-ICE Planning Service is provided, representing a new phase in the flight planning process, where a preliminary flight plan is prepared and coordinated using CDM prior to FPL filing and distribution to ATSUs. PJ.07-01 is the focus for refinement of operational requirements for this Planning Service, which aims to provide additional benefits to AUs through improved outcomes from ATFCM/ASM processes.

- Preliminary flight planning. AUO-0207
In the flight planning phase, SBT management starts with provision by the AU of the Preliminary Flight Plan (PFP), which triggers trajectory negotiation processes and ATM constraints information along the planned trajectory. Provision of PFPs will result in more efficient ASM and DCB planning processes from D-6 onwards.
- Use of simple AU Preferences in DCB processes. AUO-0208
Provided by the AU, this information can be taken into account in DCB processes, reducing the risk of application of penal constraint to critical flights.
- Use of enriched DCB information and enhanced what-ifs to improve AU flight planning. AUO-0219
Information relating to DCB constraints and measures with a view of hotspots is available to AUs to improve their decision-making processes when planning or replanning trajectories.

PJ.07-02 AU fleet prioritisation and preferences (UDPP):

AUs are able to recommend a priority order to NM and Airport for flights affected by delay at departure, arrival, and En Route, with the ability to share preference with other ATM stakeholders in Capacity Constrained Situations (CCS).

- Reprioritising flights during execution. AUO-0106
AU request for change in the priority order for flights within the scope of dDCB or sequencing tools

can be managed at NMF or airport-level. Such reprioritisation request can be driven by AU Business interest, and is not restricted to demand/capacity imbalance

- Prioritisation for Low Volume Users in a Constraint. [AUO-0107](#)
Enables AUs with few flights affected by an airport arrival/departure constraint in planning to recommend a priority order outside the UDPP process.
- UDPP for Airport constraints. [AUO-0109](#)
Prioritisation of an AU's group of flights within a single airport arrival/departure constraint, addressing also the need to integrate with the reconciliation process for the wider network.
- UDPP for Network constraints. [AUO-0110](#)
Prioritisation of an AU's group of flights within one or multiple network constraints.

PJ.07-03 Mission Trajectory driven processes:

Through the full integration of the WOC within the ATM system, this refers to the processes to manage the SMT/RMT. These processes reflect the need to accommodate individual military AU needs and priorities, without compromising optimum ATM system outcome and performance of all stakeholders.

- Pan-European OAT Transit Service. [AOM-0303](#)
Provides timely and flexible availability for routing and services to military missions transiting to training/exercise areas and to overflights across states.
- Improved and harmonised OAT flight plan. [AOM-0304-A](#)
The iOAT FPL constitutes the initial description of the MT (the iMT) and once delivered for integration into the ATM systems for processing and integration, becomes the iSMT.
- Participation in CDM through iSMT and Target Time (TTO) negotiation. [AUO-0210](#)
Data describing the iMT profile is shared via the iOAT FPL in medium/short term planning and represents the iSMT. The sharing of this data may trigger CDM processes to balance requirements – realised as target times.
- WOC Management of iRMT via improved OAT FPL. [AUO-0211](#)
The iRMT is published in the NOP, and contains the 3D profile, ATM-specific data, CDM-agreed target times and associated tolerances. The WOC may revise the iRMT in the execution phase through the iOAT FPL.
- Sharing iSMT through improved OAT flight plan. [AUO-0215](#)
The iSMT, represented by the iOAT FPL, is shared through ASM processes in medium/short term planning permits focus on ARES demand. Any changes subsequent to filing of the iOAT FPL - until transformation into the iRMT – will follow flight plan management processes.
- Agreed iRMT. [AUO-0228](#).
The iRMT is the partial implementation of the MT, used as the reference by all ATM partners during flight execution for flights using ARES.

PJ.08 - Advanced Airspace Management (ASM):

PJ.08-01 Management of Dynamic Airspace Configurations:

Refers to development of process, procedures, and tools related to DAC, supporting Dynamic Mobile Areas of Types 1 and 2. It aims to activate airspace configurations through integrated CDM at local to regional levels from planning through to execution.

- DMA Types 1 and 2. [AOM-0208-B](#)

DMA Type 1 is a defined-dimension airspace volume described as an integral part of a MT – flexibly geographically located satisfying AU requirements in terms of time/distance from an AU-defined reference point.

DMA Type 2 is a defined-dimension airspace volume described as an integral part of a MT – planned and used at any geographical location along the trajectory.

- Collaborative Airspace Configuration. [AOM-0805](#)
Airspace configurations are activated through integrated CDM processes from local to regional level, with airspace configurations managed as a continuum to meet users' expectations.
- Initial Sector design and configurations are unconstrained by predetermined boundaries. [AOM-0809-A](#)
Design principles use dynamic definition and delineation in addition to traditional sectorisation principles for en-route sectors.
- Dynamic Airspace Management based on complexity. [CM-0102-B](#)
Airspace is managed as a continuum, with dynamic configurations able to accommodate ENR/TMA environments at all complexity levels.

PJ.08-02 Dynamic Airspace Configurations supporting moving areas

Dynamic Mobile Areas (DMA) of Type 3:

DAC supporting moving areas extends to DMA Type 3. This includes an impact assessment and integration into the DAC process of areas that are potentially unsafe due to weather phenomena that can evolve in 4-D.

- Use of DMA of type 3. [AOM-0208-C](#)
DMA Type 3 is a defined-dimension airspace volume described as an integral part of a MT and is aircraft-centric - designed around moving aircraft, resulting in specific separation criteria for the aircraft configuration.
- Integrate Hazard Zones in DAC process. [AOM-0209](#)
Describes the dynamic airspace volume within which, e.g. met phenomena hazardous to traffic may occur. The hazard zone is available for use as an input into DAC process.

PJ.09 - Advanced DCB

PJ.09-01 Network Prediction and Performance:

Improved traffic and demand forecast based on SBT, with computation of confidence indexes, allows increased accuracy in diagnosis of complexity and associated DCB constraints. Network Operations are monitored through network Performance KPA/KPI, whilst Network Impact assessment provides capability to analyse trade-offs and facilitates CDM processes.

- Automated support for traffic complexity assessment. [CM-0103-B](#)
Automated tools monitor and evaluate traffic workload and complexity in defined airspace volumes, and provide accurate and timely prediction of upcoming congestion and input to tools handling hotspot/complexity resolution.
- Traffic and demand forecast in 4D trajectory management context. [DCB-0211](#)
Enhanced and continuous demand forecasting is possible through the exploitation of the shared and iteratively developed SB/MT. It includes enrichment of traffic demand enrichment through e.g. probabilistic demand and consideration of planned DCB measures.
- Network Performance assessment for distributed network operation. [DCB-0212](#)

Network Performance KPA/KPI allows network operations at all levels to be monitored continuously. Stakeholders are able to evaluate the network impact of their intent and decisions on capacity and QoS performance (i.e. flight efficiency, predictability, flexibility) using what if tools and Network Impact Assessment functionality.

PJ.09-02 Integrated Local DCB Processes:

Addresses the integration of Local Network Management with extended ATC planning and arrival management activities in short term to execution in a seamless process.

- Automated support to INAP (Integrated Network Management and Extended ATC Planning) function. [CM-0104-B](#)
Local roles within INAP are able to assess and resolve local complexity issues, utilising best performing option from measures available from DAC, ATFCM and trajectory adjustment. Automated interface between NMF and ATC Planning provides opportunity to deliver solutions with best performance effects in terms of service and use of ATM resource.
- Ground based automated support for managing traffic complexity across several sectors. [CM-0302](#)
Note – this OI relates primarily to ATC, so is described in Advanced ATS chapter, section 2.4.3
The system supports INAP actors in smoothing of traffic flows and deconflicting flights across the INAP Area of Responsibility (AoR).
- Full integration of Dynamic Airspace Configurations into DCB. [DCB-0210](#)
The incorporation of DAC into the complete DCB solution, that includes 4D constraints, to optimally adapt capacity to meet demand and minimise any adjustment to that demand.
- Consolidation and facilitation of Target Times between local DCB, Airport CDM and Extended Arrival Management. [DCB-0213](#)
Where constraint from Airport CDM or traffic sequencing activities (E-AMAN) overlay local network constraint (STAM), collaborative trajectory revision may be required to ensure effective reconciliation and minimised impact on AU operations.

PJ.09-03 Collaborative Network Management Functions:

Addresses the functions required to allow the network to operate transparently, to performance targets. Real-time evolution of the AOP/NOP supports AU and local planning activity. Network operations in planning and execution utilise agreed rules and procedures to manage trade-offs within the context of network performance targets. Collaborative 4D constraint management employs multiple constraint resolver process.

- Most penalising delay based on reconciliation between DCB and Airport CDM. [AUO-0108](#)
Reconciling DCB measures with airport sees A-CDM most penalising delays taken into account as per other congestion points within the network.
- Collaborative NOP. [DCB-0103-B](#)
The NOP is developed as a continuum from planning to execution, providing common awareness on imbalance, hotspots, DCB measures, and CDM status, with improved accuracy and reliability of shared data and prediction.
- DCB what-if network assessment. [DCB-0214](#)
Network impact assessment relies on what-if capability to support decision-making processes in the context of UDPP, A-CDM and other APOC processes.
- Consolidation of imbalances and arbitration of Trajectory Management solutions. [DCB-0215](#)
Integration of 4D constraints issued from multiple sources are managed through new rules and

mechanisms (including AU priority and preference, 4D targets and tolerance windows, arbitration of TM solutions) that replace current first planned-first served slot allocation.

- DCB support to FF-ICE. [DCB-0217](#)
NOP provides evolving DCB information to AUs to support them in calculating optimal trajectories (or less-constrained trajectories). The information is provided earlier than currently, supporting the preliminary flight plan concept in FF-ICE.

Wave 2

PJ07 Optimised AU Operations:

PJ.07-W2-38 Enhanced integration of AU trajectory definition and network management processes:

The impact of ATM planning on AU costs of operations is reduced by providing them a better access to ATM resource management and allowing them to better cope with ATM constraints. Enhancing the CDM planning between the providers and the users ensures a better adherence to the agreed trajectory during execution, hence a better predictability on the traffic demand.

This addresses improved FOC participation into the ATM network collaborative processes in the context of FF-ICE and its potential evolutions. The focus is on the planning phase and SBT management, but also considers the execution phase regarding NMF/FOC trajectory negotiation/information exchange. The following aspects are included:

- Provision of enriched DCB information
- Automation of AU flight planning/network management interactions
- Anticipated trajectory planning and negotiation to support pre-tactical phase processes
- Provision by the AU of preliminary flight plans as early as possible
- AU involvement in the application of STAM via extended collaborative advanced planning
- Assessment of interaction with the NOP
- Inclusion of AU priorities and preferences in the flight-plan
- Development of working methods AU trajectory definition/negotiation together with DCB
- Consideration of business aviation specificities (not always having a FOC)
- - Preliminary flight planning. [AUO-0207](#)
 - Use of Simple AU Preferences in DCB Processes. [AUO-0208](#)
 - Use of Enriched DCB Information and Enhanced What-Ifs to Improve AU Flight Planning. [AUO-0219](#)
 - DCB Support to FF-ICE. [DCB-0217](#)
 - FOC Management of the Reference Business Trajectory. [AUO-0206](#)

PJ.07-W2-39 Prioritisation strategies for arrivals including UDPP

The Collaborative framework is applied to the coordinated management of arrival delay constraints resulting from capacity issues, with streamlining of all prioritisation processes occurring in the planning phase: airports APOC, LTM/ATC at airport and in TMA, extended arrival management, en-route FMP and Network.

Arrival demand/capacity imbalance may use target times allocated to specific flights instead of using a classical regulation. The priority that AUs assign to every flight involved in the imbalance is key to the process of selecting flights for target time allocation, and the aim is to bring an improved integration these priorities in the target time management process.

UDPP is a key process within this solution, whereby AUs are able to express their prioritisation within the context of their own operation and business.

- UDPP for Airport constraints. [AUO-0109](#)
- UDPP for network constraints. [AUO-0110](#)
- Consolidation and facilitation of Target Times between local DCB, Airport CDM and Extended Arrival Management. [DCB-0213](#)

PJ.07-W2 -40 Mission Trajectory management with integrated DMAs Type 1 and 2

Addresses the improvement in use of airspace capacity for both civil and military AUs with increased efficiency of airspace management with introduction of automation and increased flexibility in the civil-military coordination. The focus is on the planning phase, using the WOC as a key actor (specifically the coordination with NM) to connect MT management with ARES booking (DMA Type 1 and Type 2).

Both system development and further automation are required to support the AUs to define DMAs and develop their trajectories. This includes “What-if” tool functionality and automated tool functionalities for the Airspace Management process, as well as the potential use of AI, machine learning techniques to improve this process.

- Pan-European OAT Transit Service. [AOM-0303](#)
- Improved and Harmonised OAT Flight Plan. [AOM-0304-A](#)
- Participation in CDM through iSMT and Target Time (TTO) negotiation. [AUO-0210](#)
- WOC Management of iRMT via improved OAT FPL. [AUO-0211](#)
- SMT including military user preferences associated to meteo. [AUO-0213](#)
- SMT including user preferences and mission trajectory information for DCB processes. [AUO-0214](#)
- Sharing iSMT through improved OAT flight plan. [AUO-0215](#)
- Agreed iRMT. [AUO-0228](#)
- Dynamic Mobile Areas (DMA) of types 1 and 2. [AOM-0208-B](#)

PJ09 Digital Network Management Services

PJ.09-W2-44 Dynamic Airspace Configurations:

DAC will allow managing dynamically all capacity elements and constraints in one single, seamless process, increasing the granularity and the flexibility in the airspace configuration within and across ANSPs’ areas of responsibility and ensuring DAC integration with DCB within the INAP (Integrated Network Management & ATC Planning) processes. This fully integrated civil –military ASM/ATFCM/ATS CDM layered process will ensure the best possible balance of the different performance targets and operational requirements.

- Collaborative Airspace Configuration. [AOM-0805](#)
- Initial Sector Design and Configurations Unconstrained by Predetermined Boundaries. [AOM-0809-A](#)
- Advanced Sector Design and Configurations Unconstrained by Predetermined Boundaries. [AOM-0809-B](#)
- Dynamic Airspace Management based on complexity. [CM-0102-B](#)
- Full integration of Dynamic Airspace Configurations into DCB. [DCB-0210](#)

Note: The scope of the solution includes as well the integration with other SESAR solutions e.g. DCB, sector team operations, and therefore the Solution may need updates in terms of the OI steps.

PJ.09-W2-45 Enhanced network traffic prediction and shared complexity representation:

Addresses improvement in accuracy of the network manager traffic prediction from medium-term planning phase to, and including, execution. Using on new trajectory management features such as the preliminary FPL, existing methodologies and algorithms for demand prediction and regional complexity assessment are adapted.

Calculation of Predicted flight data (PFD) is enhanced, with advances to the PFD route assignment based on AI. These elements support the concept of the dynamic/ rolling predicted flights that represents a step ahead from the more static today's approach. Integration of preliminary FPL data from FF-ICE on day of operation, to which imbalances/complexity from local are added to enable relevant actors to access accurate and timely predictions of imbalance. Accuracy of predictions are further improved in execution through integration of the downloaded trajectory with ADS-C EPP.

- Automated Support for Traffic Complexity Assessment (the part related to complexity assessment at network level). [CM-0103-B](#)
- Traffic & Demand forecast in 4D trajectory management context. [DCB-0211](#)
- Collaborative NOP. [DCB-0103-B](#)
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PJ.09-W2-47 Network optimisation of multiple ATFCM time-based measures

Addresses improvement to efficiency and reduction in adverse impact of multiple regulations affecting the same flight or flows.

Use of Big Data and machine learning to identify best solutions to manage particular traffic load patterns based on historic data – coupled with use of network simulations in order to develop optimised strategies for the European ATFCM network.

- Most penalising delay based on reconciliation between DCB and airport CDM. [AUO-0108](#)
- Consolidation of imbalances and arbitration of trajectory management solutions. [DCB-0215](#)

PJ.09-W2-48 Digital Integrated Network Management and ATC Planning (INAP):

Addresses the provision of an automated interface between local NM and ATC planning to assist controllers in alleviating traffic complexity, traffic density and traffic flow problems. In particular:

- Enhance network performance through coordination between DCB and DAC;
 - Provision of support to Multi Sector Planner (MSP)/Extended ATC Planner (EAP) to better manage traffic in a multi-sector/multi-unit and dynamic airspace management environment;
 - Provision of support to INAP actors for the identification and resolution of local hotspots in a fully integrated way;
 - Integration of FMP and Extended ATC Planning activities into a single working environment;
 - Enhancement of coordination between all involved actors – including AUs – with CDM;
 - Taking into account additional information from CDM to provide the best trade-off between optimising individual trajectories and ATM resources;
 - Development of a shareable Congestion Level Indicator.
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- Automated support to INAP (Integrated Network Management and Extended ATC Planning) function. [CM-0104-B](#)
 - Ground based Automated Support for Managing Traffic Complexity Across Several Sectors. [CM-0302](#)
 - Automated Support for Traffic Complexity Assessment. [CM-0103-B](#) (scope limited to the local tools, since network aspects are covered in candidate solution 45).

PJ.09-W2-49 Collaborative network performance management

Addresses the provision of a common framework and toolbox to allow assessment of Network Performance in the pre-tactical and tactical phases, in order to further establish a performance- driven network management culture.

Current network performance management process (focused on delay reduction) is complemented by a shared situational awareness and an agreed set of shared performance indicators to be used for real time performance monitoring in support of network operations - including trade-off indication and “what-if” impact assessments.

The concept of Network resilience and network states is developed, with a clear focus on more data driven prediction of network state evolutions, and includes development of advanced performance monitoring tool to improve awareness of network performances.

The consolidation and performance trade-off analysis aims to regularly balance stakeholder needs by an arbitration process among all stakeholders to identify most pressuring performance needs. The fair spread of advantageous and disadvantageous decisions for individual stakeholders shall be one of the major criteria of the envisaged collaboration concept.

- Network Performance Assessment for Distributed Network Operation. [DCB-0212](#)
- DCB What-if Network Impact Assessment. [DCB-0214](#)

Expected Benefits

Wave 2 aspects/solutions represented in the table at the end of each PJ. The list is done per project and KPAs are set according to the PJ.

Wave 1**Optimised AU operations**

- **Environment/Fuel efficiency** (In SESAR 2020 this KPA includes Fuel Efficiency and only partially reflects the AU driver “Flight Efficiency”):
- - Increased efficiency due to AUs ability to plan optimised trajectories that can best consider their efficiency requirements, in tune with other ATM stakeholders, enabling optimised flight trajectories and profiles with the end result being reduced fuel burn and CO2 emissions;
 - Increased network efficiency from AU participation in a new planning service, delivering greater accuracy and ensuring ATM constraint is applied only when required. This increases the possibility that flown trajectories will replicate preferred trajectories – thereby reducing overall fuel burn and CO2 emissions.

Access & Equity:

- Trajectories calculated and planned by the AU are Integrated and shared for inclusion within trajectory management processes that feed into enhanced DCB processes;
- Priority rules are always applied in a transparent, equitable manner;
- More options will be available than just the ‘first come first serve’ rule;
- All types of AUs are able to participate in the DCB process and have an influence in the allocation of ATM constraint.

Network

Flexibility: ○ Flexibility for the AU to plan and re-plan a flight throughout the whole lifecycle in order to take advantage new opportunity is increased; ○ Flexibility is increased through AU-defined preference and priority order for flights, allowing slot swapping. – <u>Advanced Airspace Management</u> Capacity: ○ Capacity is increased as the result of an optimised ASM process integrated into the DCB process, able to identify the best capacity configuration to meet demand with resources available; ○ Capacity is increased through integration of Mission Trajectories – which includes ARES - into network demand, permits more efficient use of available network capacity; ○ Capacity is increased as a result of greater accuracy of traffic demand tied to complexity and ATCO workload assessment in the prediction of ATC capacities, with sharing of this data via NOP and through integrated and holistic planning. Cost Efficiency: ○ With reference to benefits to military AUs (whose performance aspects are not shared with the wider ATM network), the overall benefits primarily arise through integration of military flights into the ATM network operations via the MT concept, and bring: – training flight efficiency; – flexibility to accommodate mission requirements and short notice changes; – optimisation and synchronisation of combined missions; – easy de-confliction of military demands; – optimised allocation of En-Route air-to-air refuelling areas; – effective cross-border operations. <u>Advanced DCB:</u> Environment/Fuel efficiency: ○ The average fuel burn/CO2 per flight is reduced thanks to a full integration of Dynamic Airspace Configurations into DCB processes. Cost Efficiency: ○ Costs per impacted stakeholders associated with enhanced ATFCM Processes are reduced. The interaction of dDCB with ATC through the INAP function, facilitated through tools to manage complexity and coordination between the actors involved, allows for an optimised use of the available capacity. ○ Complexity and ATCO workload are better predicted and managed, enabling more efficient use of resource. ○ Integrated planning processes allows improved ATM resource planning and better use of existing capacities leading to a reduction in ATC cost-per-flight. From an AU perspective, this integration provides opportunity to increase their cost-efficiency. ○ Performance parameters in network operations allow a performance-driven resolution of network constraints, improving the efficiency of network operations. Capacity: ○ Increased accuracy of capacity prediction together with adaptive capacity	
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capabilities will contribute to minimise demand modification. ○ Increased capacity results from greater accuracy of traffic demand tied to complexity and ATCO workload assessment in the prediction of ATC capacities, with sharing of this data via NOP and through integrated and holistic planning ○ En Route Throughput (in busy period and high capacity, high complexity En Route airspace) is increased thanks to a full integration of Dynamic Airspace Configurations into DCB processes as well as a coordinated 4D constraints management. In addition, the interaction of dDCB with ATC through the INAP function, facilitated through tools to manage complexity and coordination between the actors involved, allows for an optimised use of the available capacity. ○ Lower performance aircraft e.g. RPAS operations where appropriate, are fully integrated into the network via 4D trajectory management processes, and are consequently absorbed in a way to optimally manage capacity. Punctuality and Predictability: ○ Punctuality and Predictability - are improved as a result of better planning and sharing of intentions, thanks to the 4D Trajectory concept in a TBO context. ○ Punctuality and predictability are increased thanks to a full integration of DAC/ASM into DCB processes, allowing an optimised use of airspace to support preferred trajectories. ○ Predictability of the network is improved with the latest, most accurate information (including MET, constraints, hotspots...) relating to the Network Situation status available via the NOP and its integration with AOP. ○ The number of flights arriving on time (as planned) is increased thanks to a full integration of dynamic airspace configurations into DCB processes as well as a coordinated 4D constraints management. <u>Wave 2 Performance Expectation</u> PJ07 W2 Optimised Airspace Users Operations This topic will develop solutions that are expected to have a positive impact on the Network improving: ○ Capacity - based on a better integration of the AUs trajectory definition and the network demand and capacity function; ○ Cost-efficiency – reduction in AUs' operating costs with increased automation; ○ Flexibility - by allowing the airspace users to recommend to the network management function a priority order for flights; ○ Environmental sustainability (fuel efficiency) - thanks to the use of the preferred trajectory taking into consideration the Airspace users' needs; ○ Punctuality and predictability of individual flights - better adherence to the agreed trajectory thanks to the collaborative framework of the trajectory management. PJ09 W2 Digital Network Management Services This topic will develop solutions that are expected to have a positive impact on the Network improving: ○ Capacity - through a dynamic airspace management responding with flexibility to the Airspace Users' flight trajectory needs; ○ Safety - through better anticipating and managing potential overloads; ○ Efficiency - thanks to the monitoring of the DCB measures and network performance and the implementation of corrective actions;	
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○ Cost-efficiency - DAC allows improved ATM resource planning and better use of existing capacities leading to reduced ATC and airport capacity costs as well as reduced impact of DCB on AUs' operations. More accurate and appropriate definition of DCB solutions induces as well reduction in DCB impact. Increased automation involves reduction of operating costs; ○ Predictability - the NOP will provide the planned network situation considering all known constraints and time deviation will be managed by anticipating demand/capacity imbalance detection and improving the implementation of DCB solutions; ○ Flexibility - common awareness to all stakeholders through the NOP and access to opportunities in case of late changes in capacity or demand.	
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2.3 Advanced Air Traffic Services

2.3.1 Advanced Air Traffic services and Trajectory Management

Trajectory Based Operations in the execution phase will rely on accurate and predictable trajectories shared with all stakeholders involved in the planning process and in the execution of the flight.

Thanks to the introduction of new cutting-edge ground-ground trajectory exchange mechanism (e.g. Interoperability (IOP) and the SWIM infrastructure), high quality 4D trajectories data with all relevant information are automatically made available to all relevant stakeholders, improving common situation awareness, automation and global performance and facilitating the ATM Processes in the flight execution phase.

Increased automation will allow substantially reducing controller task load per flight, while meeting safety and environmental SESAR goals (including fuel efficiency), thus contributing to a reduction in ATC cost per flight and increase in performance via optimisation of network and individual trajectories. However, human operators will remain at the core of the system (in the role of overall system managers) using automated systems with the required degree of integrity and redundancy.

Meaning and Scope of the Major Trajectory Management Processes

Trajectory Management, as the backbone of future ATM operations, reflects on-line the progress of the flight whilst concatenating and supporting the optimisation of the various activities of the stakeholders, be it the Airports, the Airspace Users (all categories), the ANSPs and the overall network itself.

Once appropriate for all stakeholders, the trajectory is published by the Airspace Users and becomes the reference for the flight – the Reference Business Trajectory (RBT). This corresponds to the start of the execution phase⁶. A detailed description of the Trajectory Management process relevant to the Planning Phase is contained in the beginning of Section 2.2 Optimised ATM Network Services, whilst the elements relevant to the Execution Phase are taken for description here.

In order to appropriately manage the progress of the flights, a series of data specific to each flight is created and shared amongst concerned stakeholders according to specific rules and access rights⁷.

This data represents the shared knowledge of the Trajectory and provides common flight and trajectory information across multiple stakeholders, ensuring that all systems have a consistent view of the flight, and that the data is widely and easily available, subject to appropriate access controls. The fundamental idea is that a single logical entity is kept up to date by all parties wishing to share

⁶ The “execution phase” in this document is seen from an ATM perspective. It has to be distinguished from the flight execution, which starts at the “start-up/push back” and ends at “engine shutdown”; as well as from the “in flight phase” extending from Take-off until landing.

⁷ The detailed content of such data set will be specified during the research activity, preferably in liaison with the global initiatives at ICAO level - FF-ICE (Flight and Flow Information for a Collaborative Environment) and Flight Information Exchange Model (FIXM).

information about a flight. All parties use it as a reference and all keep it updated with the latest information.

In addition, local entities may possess additional/complementary local data in order to manage the flight within their own environment.

At the end of the planning phase the trajectory is consistent with constraints and related tolerance windows. When the AU explicitly accepts this final trajectory it becomes the reference for the flight. Early stability of the trajectory is important for stable Network planning; partners contributing to enhanced predictability shall be encouraged by the network planning system (e.g. through prioritisation, special access to ATM resources). This will also encourage accurate planning and avoidance of last minute changes. For Mission Trajectories the stability is subject to technical and weather related changes close to execution as well as re-tasking of missions.

The Flight Crew is made aware in particular of specific conditions leading to the selection of the agreed trajectory.

Execution Phase

The first instantiation of the RBT contains in particular the trajectory respecting the planning parameters (and TSAT/TTOT with related tolerance windows at ADEP). For the airport the operational information affecting one aircraft is reflected in the Airport Transit View (ATV).

SESAR ultimate objective is to manage the trajectories as close as possible to AU optimal trajectory including lateral, vertical and time elements, using as few ATC clearances as possible. This optimum trajectory is to be facilitated by all actors, with as little intervention as possible.

SESAR 2020 will allow significant progress towards the full achievement of this goal, thanks to advanced systems, tools and procedures for coordination and data sharing between stakeholders and the collaborative development and management of optimised trajectories.

Throughout the execution phase, the monitoring and management of changes to the trajectory, as well as their distribution to all relevant partners are performed through several key processes described below.

Trajectory monitoring during the flight is performed, besides current modes of surveillance, through the following new means:

- Automatically, through Trajectory Management Requirements (TMRs) generated by the ground system and communicated to the airborne system as part of the surveillance contract. Whenever the on board predicted trajectory differs from the previously shared predicted trajectory by more than any of the 4D thresholds specified via these TMRs, the airborne systems automatically downlinks an update of its trajectory prediction.
- 4D Trajectory Management parameters (Target Times at Target Points (hotspots in particular), Target Altitudes or any combination thereof. Associated tolerances are “windows” within which the actual trajectory may be executed without affecting the capacity or safety of the associated environment.)

Following a downlink of the FMS predicted trajectory to the ground system (e.g. through ADS-C EPP), a **consistency check** is performed, which ensures that the trajectory in the Airborne Navigation

System and the cleared trajectory data held in the ground system are consistent. This is done in order to ensure safety and to enable the use of the downlinked data into the ground systems.

An **update of the trajectory** is automatically triggered when the local trajectory predictions continuously computed by the ground and/or aircraft system(s), differs (according to specific conditions) from the previously shared trajectory predictions. The sharing of the update with other ground stakeholders will then be triggered according to interoperability requirements.

The **revision of the trajectory** is a process to address a need or request for change in specific elements of the trajectory data currently shared in the SWIM environment. This process is associated to a need for a decision to be made by two or more stakeholders.

When there is a need for revision of the RBT and time/workload permits and specific conditions are met, a CDM process is triggered based on a proposal made by the initiator of the revision request. An impact assessment on the proposed change is then performed through CDM, including possible adjustments through negotiation amongst the interested stakeholders, leading to an agreement on the change.

A revision process ends when the final decision is made and in case of modification of related Trajectory data, when the Flight Crew acknowledge the trajectory change.

If the final decision implies a modification of the trajectory data, a **Trajectory Synchronisation** process will be performed. Trajectory Synchronisation is the automatic process, which ensures the unicity and consistency of a change modifying the trajectory data and its sharing amongst all (relevant) ground stakeholders.

After the Trajectory Revision process is terminated and when a change modifies the Cleared Trajectory Data, a new aircraft predicted trajectory might be downlinked for consistency purpose and for update of related supporting data.

ATC Clearances in the context of TBO

As more strategic management of ATM processes and trajectories is possible, a certain trajectory may be modified in advance (up to several hours), resulting in a Ground Agreed Trajectory, not yet communicated and/or cleared to the Flight Crew. Already at this stage, the Ground Agreed Trajectory may be used as input to perform Network Management, Airspace Management and other activities.

Better knowledge of airborne data and appropriate processes to align airborne and ground based trajectory information, allows the introduction of wider set of clearances, which can also be given longer in advance. In some instances, this allows a certain trajectory modification to be cleared in advance with the Flight Crew by an upstream ATC sector – i.e. an ATC sector currently managing the flight but not responsible for the portion of the airspace where the trajectory change occurs. Negotiation of proposed trajectory modifications may take place between sectors and/or Network Management Functions when necessary.

Different roles, such as e.g. the Extended ATC Planning, may also be able to issue clearances, provided technical means and appropriate procedures are in place to maintain Executive and Planning ATCOs aware of the clearance execution and able to intervene when necessary.

In some cases, the level of automation support to controllers, allows a safe transition towards the automatic uplink of clearances with or without previous controller validation. In those instances,

situational awareness is ensured through appropriate system support, allowing the ATCO to intervene and take over the execution of the clearance at any point in time. Safety is further ensured by ground tools, which scan for any potential conflicts of the new trajectory and alert the ATCOs appropriately.

In TBO, enhanced knowledge of the predicted vertical profile of the aircraft is available (e.g. through the eFPL or ADS-C data, such as EPP). This allows automation to be developed which supports ATCOs in issuing enhanced vertical clearances to facilitate continuous climb and descent, while ensuring separation from other traffic.

The implementation in the FMS of the above-mentioned lateral and vertical clearances is done by the flight crew, as much as possible through the usage of loadable CPDLC messages, which allow them to be aware of the ATC intent while minimising the risk of errors. ADS-C EPP and additional Aircraft Derived Data are used by the ground for consistency check and conformance monitoring. Any discrepancies are automatically identified by the system and highlighted to the ATCO, to take appropriate action.

Via CPDLC, flight crews may also be able to request revised routing directly to ATC, which the latter will aim to accommodate, after appropriate ground coordination processes have taken place.

Military specific aspects

In the execution phase, trajectory management focuses on MT monitoring and adherence to the planned and agreed ground trajectory. Trajectory monitoring functionality is widely supported by automation and relies on surveillance data collected from different sources. Considering the variety of military aircraft types and different flight performance, the trajectory management in execution phase relies on the ground-to-ground data exchange on the aircraft 2D position, the vertical profile, speed and other relevant parameters. The SWIM technical profiles facilitate ground-to-ground MT data sharing between WOC and ATC for trajectory revision and update.

Air-to-ground data exchange⁸ is limited due to the military aircraft equipage considering that available ATM/CNS and data link capabilities are not used for ATM purpose.

Nonetheless, the fighter aircraft is a primary participant in the MT concept and must meet the common ATM/CNS requirements while operating in TBO environment, to mitigate adverse effect on the ATM network performance and minimise use of exemptions and/or derogations.

For transport type aircraft, the trajectory management air-to-ground data link capabilities are the same as for civil aircraft.

⁸ The MT data model will be further developed within horizon of SESAR 2020 with respect to the confidentiality of mission specific data. The further harmonisation of MT data regards FF-ICE and eFPL data models as future reference models to substitute improved OAT FPL allowing military AUs to perform coherent and efficient trajectory management.

SESAR 2020 Significant Changes

Wave1

PJ18 4DTM 4D Trajectory Management

PJ18.02a Trajectory Based Operations

PJ.18-02a will define, in coordination with other solution projects, procedures needed to establish and manage a synchronised view of the trajectory during execution phase - that is the Reference Business Trajectory (RBT). The new procedures will describe operational scenarios where each stakeholder is expected / allowed to review / update the RBT, and when relevant, any potential collaborative decision-making process. A dedicated subsection will look at Air / Ground exchanges and will propose new or improved working methods and operational procedures especially for the new roles in the Trajectory Based Operations including the use of the existing enablers (e.g. datalink / CPDLC) and standards (e.g. ATM Baseline 1 and 2, FANS) and demonstrate benefits.

- Enhanced A/G data exchanges for Trajectory Based Operations (TBO). SDM-0213
Trajectory Based operations through automated support and enhanced procedures for A/G data exchanges including CPDLC and use of aircraft trajectory data on the ground (e.g. for A/G conformance monitoring and probing)
- Improved trajectory management based on enhanced G/G data sharing among ATC and Network Management. SDM-0214
Improved trajectory management including revision through improved procedures and automated support to enhanced ATC operations and Collaborative Decision Making based on G/G trajectory data sharing among ATC (including ENR, APP and Airport ATC) and Network Management leading to improved ATM operations supporting TBO. Decisions are shared among ATC and ATFCM/ASM actors.
- Enhanced ATC and airborne operation using datalink for Trajectory Based Operations (TBO). SDM-0215
Enhanced ATC and airborne operations for TBO, by providing 2D planning information in advance with/without clearance to execute and complex clearances through automatic support using data link standards.

Wave 2

PJ18 W2 4D Skyways

PJ.18-W2-53 Improved Ground Trajectory Predictions enabling future automation tools⁹

This SESAR Solution follows-up the work performed by SESAR Solution #10-02a and #10-02b in Wave 1 and covers the following OI steps (subject to be reviewed in future datasets):

- Conflict Detection and Resolution in the TMA using trajectory data – CM-0206.
- Automated Ground Based Flight Conformance Monitoring in the TMA – CM-0208-A.

⁹More detail on this topic is provided in the section Conflict Management.

- Conflict Detection and Resolution in En-Route using enhanced ground predicted trajectory in Predefined and User Preferred Routes environments – CM-0209.
- Conflict Detection and Resolution in En-Route using aircraft data in Predefined and User Preferred Routes environments – CM-0209-B.
- Ground Based Flight Conformance Monitoring in En-Route using enhanced ground predicted trajectory – CM-0210.
- Ground Based Flight Conformance Monitoring in En-Route using aircraft Data – CM-0210-B.
- Advanced Support for Conflict Detection and Resolution for ATC planning in En Route – CM-0211.
- Separation Management in En-Route using Pre-defined or User-preferred Routes with 2D RNP Specifications – CM-0605.
- Separation Management in the TMA using Pre-defined Routes with 2D RNP Specifications – CM-0606.
- Automated Ground Based Flight Conformance and Intent Monitoring in En-Route improved With Aircraft Downlinked Data – CM-0207-B.
- Automated Ground Based Flight Conformance and Intent Monitoring in the TMA – CM-0208-B.
- Enhanced Conflict Detection and Resolution in En-Route – CM-0407.
- Enhanced Conflict Detection and Resolution in the TMA – CM-0408.
- Separation Management in En-Route using RBTs with 2D RNP Specifications – CM-0607.
- Separation Management in the TMA using RBTs with 2D RNP Specifications – CM-0608.

PJ.18-W2-56 Improved vertical profiles through enhanced vertical clearances

This is a new SESAR Solution that needs to be created together with the required Operational Improvements steps (OIs) for future datasets. Safe vertical clearances have been initially covered by solution #10-02b in SESAR 1 at a low level of maturity, relevant OI steps would need to be revisited to cover this SESAR Solution.

PJ.18-W2-57 RBT revision supported by datalink and increased automation

This is a new SESAR Solution that needs to be created together with the required Operational Improvements steps (OIs) for future datasets. The scope of this solution has been initially covered by solution #18-02a in SESAR 1 at a low level of maturity, relevant OI steps would need to be created/revisited to cover this SESAR Solution. The initial list includes:

- Enhanced A/G data exchanges for Trajectory Based Operations (TBO) – SDM-0213.
- Improved trajectory management based on enhanced G/G data sharing among ATC and Network Management – SDM-0214.

Expected benefits

The project will contribute to increase:

- **Safety** by generating and proposing conflict free clearance that considers aircraft performance
- **Interoperability** by sharing the same trajectory view between air and all ground actors;
- **Predictability** by improved trajectory accuracy and sharing information updates throughout the flight;
- **Flexibility** by matching the trajectory with the Airspace User needs;
- **Flight efficiency**: flight is managed closer to its optimal profile.

2.3.2 Conflict Management

The 4D Trajectory Management concept introduces a shift from “management by tactical intervention” to a more strategic “management by planning and intervention by exception”, however not questioning tactical actions for safety and flexibility/optimisation purposes.

In both En Route and TMA environment, the working methods will be adapted to an environment of more flexible structures, tailored to accommodate main traffic flows through applying ATC complexity/workload balancing principles. Instead of constraining the traffic to respect rigidly delimited structures for sake of capacity, it is the ATC configuration, which will adapt to the needs and specificities of the traffic, thus preserving both capacity and users’ needs in the most optimised way. ATC tools and working methods shall be adapted accordingly. Within all these ATM components, the human will still play a central role, but tasks will be more oriented towards managing and decision making, while routine tasks will be supported by system automation (on board and on the ground). The improvement of planning activities will reduce the amount of tactical actions to be managed. The system will provide ATCOs with efficient support for decision-making, tailored to the role’s specific needs in terms of time horizon and, at the same time, allowing appropriate situational awareness to be maintained. Better opportunities for optimised resolution will be available, including altogether a larger geographical area, a longer look-ahead time and needs from various stakeholders, while reducing ATCO workload per flight.

Complexity and separation management include revision of the trajectory. This may be done using additional or amended lateral routes/waypoints and/or vertical constraints on specified points to ensure separation in execution phase. ATCOs may uplink to the cockpit vertical constraints on points where both the geographical position of the point (expressed in distance/bearing from a known point or a lat/long) and the vertical constraint (expressed in Flight Level) are dynamically calculated.

After a Revision of the trajectory has been agreed, the resulting trajectory is transmitted for acceptance by the Flight Crew preferably through closed-loop clearances; open-loop clearances may still be used for safety, technical (e.g. non-equipped aircraft) or operational reasons but will be superseded by closed-loop clearances as soon as possible. The usage of a closed-loop clearance allows issuing one, more strategic clearance, in opposition to several short-term ones. This allows a reduction of ATCO workload per flight, as well as a more precise input to the ground and airborne systems. This also gives opportunities to fly the aircraft in FMS managed mode providing a more representative EPP reports.

For the complete benefit of Trajectory Management, ATCOs and Flight Crews are supported by a set of advanced support tools aiming at reducing their workload, and allowing monitoring and ensuring trajectory compliance. Systematic synchronisation of air and ground plans with minimal or no human intervention will be supported by new A/G information exchanges based on evolutions of air and ground systems.

Controllers will be assisted in both their planning and tactical separation tasks by the following functionalities:

- Automated Detection and Resolution support based upon ground trajectory prediction taking into account effects on flight efficiency and ATCO workload in order to allow selection of the optimal resolution. Digitalisation allows for improved detection and resolution algorithms, which calculate multiple resolution options and provide the Planner and Executive ATCOs with

proposals, based on selectable criteria. Such criteria are optional and, when applied, may include improving flight efficiency in the longer time horizon of the Planner, or reducing ATCO workload in the shorter time horizon of the Executive ATCO;

- Resolution proposals for safe vertical clearances, using ATC-defined points;
- Enhanced tools facilitating enhanced situational awareness and the decision making of the ATCO by highlighting relevant traffic for a selected flight;
- PBN capability of fixed wing aircraft and rotorcraft will be taken into account;
- Conformance Monitoring: Indication if aircraft deviate or will potentially deviate from the cleared trajectories, using aircraft derived data;
- Complexity Management Tools, used in the context of INAP;
- System supported screen-to-screen electronic coordination tools, supporting intra and inter-centre electronic coordination;
- Data Link as the predominant means of communication between airborne and ground side, except for time critical messages;

These functionalities rely on ground trajectory prediction, using the following input data:

- ATC intent based on existing lateral and vertical clearances in the form of complete trajectory information as far as feasible;
- Full integration of ground trajectory constraints within the trajectory prediction computed on-board;
- Airborne trajectory predictions (e.g. EPP data);
- Aircraft derived data (e.g. Mode S and ADS-B data);
- Integration of input data derived from meteorological services (e.g. advanced wind consideration and consideration of "no fly" zones provided by meteorological services).

ATC automation is adapted to a new operational environment (e.g. increased traffic demand, trajectory based operations, new separation and complexity management modes) while avoiding generating an increase in controller workload, whether at individual or teamwork level. Automation has to be designed to facilitate operations within the Free Route environment, which is extended to airspace of higher complexity and results in management of "random" instead of usual crossing points, and unusual traffic patterns.

RPAS IFR operations will take account of RPAS performance, control latency, reduced crew awareness and sensitivity to severe weather conditions. RPAS will also often operate along specific trajectories, such as long loiter patterns, that are unlike those of manned aircraft. Appropriate level of awareness to ATC will be provided with regards to the intent and performance of RPAS traffic, to allow their safe handling in an environment of predominantly manned aircraft.

Safety nets for ATCOs will be improved through the more precise knowledge of aircraft intent and performance. This will be enabled by the usage of Downlinked Aircraft Parameters (e.g. via Mode S and ADS-B) and will allow reducing nuisance alerts and increasing warning time in case of unsafe aircraft manoeuvres.

Airborne Collision and Avoidance Systems (ACAS) operations will be adapted to SESAR future Trajectory Management and new separation modes through the optimisation of ACAS resolution advisories and the use of enhanced surveillance. Inappropriate alerts will be reduced.

Airborne collision avoidance functions will benefit from additional and more accurate surveillance sources, using aircraft information, in particular current maximum possible vertical rate, in order to select the most appropriate RAs (Resolution Advisories). Active interrogations to establish the range of intruders is part of the ACAS development.

Airborne Collision Avoidance for General Aviation (GA) and Rotorcraft (RC) provides Airborne Collision Avoidance to GA and RC, taking into account their limited capability to carry equipment and their operational specificities.

Performance improvements related to ACAS are not expected to change the operating methods and will be transparent for Pilots and Controllers. Airborne Collision Avoidance algorithms are anticipated to be integrated in the Detect-and-Avoid logic of RPAS. The airborne Remain Well Clear procedure needs to take place before, in order to prevent any separation infringement which may present a challenge with regards to situational awareness.

Advanced operating methods, ATCO team organisation and Generic ATCO validations

Advanced tools and operating methods, as well as the extensive usage of CPDLC, influence the task allocation between Controllers (Extended ATC Planner, MSP, PC&EC), thus team structure and communication are also affected.

Subject to the definition of clear procedures, roles and responsibilities, some tasks may be re-distributed between Planning and Executive Controllers (Incl. MSP). At an initial stage, those may be limited to simple routine tasks such as change of frequency. However, to take advantage of their spare capacity, the Planner may issue clearances for e.g. conflict resolutions to traffic still with upstream sectors. Usage of closed-loop clearances instead of heading instructions would provide additional workload reduction per flight to the Executive.

As well as the traditional Planner-Executive (1P-1E) two-person ATC sector team, sectors will operate with a combined role of Single Person Operations (SPO) or with the distinct role of Multi-Sector Planner (MSP) where the Planning Controller is responsible for the airspace that is under the executive control of two or more independent Executive Controllers (1P-nE)¹⁰.

The Planning Controller will provide support to several Tactical Controllers (or Executives) operating in different AORs, including ensuring suitable coordination agreements between them and assisting in managing the workload of the Tactical Controllers. These roles will be applied in both En Route sectors and in the TMA.

In the Collaborative Control with unplanned boundaries concept, the traditional requirement to coordinate traffic at all sector boundaries is waived for an area covering two or more sectors, and constraints are only applied when it is necessary to fulfil a specific ATM function, e.g. separation, or traffic synchronisation. Controllers of the unplanned boundaries area work in a collaborative way and

¹⁰ Validations in Wave 1 PJ10-01A have concluded that MSP in En Route for more than 2 Executive Controllers is not feasible. This does not invalidate the already proven concept whereby MSP is capable of working in geographically separated sectors with 2 Executive Controllers. The expectation is that CM-0303 will be modified to reflect the validation results.

are able to issue clearances into or operate traffic within all of the sectors in the area without the requirement to pre-coordinate. This concept is envisioned as being particularly well suited for groups of sectors for which there is a single MSP and traffic is not very high, but the research may explore additional team set-ups, e.g. groups of sectors with SPO at times of low traffic demand.

Additionally, the concept may be useful to reduce coordination between adjacent AoRs, operated as Flight-centric Areas, or in a setup where sector-based ATC is adjacent to Flight-centric, as well as between AoRs within a Free Route Airspace.

The above-mentioned procedures enabled by advanced and highly automated support tools allow controllers to hold more generic validations rather than validations based purely on specific (geographic) sectors. Generic validations will allow greater flexibility to match ANSP resources to predicted demand. A prerequisite is the implementation of more consistent operating techniques less reliant upon detailed knowledge of specific local systems, airspace and procedures.

Above a certain Flight Level or Altitude, depending on operational constraints and applicable working methods, a non-geographical flight-centric concept of ATC could also apply. In the flight-centric area, a flight (trajectory) remains under the control of the same ATCO throughout the whole, or a significant part of its En Route segment, within the designated Airspace, (including FRA). A number of flights are assigned to an ATCO, in function of his/ her workload unconstrained by the geographical location, sector or national boundaries.

The solution requires flight-centric specific complexity and workload algorithms, as well as allocation, visualisation (traffic filtering) and coordination tools (e.g. in the event of a conflict, establish which controller is responsible for its resolution). For high traffic densities, the concept also requires advanced CD&R tools that are not flight-centric specific, but may have specific functions to support flight-centric ATC (e.g. allocation of conflict to a controller, consideration of potential conflicts with a long look-ahead time to allocate when possible aircraft that may be in conflict to the same controller, etc.).

In a flow based allocation within the flight-centric concept a flow (multiple trajectories), remain under the control of the same ATCO (or ATCO team), throughout the whole, or a significant part of their flight. This is a refinement to the flight-centric concept where the allocation of trajectories to ATCOs takes into account interactions between them, aiming to reduce potential coordination between Executives.

Dissolving sector boundaries and having one controller in charge of a single flight to guide it through a larger airspace could alleviate Flight Crew workload by reducing frequency changes and the number of ATC messages.

Integrated Network Management and Extended ATC Planning (INAP) function is introduced. To bridge the gap between the management of traffic flows at European level and the control of flights in individual sectors, support is provided to the Multi Sector Planning and Extended ATC Planning roles for reducing complexity and smoothing flows of traffic in a multi-sector/multi-unit environment. Procedures and tools assist the Controllers in alleviating traffic complexity, traffic density, and traffic flow problems across several sectors through acting on individual trajectories. The area of responsibility is established to be sufficiently large to enable INAP function activities to be performed.

The scope of the INAP function is to address the overlapping period where the Network Management function runs DCB and dynamic DCB processes at all geographical levels, while ATC planning starts preparing early strategic de-confliction and conflict detection within the appropriate look ahead time horizon and within its defined local area of responsibility.

The objective is to enable a seamless ATM layered planning process, which provides optimum solutions (airspace configuration and trajectory/flow management) to solve workload imbalances. These solutions are assessed for compatibility with traffic synchronisation activities and strategic conflict management under the responsibility of the ATC planning function. Additional opportunities to optimise individual trajectories and/or traffic flows are identified and implemented.

Resolutions in INAP are automated, based on large number of resolution scenarios with different levels of granularity, synchronised and applicable at different levels in the ATM layered planning. These performance-based, pre-defined and pre-agreed scenarios would be taking Airspace Users' needs into consideration when appropriate through UDPP priorities and preferences and Airspace Users' preferred trajectories.

SESAR 2020 Significant Changes

Wave 1

PJ10 PROSA Controller tools and team organisation for the provision of separation in air traffic management

PJ.10-02a Improved performance in the provision of separation

- Conflict detection and resolution in the TMA using trajectory data (CM-0206)
- Automated ground based flight conformance monitoring in the TMA (CM-0208-A)
- Conflict Detection and Resolution in En-Route using enhanced ground predicted trajectory in Predefined and User Preferred Routes environments (CM-0209)
- Conflict Detection and Resolution in En-Route using aircraft data in Predefined and User Preferred Routes environments (CM-0209-B)
- Ground Based Flight Conformance Monitoring in En-Route using enhanced ground predicted trajectory (CM-0210)
- Ground Based Flight Conformance Monitoring in En-Route using aircraft Data. CM-0210-B
The system provides the En-Route controller with warnings if aircraft behaviour or aircraft intent deviate from the flight intent (RBT/MT) in ground system.
- Advanced Support for Conflict Detection and Resolution for ATC planning in En Route. CM-0211
The system provides real-time assistance to the En route ATC Planning in conflict detection and resolution. It is based on new approaches that enhance and refine the existing tools yielding more efficient and usable services.

PJ.10-02b Advanced separation management

- Enhanced Conflict detection and resolution in TMA (0408))
- Enhanced Conflict detection and resolution in En Route (CM-0407)
- Automated Ground Based Flight Conformance and Intent Monitoring in the TMA (CM-0208-B)
- Automated Ground Based Flight Conformance and Intent Monitoring in En Route improved With Aircraft Downlinked Data (CM-0207-B)
- Separation Management in the TMA using RBTs with 2D RNP specifications. CM-0608
The Reference Business Trajectory (RBT) is a 'precision trajectory' in that it may include required navigational performances associated to the Pre-defined or User-preferred routes of RBT e.g. 2D

RNP specifications and/or altitude constraints on specified points to ensure the de-confliction of 3D profiles. The RBT may be revised using additional or amended lateral routes/waypoints and/or altitude constraints on specified points to ensure separation in execution phase.

- Separation Management in En Route using RBTs with 2D RNP specifications. CM-0607
The Reference Business Trajectory (RBT) is a 'precision trajectory' in that it may include required navigational performances associated to the Pre-defined or User-preferred routes of RBT e.g. 2D RNP specifications. The RBT may be revised using additional or amended lateral routes/waypoints and/or level to ensure separation in execution phase; ATC clearances are included as part of the RBT Revision process.

PJ.10-01b Flight centric ATC

Flight-Centric ATC sees the provision of ground-based automated support for managing separation provision across several sectors in order to enable larger sectors to be used. Rather than managing the entire traffic within a given sector, with this solution ATC is responsible for a certain number of aircraft throughout their flight segment within a larger airspace or along flows of traffic.

- Flight-centric ATC in Non-Geographically-Constrained, Low and Medium complexity environments. CM-0200-B
In Low and Medium complexity environments above a certain Altitude / Flight Level, depending on local organisation and applicable working methods, non-geographical flight-centric method of ATC could be applied. A flight (trajectory) remains under the control of the same ATCO throughout the whole, or a significant part of its route segment, within the designated Airspace
- Flight-centric ATC in Non-Geographically-Constrained, High and Very High complexity environment. CM-0200-C.
In High and very High complexity environments, depending on local organisation and working methods, a flight-centric method of ATC could be applied. In performance-driven ATM operations, different service providers will decide of application of this method in different environments, according to their operational needs and business model.

PJ.10-01a High Productivity Controller Team Organisation

High Productivity Controller Team Organisation sees the extension of sector team operations beyond team structures of one planning ATCO and two tactical ATCOs both in En Route and TMA in order to optimise flight profiles, minimise delays and improve ANSP cost efficiencies while taking into account intrinsic uncertainty in the trajectory.

- Sector Team Operations Adapted to New Responsibilities in En Route, 1 Planning to several Tactical Controllers team structure. CM-0303¹¹
- Sector Team Operations Adapted to New Responsibilities in TMA, 1 Planning to several Tactical Controllers team structure. CM-0304

PJ.10-01c Collaborative Control

Collaborative Control refers to coordination by exception rather than coordination by procedure and is

¹¹ Reference MSP - see footnote section 2.3.2 regarding impending change of scope for MSP

facilitated by advanced controller tools, supporting reduced need for coordination agreements, fewer boundary constraints and the ability to combine sectors into multi-sector planner teams

- Sector Team Operations Adapted to New Responsibilities and Operating Procedures involving reduced Coordination in En-Route and TMA-En Route interface airspace. [CM-0306](#)
- Sector Team Operations Adapted to Collaborative Operating Procedures involving reduced tactical co-ordination in the TMA. [CM-0309](#)
- Sector Team Operations Adapted to Collaborative Operating Procedures involving reduced tactical co-ordination in En-Route. [CM-0310](#)

PJ.10-06 Generic ‘non-geographical’ controller validations

Generic (non-geographical) Controller Validations refer to the development of advanced tools and concepts that will help to remove the qualification constraints imposed on ATCOs for controlling a single volume of airspace. This approach would allow ATCOs to operate in any airspace classified as a particular type.

- Generic ‘non-geographical’ Controller Validations. [SDM-0203](#)
Advanced automation and other future concepts (e.g. 4D Trajectory management), harmonising ATC working methods and decreasing ATCOs tactical tasks, allow controllers to hold more generic validations rather than validations based purely on specific (geographic) sectors.

PJ.10-05 IFR RPAS Integration

IFR RPAS Integration provides the procedural and technological means to safely integrate RPAS traffic in a non-segregated controlled airspace (En-Route and Terminal ATM areas), complying with ATC instruction.

- Enabling integrated RPAS IFR operations. [AUO-0618](#)
Remotely Piloted Aircraft Systems (RPAS) that are to operate IFR operations in non-segregated airspace will have to integrate into an environment, which is dominated by manned aviation. RPAS will have to comply with flight rules, airspace requirements/procedures and be interoperable with the CNS/ATM system.

PJ.09 DCB Advanced Demand and Capacity Balancing

PJ.09-02 Integrated local DCB processes (ATC perspective)

Integrated Local DCB Processes see the seamless integration of local network management with extended ATC planning and arrival management activities in short term and execution phases. It represents the core functionality for the Integrated Network ATM Planning (INAP) process through an enhanced Local DCB tool set.

- Ground based Automated Support for Managing Traffic Complexity across Several Sectors. [CM-0302](#) The system supports INAP actors for smoothing flows of traffic and de-conflicting flights across INAP AoR.

PJ.11 Enhanced Air and Ground Safety Nets

PJ.11-G1 Enhanced Ground-based Safety Nets adapted to future operations

The use of DAPs information via ADS-B and Mode-S technology in the Short Term Conflict Alert (STCA) and in the Non Transgression Zone (NTZ) takes advantage of the greater frequency of availability and updating of

the data received in order to obtain different improvements in both safety nets.

- Non Transgression Zone (NTZ) ground safety net using DAPs information via ADS-B technology. [CM-0812](#)
- Non Transgression Zone (NTZ) ground Safety Net using DAPs information via Mode-S technology. [CM-0813](#)
- Enhanced Short Term Conflict Alert (STCA) ground Safety Net using DAPs information via ADS-B technology. [CM-0814](#)
- Enhanced Short Term Conflict Alert (STCA) ground Safety Net using DAPs information via Mode-S technology. [CM-0815](#)

PJ.11-A1 Enhanced Airborne Collision Avoidance for Commercial Air Transport normal operations - ACAS XA

Enhanced Airborne Collision Avoidance for Commercial Air Transport normal operations- ACAS Xa refers to the use of ACAS Xa, an airborne collision avoidance system which takes advantage of optimised resolution advisories and of additional surveillance data, without changing the cockpit interface (i.e. same alerts and presentation in the current TCAS)

- Improved collision avoidance for commercial air transport in standard operations. [CM-0808-a](#)

PJ.11-A2 Airborne Collision avoidance for remotely piloted aircraft (ACAS Xu)

Airborne Collision Avoidance for Remotely Piloted Aircraft Systems - ACAS Xu provides airborne collision avoidance to RPAS, building on optimised resolution advisories and additional surveillance data, while taking into account the operational specificities of RPAS (the additional surveillance sources could be ADS-B but also any other sensor installed on the RPAS).

- Collision avoidance for remotely piloted aircraft. [CM-0808-u](#)

PJ.11-A3 ACAS for commercial air transport specific operations – ACAS Xo

ACAS for Commercial Air Transport specific operations - ACAS Xo improves ACAS (Airborne Collision Avoidance System) building on optimised resolution advisories and additional surveillance data, while avoiding unnecessary triggering of resolution advisories (RAs) in new separation modes (e.g. ASAS), in particular if lower separation minima are considered

- Collision avoidance for commercial air transport adapted to new separation modes. [CM-0808-o](#)

PJ.11-A4 Airborne Collision avoidance for general aviation and rotorcraft (ACAS Xp)

Airborne Collision Avoidance for General Aviation and Rotorcraft - ACAS Xp provides Airborne Collision Avoidance to GA/RC, taking into account their limited capability to carry equipment and their operational specificities.

- Collision avoidance for general aviation and rotorcraft (ACAS Xp). [CM-0808-p](#)

Wave 2

PJ10 W2 Separation Management and Controller Tools¹²

¹² PJ.10-02a and PJ.10-02b from SESAR Wave 1 have been transferred to Wave 2 Solution PJ.18-W2-53 Improved Ground Trajectory Predictions enabling future automation tools. For the list of OI steps - see chapter 2.3.1.

PJ.10-W2-73 Flight-centric ATC and Improved Distribution of Separation Responsibility in ATC

This SESAR Solution follows-up the work performed by SESAR Solution #10-01b01b in Wave 1 and covers the following OI steps:

- Flight-centric ATC in Non-Geographically-Constrained, Low and Medium complexity environment – CM-0200-B.
- Flight-centric ATC in Non-Geographically-Constrained High and Very High Complexity environment – CM-0200-C.
- CM-0200-D -

This SESAR Solution also follows-up the work performed by SESAR Solution #10-01c in Wave 1 related to collaborative control

- Sector Team Operations Adapted to New Responsibilities and Operating Procedures involving reduced Coordination in En-route – CM-0306.
- Sector Team Operations Adapted to Collaborative Operating Procedures involving reduced tactical co-ordination in the TMA – CM-0309.
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- Sector Team Operations Adapted to Collaborative Operating Procedures involving reduced tactical co-ordination in En-Route – CM-0310.

PJ.10-W2-93 Delegation of airspace amongst ATSUs

This SESAR Solution follows-up the work performed by PJ.16-03 and PJ.15-09.

PJ.10-W2-70 Collaborative control and Multi sector planner (MSP) in En Route¹³

This SESAR Solution follows-up the work performed by SESAR Solution #10-01a in Wave 1 and covers the following OI steps:

- Sector Team Operations Adapted to New Responsibilities in En-route, 1 Planning to several Tactical Controllers team structure – CM-0303.¹⁴
- Sector Team Operations Adapted to New Responsibilities in En-route, Single Person Operation – CM-0307.
- Sector Team Operations Adapted to New Responsibilities and Operating Procedures involving reduced Coordination in En-route – CM-0306.
- Sector Team Operations Adapted to Collaborative Operating Procedures involving reduced tactical co-ordination in the TMA – CM-0309.

¹³ Solution referenced in the call from December 2018 - H2020-SESAR-2019-1 IR VLD WAVE 2 [10] has consequently been merged with Solution PJ.10-W2-73 Flight-centric ATC and Improved Distribution of Separation Responsibility in ATC.

¹⁴ Reference MSP - see footnote section 2.3.2 regarding impending change of scope for MSP

- Sector Team Operations Adapted to Collaborative Operating Procedures involving reduced tactical co-ordination in En-Route – CM-0310.

PJ13 IFR RPAS

PJ.13-W2-111 Collision avoidance for IFR RPAS

New solution looking for a global Collision avoidance with RPAS in Classes A, B and C or airspace. It will encompass Remain Well Clear (RWC) processes, Detect & Avoid (D&A) including with non-cooperative A/C, and will go beyond the work conducted in Wave 1 by PJ.11-A2 on ACAS Xu and of PJ.10-05 as regards Conflict Management. Relevant links to the ATM Master Plan Level 2 (DS19):

- CM-0808-u — Collision Avoidance for Remotely Piloted Aircraft

The solution will require additional Operational Improvements steps (OIs).

PJ.13-W2-115 IFR RPAS accommodation in Airspace Class A to C

This is a new SESAR Solution that needs to be created together with the required Operational Improvements steps (OIs) for future datasets, in the continuation of OI step AUO-0618 in wave 1.

PJ.13-W2-117 IFR RPAS integration in Airspace Class A to C

This is a new SESAR Solution that needs to be created together with the required Operational Improvements steps (OIs) for future datasets, in the continuation of OI step AUO-0618 in wave 1.

Expected Benefits

Capacity - Enhancing the situational awareness of the ATCOs and providing them with decision support tools and some proposed resolutions implies less workload in their decision-making processes. This can facilitate the process to generate clearances, which will be more strategic and/or precise, and can be issued via Data Link by both Planner and Executive ATCOs, bringing an increase in capacity.

Cost efficiency - ATCO support tools enable an increase in ATCO productivity and allow staffing optimisation. Generic Controller validations enable more flexible staff allocation and benefit cost efficiency.

Safety will be improved through the enhanced performance of air and ground based safety nets reducing nuisance alerts and increasing warning time in case of unsafe aircraft manoeuvres. Additionally, better knowledge of predicted trajectory using aircraft derived or other data, shared through SWIM, allows some conflicts to be solved earlier and/or with more precise clearances. Those clearances are in turn more likely to be achieved by the aircraft. Once clearances are issued the remaining ATCO tasks are related to conformance monitoring, supported by tools. Knowledge of the predicted trajectory on-board the aircraft allows to perform conformance monitoring with a longer look-ahead time and intervene earlier. This strategic management of the traffic situation allows for a reduced number of situations to be resolved tactically and interventions by the ATCO are by exception. Additionally those interventions are likely to be more precise since better data is taken into account. The above functionalities, designed following a “human centric” approach, bring an improvement in safety.

Fuel efficiency In the En Route and TMA environment, through the usage of data from the aircraft and shared through SWIM, controller tools enable earlier and more precise detection and resolution of conflicts. This reduces the need for vectoring and enables de-confliction action to be taken earlier and through the usage of closed-loop clearances. Those can be managed more proactively on-board, thus reducing variability of flight times, particularly in the climb and descent segment of the flight. Clearances issued by the ATCOs may, in some situations, take into account aircraft derived data related to airline preferences, bringing an improvement in fuel efficiency and potentially a small improvement in predictability.

Wave 2 Performance expectations

The solutions under the scope of this topic are expected to have a positive impact on the Network improving:

- Safety through managing trajectories more strategically, therefore reducing the number of tactical conflicts
- Capacity through reducing ATCO workload per flight
- Cost-efficiency through a further improvement in ATCO productivity and staff optimisation.
- Accessibility of certified IFR RPAS in Airspaces classes A, B, C

2.3.3 Enhanced Arrival and Departures (W2)

2.3.3.1 Traffic Synchronisation

Traffic Synchronisation refers to the establishment and maintenance of a safe, orderly and efficient flow of air traffic. It concerns the management and execution of 4D trajectories combined with integrated queue management and handling, aiming to develop a sequence, both in the air and on the ground. It operates on individual flights but for the benefit of the global ATM network performance, aiming to facilitate the highest achievable capacity of the ATM System and to manage constraints, if needed, in a fuel-efficient and environmentally acceptable manner.

Arrival management processes will be enhanced using automation to cover a series of improvements to the E-AMAN solution, potentially including overlapping AMAN horizons, airspace design for independent parallel runway operations, Pair Wise Separation (PWS) and Weather Dependant Separation (WDS), potential interaction with dDCB, ground based ROT prediction, improved information exchange (i.e. SWIM), improved Trajectory Prediction and improved sequencing algorithms. This may include the use of artificial intelligence, in particular to build more robust arrival sequences supporting airspace users' preferred trajectories.

The arrival sequence benefits from improved trajectory information for each flight. A modified STAR or custom arrival route (including any associated vertical or speed constraints) may be uplinked already in the En Route sectors. Additionally, trajectory prediction is improved through downlinked aircraft data such as EPP, which takes into account optimum aircraft performance (e.g. maximum descent speeds), in turn enhancing the prediction algorithms.

In addition to better input from trajectory information, digitalisation brings improvements to arrival sequencing through the usage of machine learning for the refinement of AMAN algorithms including use of artificial intelligence.

Extended Arrival Management with overlapping AMAN operations

In order to manage the inbound traffic flow efficiently, the Extended AMAN allocates a time to establish a sequence. The integration of this AMAN Allocated Time (AAT) with constraints applied for DCB/dDCB takes place in order to ensure the effectiveness of either measure is not compromised. This takes into consideration the fact that En Route sectors may be affected by concurrent arrival management strategies due to the overlapping AMAN horizons of several independent TMAs, and will take into consideration constraints associated with solving DCB imbalances at an airport.

Where appropriate, the AAT can be uplinked to the affected flight as a CTA – providing the advantage of high levels of accuracy where there is sufficient stability of the traffic sample. The use of ETA min/max data by the ground system in determining the CTA and the accurate management of flights to CTA improves the performance and reliability of the arrival sequencing. Together with the airspace design and overall traffic management, this has the potential to enable the use of more closed-loop procedures to continue in higher density traffic situations, thereby allowing a high proportion of traffic to maintain optimum profile, ideally from the cruise phase of flight.

Enhanced ATC tools may allow the automatic uplink of AMAN constraints from the ground to the aircraft, while ensuring situational awareness of ATCOs and allowing them to monitor their impact and to intervene if necessary.

ATM-imposed time constraints can be allocated during the flight in order to manage airspace access, to sequence for positioning over intersecting points over metering fixes in TMAs/E-TMAs. As these constraints can be achieved by the flight with high accuracy (i.e. a few seconds), they enable performant metering and precise sequencing. The Trajectory Management of merging traffic flows at metering points supports new terminal airspace designs and optimises multiple airport arrival and departure services.

Whenever Airport and Extended Arrival sequencing induced constraints are overlaying Network constraints, local actors operate within an INAP context, reconciling DCB measures with Arrival Management measures where required, based on common view and collaborative rules and decision approach (see also chapter 2.2 Optimised ATM Network Services).

Departure Management is enhanced by increase of automation handling in-horizon departures via TTOT (which allow in-horizon departures to absorb delay on the ground).

The advanced traffic synchronisation tools shall consider departure synchronisation from multiple nearby airports. This may require the synchronisation of TTOTs between different airports in order to regulate in real time the flow of aircraft through a waypoint located a few minutes after take-off, as well as the synchronisation of AMAN planned times and TTOTs when arrival and departure routes must cross at similar altitudes. The concept should aim at regulating departures' TTOTs as far in advance as possible, in order to allow as much delay as possible to be absorbed at the gate. However, the use of short ground holdings (at the runway holding point) will probably also need to be considered in order to improve adherence to the TTOT when deemed necessary for synchronisation purposes.

Use of Arrival and Departure Management Information for Traffic Optimisation within the TMA

In addition to the management of inbound flows for each airport, traffic in the TMA can be effectively managed in near real-time, resulting in coupling of arrival and departure management processes. The system will provide automated support to departure metering to the Tower ATCOs from multiple airports within the TMA to enable adjustment of the departure sequence and the integration of arrival sequence impact, enabling a more consistent and manageable delivery of departures into En Route. Enhanced Trajectory Predictors using accurate 4D data including predicted climb performance may be used as a component in the process to assess any intervention needs.

When considering queue management for the same runway (or for dependent runways), both Arrival and Departure Management systems are coupled with the Advanced Surface Movement Guidance and Control System (A-SMGCS). This permits the setting up of a throughput-optimised sequence of arrivals and departures at the same time as taking into account AMAN Allocated Times/CTAs associated to incoming flights in addition to TSAT/TTOTs associated to departing flights (see chapter 2.4 High Performing Airport Operations).

The performance of the Arrival and Departure Management systems may be improved through availability of more precise ROT for arriving and departing aircraft (see chapter 2.4 High Performing Airport Operations). Predicted demand information from both Arrival and Departure Management systems within a TMA provides information to manage the TMA traffic in an optimal manner, allowing the identification and resolution of complex interacting traffic flows through definable TMA 'hotspots' (where typically two or more SIDs or STARs converge) into the En Route phase of flight.

Real time synchronisation of arrivals and departures is introduced, addressing the synchronisation of the use of routes in the TMA; runway use times at one airport or at two or more nearby airports; and start-up and ground operations for regulating the flow of air traffic through a waypoint.

Wherever feasible, capacity will be adjusted to match predicted demand e.g. through use of dynamically allocated 2D departure/arrival routes to accommodate high departure and arrival demand along particular flows.

Airborne Spacing Flight Deck Interval Management

Following the sequencing by the queue management processes, or following weather avoidance manoeuvres, spacing may be delegated to the Flight Crew in order to ensure a regular flow to the runway, an increase in runway throughput, and a reduction in controller workload. Whilst separation provision remains the responsibility of the controller, spacing can at certain instances be delegated to the Flight Crew, who are assisted by on-board functionalities to monitor and ensure spacing of their aircraft from one or more aircraft, and provides flight information concerning surrounding traffic

2.3.3.2 Dynamic E-TMA for Advanced Continuous Climb and Descent Operations and improved Arrival and Departure Operations

Dynamic accommodation of Continuous Climb and Descent Operations includes the transition between the TMA and both Free Route Airspace and fixed route airspace. This ensures, as much as possible, the facilitation of a broad range of climb and descent profiles, which are fuel-efficient for the specific flight, with Continuous Climb Operations (CCO) to Top of climb (ToC), and Continuous Descent Operations (CDO) from Top of Descend (ToD). New controller tools using enhanced knowledge of the predicted vertical behaviour of the aircraft help to ensure separation to other traffic and facilitate CCO and CDO¹⁵. Possible usage of CPDLC below FL285 may allow reducing ATCO workload.¹⁶ Enhanced Airborne functionalities are used to optimise climb and descent paths and support the crew in the management of aircraft energy.

Continuous Climb Operations at optimum thrust and Continuous Descent Operations at near idle thrust are environmentally friendly, because they are more fuel-efficient and minimise the time that aircraft have to be at low altitude, thereby reducing their noise impact. Descent and climb profiles in busy airspace (as close as possible to near idle CDOs and CCOs) are improved, as well as the horizontal flight efficiency of arrivals and departures, while at the same time ensuring traffic

¹⁵ See also section 2.3.2 Conflict Management.

¹⁶ The usage of CPDLC below FL285 and the different operational and technical aspects for the ground and the airborne side are subject to further research by a separate SESAR project. The future of controller-pilot communications requires a new concept of operations that includes voice and datalink. It is expected that this new “Controller-pilot communications CONOPS” will be developed by a separate SESAR exploratory research project. The expectation is that this will require considerable research in the airborne domain, both in terms of procedures and in terms of development of improved on board HMI and auto-load capabilities.

synchronisation (for optimum use of runway capacity), short-term DCB (to ensure best use of ATC capacity) and separation.

With minimal published restrictions in the planning phase, Advanced CCO and Advanced CDO can be planned as part of the trajectory and will represent the nominal case of operations. Both the trajectories in planning and in the execution phase include a continuous climb and descent, which delivers the most flight cost efficient solution for the AU, and will be facilitated, to the greatest extent possible, from cruise ToD to final approach. A-CDO sets out to deliver the optimum vertical profile calculated by the FOC in the planning stage and is revised as required in the execution phase. ToD will be a part of the trajectory taking into account all aspects of an optimum flight operation from the AU perspective.

The interface between TMA and En Route Free Route Airspace will see dynamic use of lateral routes and the application of 4D trajectory constraints only when required; making use of high accuracy and high reliability RNP capability (including Tactical Parallel Offsets), speed and various climb/descent profiles in order to provide the optimal link between the two environments. The objective is to dynamically allocate arrival and departure routes to aircraft that support traffic synchronisation and short-term DCB. The dynamically allocated routes could be either pre-defined SIDs or STARs or, in a more advanced concept, be flight-specific arrival or departure routes (a 2D route created by the controller specifically for one flight, which could have altitude constraints along the way). The dynamically allocated routes will support the transition from FRA into the TMA where FRA is in operation.

In low density TMAs, fixed SIDs/STARs will be complemented with the ability to dynamically utilise the TMA airspace with minimal use of constraining fixes. In higher density TMAs a constraining route structure may be required, but it will be used as necessary (e.g. through dynamic use of alternative SIDs for different aircraft performance/traffic mix and in different weather conditions). At multi-airport TMAs with major hub airports, the trade-offs and potential conflicts between arrivals and departures are managed through adapting the flows and allocation of controllers as appropriate.

DMAN, enhanced with automation and digitalisation, will take into account PWS and WDS optimised departure separation rules, surface constraints (taxi out, runway use), etc. This includes the use of improved algorithms and artificial intelligence.

Utilisation of RNP procedures provides an additional level of assurance in aircraft adherence to the trajectory, which is important in an environment where constraints are dynamically managed.

Improved Parallel approach operations are enhanced through the application of RNP navigation specifications (e.g. RNP AR APP), which may be used to replace, in particular, vectoring to final intercept in order to improve operations on the approach.

Improved parallel operations are in place, addressing the reduction of separation minima applicable between aircraft landing on parallel runways using Baro V-NAV, SBAS (EGNOS) or GBAS approaches, including use of RNP containment to avoid the need for vertical separation until both aircraft are aligned to the runway.

Advanced geometric GNSS based procedures encompass transition from RNP to xLS, the transition from barometric to geometric guidance and the containment of the lateral path using RF turns.

Planning and tactical information is used to manage the traffic flows of rotorcraft to major hubs and/or smaller airfields with the more consistent traffic flows of the major hub airports.

There will be improved access to landing locations, with rotorcraft procedures designed to allow IFR access to VFR FATOs, in particular when weather conditions are adverse. Advanced (e.g. curved) Satellite-Based Augmentation System (SBAS)-guided Point-in-Space RNP approaches towards landing locations and Point-in-Space departures from landing locations are created with connections to/from Low Level IFR route network. This will facilitate Simultaneous Non-Interfering (SNI) operations.

Advanced GNSS-based navigation based on vertical positioning will enable arrival procedures (including curved procedures) through the terminal area and the use of geometric altitude will enable optimised 3D routes that minimise levelling off.

Enhanced Terminal Area for efficient curved operations is developed, allowing curved approaches and curved departures where the turn is closer to the runway than it is possible today, in order to optimise TMA procedures in terms of fuel consumption or noise abatement, and allow segregation of traffic according to wake category. This includes the use of geometric vertical guidance to link the vertical trajectories in the STAR with the final approach.

Advanced navigation capabilities allow for an improved access to poorly equipped runways in reduced visibility conditions, avoidance of noise-sensitive areas and facilitate the separation task between aircraft with various vertical and lateral profiles.

The increasing traffic flows in the lower airspace, with a traffic mixture of GA, RC, and drones/RPAS and mixed operations with a wide range between IFR, VFR, VLOS and BVLOS flights, call for considering concepts around assigning the responsibility for ensuring separation provision to the vehicles. The technology on board could guarantee separation provision between peers in the air, obstacles and terrain via vehicle-to-vehicle datalink communication and accurate databases.

SESAR 2020 Significant Changes

Wave 1

PJ.01 EAD Enhanced arrivals and departures

PJ.01-01 Extended Arrival Management with overlapping AMAN operations and interaction with DCB and CTA

Extended Arrival Management with overlapping AMAN operations and interaction with DCB integrates information from multiple arrival management systems, enabled by SWIM, operating out to extended ranges into En Route sectors using local traffic/sector information and balancing the needs of each AMAN.

- Arrival Management Extended to En-Route Airspace – impact of overlapping AMAN operations. TS-0305-B
En-Route sectors are expected to contribute to the arrival sequencing towards multiple TMA simultaneously with overlapping AMAN operations inducing potential issues on workload/complexity/ performance for En Route ATCOs in the execution phase of the flights. This operational improvement will identify and assess these issues, and the relevant means to solve them;
- Controlled Time of Arrival (CTA) in high density/complexity environment. TS-0109
The CTA (Controlled Time of Arrival) is an ATM imposed time constraint on a defined point associated with an arrival runway, using airborne capabilities to improve arrival management.

- Arrival Management systems integrated with DCB. [TS-0315](#)¹⁷

To improve the overall efficiency of arrival and airport operations, the Arrival Management should integrate constraints applied for DCB/dDCB purposes, including those associated with solving DCB imbalances at an airport and preference information supplied in Airport Operations Plans

PJ.01-02 Use of Arrival and Departure Management Information for Traffic Optimisation within the TMA

Use of Arrival and Departure Management Information for Traffic Optimisation within the TMA sees TMA traffic managed in near real time, taking advantage of predicted demand information provided by arrival and departure management systems from one to multiple airports.

- Departure management from Multiple Airports. [TS-0302](#)

Departure information is compiled and presented to the TMA Supervisor to allow adjustment to the departure flows and enable a more consistent and manageable delivery into the En route phase of flight.

- Integrated Arrival Departure Management for traffic optimisation within the TMA and Extended TMA Airspace. [TS-0307](#)

Traffic in the TMA and nearby sectors is managed in near real-time, taking advantage of predicted demand information provided by local Arrival and Departure Management systems to identify sector/route over-demand or additional capacity, to resolve complex interacting traffic flows in and nearby the TMA and to balance the sector/flow load by controlling sector entry times or waypoint times e.g. via TTL, AMAN Allocated Times, speed advisories, CTA, ground delay and alternate routing, or use of ROC/ROD.

PJ.01-03A Improved Parallel Operations

Parallel Approach operations are improved through the application of RNP navigation specifications and the development of enhanced ATC procedures.

- Enhanced Parallel Approach Operations using PBN/RNP transitions to xLS. [AOM-0606](#)

Independent Parallel Approach operations are enhanced through the application of PBN/RNP navigation specifications with a transition to xLS final approach segments.

PJ.01-03B Dynamic E-TMA for Advanced Continuous Climb and Descent Operations

Dynamically attributed use of routes brings together vertical and lateral profile issues in both the En-route and TMA phases of flight, with a view to creating an end-to-end optimised profile and ensuring transition between free route and fixed route airspace.

- Advanced Continuous Climb Operations. [AOM-0705-B](#)

Progressive implementation of CCO, ideally to ToC, and in high density operations, employing new controller tools (and enhanced airborne functionalities) to facilitate operations.

- Advanced Continuous Descent Operations. [AOM-0702-B](#)

Progressive implementation of CDO ideally from ToD, and in high density operations, employing new controller tools (and enhanced airborne functionalities) to facilitate operations.

¹⁷ The interrelation between Traffic Synchronisation and DCB concepts related to target time management needs to be carefully addressed in a collaborative way between PJ.01-01 and PJ.09-02

- Dynamic Management of Terminal Airspace Routes and Transition. [AOM-0806](#)
In order to manage the transition into and out of the TMA from/to airports and En-route sectors with both fixed route and free route airspace, terminal airspace is dynamic, accommodating a broad range of climb and descent profiles.

PJ.01-05 Airborne Spacing Flight Deck Interval Management

Airborne spacing flight deck interval management refers to new ASAS spacing interval management sequencing and merging (ASPA IM S&M) manoeuvres encompassing the potential use of lateral manoeuvres and involving more complex geometries where a designated target aircraft may not be flying direct to the merge point.

- Airborne Spacing Flight Deck Interval Management (speed/more-complex route geometries) in high-density environment. [TS-0108](#)
The ASPA-FIM application require the flight crew to achieve and maintain a given spacing (either precise or 'at or greater than') with a designated target aircraft which may not be flying direct to a merge point (more complex route geometry).

PJ.01-06 Enhanced Rotorcraft operations in the TMA

Enhanced Rotorcraft operations in the TMA further develop the simultaneous non-interfering (SNI) concept of operations to allow RC to operate to and from airports without conflicting with fixed-wing traffic or requiring runway slots.

- Advanced Point-in-Space RNP approaches and departures. [AOM-0104-B](#)
Rotorcraft procedures are designed to allow easier IFR access to VFR FATOs, in particular when weather conditions are adverse. Advanced (e.g. curved) SBAS/GBAS guided Point-in-Space RNP approaches towards landing locations and Point-in-Space departures from landing locations are created with connections to/from Low Level IFR route network.

PJ.01-07 Approach Improvement through Assisted Visual Separation

Approach Improvement through Assisted Visual Separation refers to cockpit display of traffic information (CDTI) assisted visual separation (CAVS) and CDTI assisted pilot procedure (CAPP) applications that enable aircraft to separate each other visually in marginal visual conditions and that facilitate transitions from IFR operations to CAVS.

- Airborne spacing monitoring under IFR (ATSAW – spacing monitoring). [AUO-0507](#)
The Airborne Spacing monitoring under IFR operation consists in enabling the Flight crew to monitor spacing ('no closer than' a time or distance spacing) with regard to a given traffic, by managing speed with the support of CDTI augmented with monitoring and alerting function.

Wave 2

PJ.01 W2 Enhanced Arrival and Departures

PJ.01-W2-01 Next generation AMAN for 4D environment

This solution represents a significant step forward regarding AMAN and will provide enhancements to the arrival management systems and procedures in the context of digitalisation in ATM. Relevant links to the ATM Master Plan Level 2 (DS18a): This SESAR Solution follows-up the work performed by SESAR Solution #01-01 in Wave 1 and covers the following OI steps:

- Controlled Time of Arrival (CTA) in high density/complexity environment (TS-0109)
- Arrival Management Extended to En-Route Airspace - overlapping AMAN operations. (TS-0305-B)
- Arrival Management Systems integrated with DCB (TS-0315)

PJ.01-W2-08 Dynamic E-TMA for Advanced Continuous Climb and Descent Operations and improved Arrival and Departure Operations

The advanced traffic synchronisation and short-term DCB concepts required for the dynamic allocation of routes have been researched in Wave 1 by solution #01-02. This Wave 2 solution will continue solution #01-02 and #01-03B work on the integration of arrival and departure management and covers the following OI steps:

- Advanced Continuous Descent Operations (AOM-0702-B)
- Advanced Continuous Climb Operations (AOM-0705-B)
- Dynamic Management of Terminal Airspace Routes and Transition (AOM-0806)
- Departure Management from Multiple Airports (TS-0302)
- Integrated Arrival Departure Management for traffic optimisation within the TMA and Extended TMA Airspace (TS-0307)

These OI steps may need to be updated (or new ones may need to be added) in future datasets to cover the enlarged scope with regards to Wave 1 e.g. in order to cover digital components, 'descend via', tools to support reduced impact on fuel of separation aspects, air-to-ground sharing of descent planning information. In addition, the descriptions of the OIs listed above need to be cleaned to remove references to 'implementation' or to 'airborne side mitigations'.

Expected benefits

Capacity – Capacity is expected to be significantly improved by:

- optimising TMA (and Extended TMA) airspace capacity
- Smoothing Traffic load
- Reducing controller workload per flight.

This will be achieved thanks to several combined processes:

- Transferring some of the delay management tasks upstream to En Route sectors from the TMA and taking some delay on the ground before Take-Off
- Coordinating departures across multiple airfields what will optimise traffic flows and contribute to the reduction of predicted congestion and imbalances, thus improving capacity.
- Fine-tuning of spacing in the TMA leads to improved consistency in flows and hence improves TMA capacity.
- Delegation of spacing to the Flight Crew leads to a reduction in ATCO workload, which may be exploited through improved capacity.

Predictability – By integrating information from several airports with extended AMAN systems with local traffic/sector information and reducing network uncertainty, predictability will be increased. The use of more accurate data and the coordination of multiple Arrival and Departure Management systems within the extended TMA, using the latest information possible, will bring predictability benefits. The set-up of a pattern-based sequence for arrivals and departures, taking into account arrival and departure demand, should improve predictability through the allocation of runway slots that aircraft are more able to meet.

Cost Efficiency – Cost efficiency is expected to be significantly improved, in particular in TMA and E-TMA, thanks to:

- the progressive use of more accurate data
- the management of airspace structure and configurations,
- the optimisation of staffing:
- Better predictability of traffic
- reduction of ATCOs workload per flight (improved situation awareness, less tactical intervention, tactical interventions facilitated by advanced ATC and pilots support tools)
-

Flexibility – Better use of Arrival and Departure Management Information in the E-TMA will allow a more

flexible flight strategy for the AUs and the ability to change/adjust the SID or STAR in a medium timeframe.

The NOP will give the common awareness to all stakeholders (incl. non-scheduled flights) and provide updates resulting from late changes to demand, capacities and influencers (e.g. weather). The NOP will provide access to opportunities in case of late changes in capacity or demand.

Fuel Efficiency – fuel burn reductions are expected from the application of extended AMAN procedures that adjust the speed of aircraft earlier in the flight (at higher altitudes) to meet Target Times enabling the aircraft to be flown more efficiently compared to later, low level manoeuvres (e.g. vectoring). Improved management of traffic flows within the TMA will reduce the impact of hotspots (e.g. significant reduction of Holding stacks), resulting in more opportunity for CDO and CCO, leading to less fuel burn and improved environmental sustainability.

Wave 2 Performance expectations

The solutions to be addressed under this topic are expected to have a positive impact on the Network improving:

Airspace capacity (improved throughput / airspace volume & time) thanks to the synchronisation of the arrival and departure sequences to and from airports using extended AMAN overlapping operations;

Predictability with the increased accuracy of the AMAN data;

Cost-efficiency (reduced direct ANS cost per flight);

Environmental sustainability and fuel Efficiency thanks to the implementation of continuous descent and climb profiles;

Safety, in particular, assisting rotorcraft pilot for landing in bad weather conditions.

2.4 High Performing Airport Operations

2.4.1 Increased Runway and Airport Throughput (PJ-02)

Increased Runway and Airport Throughput

The tactical driver for runway throughput is the ability to safely maintain a high arrival and departure flow rate – that is to avoid wake vortex turbulence while airborne and to minimise runway occupancy on the ground.

The concepts supporting increased runway and airport throughput may be broken down into the following sub-elements:

Resilient Synthetic Vision (SV) tools and Augmented Reality (AR) technologies for Tower Controllers allow them to conduct safe operations under any meteorological conditions while maintaining a high runway throughput equivalent to good visibility. The function improves runway capacity, especially during low visibility procedures due to fog and adverse weather.

[Solution PJ.02-W2-14] Evolution of separation minima for increased runway throughput

Wake turbulence separation optimisation - an incremental approach to the determination of separation minima to increase safety and potentially increase capacity whilst avoiding wake vortex turbulence in the most efficient manner. New technologies enable better measurement of vortex behaviour both from the ground and by airborne sensors. Specific separation rules, reflecting the appropriate weights, speeds and configurations of the aircraft pairs can be applied for landings and departures. -> **(W2-14)**

Traffic optimisation on single and multiple runway airports aims at providing ATC with an integrated support tool to improve single and multiple runway airport operations by increasing the predictability of runway capacity, optimising runway configuration and optimising arrival/departure spacing. Reduced wake and weather-dependent separation minima is enabled when applicable, as well as ROT information when available. -> **(W2-14)**

Wave2: More efficient arrival separation between arriving aircraft will be delivered using better prediction of runway occupancy time and final speed profile also using big data / machine learning techniques and considering the effect of separation delegation (AUO-0507) on separation delivery tool and the effect of traffic mix using Increased glide slope or displaced thresholds. A/G exchange will be used in order to better predict ROT and in order to support flight crews expedite runway vacation.

Runway safety will be improved by predicting runway exit overshoot and developing advance lighting system also supporting safe use of increased glide slope or displaced threshold.

Dynamic Pairwise Separation minima are based on flight-specific aircraft characteristics taken from the eFPL and/or downlinked from the aircraft, or from an evolution of the eFPL (e.g. aircraft weight), potentially WX-dependent.

Flight specific information (e.g. vertical climb profile for departure or speed schedule in approach) will be used (including EPP) in order to support reduced separations between arrivals, departures or between arrivals and departures.

Enhanced arrival procedures - mitigation of noise and adaptation of wake avoidance procedures through concepts such as dual thresholds; second runway aiming point; glide slope increase; and double slope approaches, which move noise into the centre of the airport. In some runway configurations this may provide additional benefits of wake vortex separation between landing aircraft on common headings or parallel flight paths.

Minimum Pair Separations based on RSP - effective monitoring of small separation distances is strictly dependent on the Required Surveillance Performance (RSP), which may allow for the application of pair-wise separation down to a minimum of 2NM for arrivals on final approach.

Independent Rotorcraft operations at the airport - independent parallel or converging Rotorcraft-specific approach and departure procedures remove IFR rotorcraft traffic from active RWYs and allow independent fixed wing aircraft and rotorcraft Simultaneous Non-Interfering (SNI) approaches, which has a positive impact on runway throughput and improves access into small/medium airports in LVC.

[Solution PJ.02-W2-17]Improved access to secondary airports

Improved access into small/medium airports - access into small/medium airports /secondary airports in all conditions, including in particular low visibility conditions is increased thanks to precision approaches (RNP and GBAS/SBAS approaches) to all runways supplemented by equivalent visual approach and landing operation using on-board vision system, or enhanced RNP approach procedures enabled by satellite based navigation allowing approaches down to CATII minima (LPV 100 ft) without requiring additional infrastructure. In the context of small airports, with limited demand, new airborne capabilities makes it possible for aircraft to make a safe approach in low visibility conditions at any location.

Wave2: The improvements may be used by all airspace users, but may be focused on Business Aviation, GA, rotorcraft and military aircraft. Performance benefits are related to percentage of equipped aircraft systems and operational procedures that allow operational credits as considered by the Performance Based Aerodrome Operational Minima (PBAOM). For example:

Low visibility procedures (LVP) for towered secondary airports with support from a low-cost surveillance system (LCS) that covers the airport surface and is based on ADS-B and/or Multilateration in combination with cameras and vehicle tracking systems (VTS);

Combined Vision Systems compensate for the lack of natural visibility in bad weather conditions and enable pilots to get operational improvement in approach to most runways down to a very low runway visual range. These systems are also beneficial for taxiing on the ground. Improved performance efficiency and availability of current equivalent visual landing operations in low visibility conditions is achieved by improving the performances of the existing enabler (Infrared and visual sensor). On-board vision systems such as Combined Vision System (CVS) and Synthetic Vision Guidance System (SVGS) enables advanced Equivalent visual approach and landing operation by using forward external view topography provided by active sensor and combination of flight guidance and high precision position assurance monitors.

Enhanced Terminal Area for efficient curved operations using geometric vertical navigation guidance in the TMA simplifies operations by removing the workload associated with barometric to geometric vertical navigation transition and improve the efficiency and predictability of individual

operations. GBAS and SBAS capability in the Terminal Area allow increased flexibility in airspace design. In addition it is expected to improve safety. (PJ2-11)

Solution PJ.02-W2-04 Advanced geometric GNSS based procedures in the TMA

Advanced GNSS-based navigation based on vertical positioning will enable arrival procedures (including curved procedures) through the terminal area down to the runway, and the use of geometric altitude will enable optimised 3D routes that minimise levelling off.

Curved departures, potentially combined with precise geometric altimetry, use curved departure routes that turn shortly after take-off, e.g. in order to avoid noise-sensitive areas or approach or missed approach routes.

Use of curved approaches and geometric altitudes for parallel runways, addressing, for example: - Reduction of the MRS between pairs of aircraft flying PBN approaches (Baro V-NAV, SBAS (EGNOS) or GBAS approaches) to parallel runways.

SESAR 2020 Significant Changes

Wave1

Wake turbulence separation optimisation (PJ02-01)

- Weather-Dependent Reductions of Wake Turbulence Separations for Departures (AO-0304)
- Wake Turbulence Separations (for arrivals) based on Static Aircraft Characteristics (AO-0306)
- Weather-Dependent Reductions of Wake Turbulence Separations for Final Approach (AO-0310)
- Wake Turbulence Separations (for departures) based on Static Aircraft Characteristics (AO-0323)
- Reduction of Wake Turbulence Risk considering Acceleration of Wake Vortex Decay in Ground Proximity (AO-0325)
- Reduction of Wake Turbulence Risk considering Wake Monitoring and Awareness (AO-0327)
- Optimised Runway Delivery on Final Approach (AO-0328)
- Optimised Separation Delivery for Departure (AO-0329)

Enhanced arrival procedures (PJ02-02)

Mitigation of noise and wake avoidance procedures through concepts such as:

- Enhanced Arrival Procedures using Dual Thresholds (DT) (AO-0308)
- Enhanced Arrival Procedures using a Second Runway Aiming Point (SRAP) (AO-0319)
- Enhanced Arrival procedures using Increased Glide Slope (IGS) (AO-0320)
- Enhanced Arrival procedures using Adaptive Increased Glide Slope (A-IGS) (AO-0321)
- Enhanced Arrival Procedures using Double Slope Approach (DS) (AO-0322)
- Enhanced Arrival Procedure using an Increased Glide Slope to a Second Runway Aiming Point (IGS-to-SRAP) (AO-0331)

Minimum pair separations based on Required Surveillance Performance (RSP) (PJ02-03)

- Reduction in separation minima using Minimum Separation based on RSP (AO-0309).

Independent rotorcraft operations at the airport (PJ02-05)

- Independent IFR/VFR rotorcraft through the use of concepts such as:
 - SBAS based Point in Space (PinS) approaches and departures to/from VFR FATOs.
 - Simultaneous Non-Interfering Operations (SNI).
- Increased Airport Performance through Independent (parallel or convergent) IFR Rotorcraft Operations (AO-0316)

Improved access into secondary airports in Low Visibility Conditions (LVC) (PJ02-06)

- Improved Capacity and Safety of Runway Operations at Secondary Airports in Low Visibility Conditions (SDM-0301)."
- Improved Approach procedures into Secondary airports in Low visibility conditions (AO-0333)

Traffic Optimisation on single and multiple runway airports (PJ02-08)

- Optimised use of runway capacity for multiple runway airports (TS-0313)
- Integrated Arrival Departure Management for full traffic optimisation on the runway (TS-0301)
- Enhanced prediction of Runway Occupancy Time (AUO-0704)

Enhanced terminal area for efficient curved operations(PJ02-11) (AOM-0607)

- Ground-Based Augmentation System (GBAS) and SBAS capabilities increases flexibility of operations in the terminal area.
- General Aviation, RPAS and rotorcraft integrated in a multi-aircraft and manned flight environment.
- Enhanced Terminal Area for efficient curved operation (AOM-0607)

Wave 2

PJ02 W2 Airport airside and runway throughput

Solution PJ.02-W2-14 Evolution of separation minima for increased runway throughput This solution builds on the work performed in Wave 1 by solutions #02-01 and #02-08

This is a new SESAR Solution that needs to be created together with the required Operational Improvements steps (OIs) for a future Integrated Roadmap dataset.

The SESAR Solution could cover the following existing OI step in DS19:

- AO-0307: Wake Turbulence separations (for arrivals) based on Dynamic Aircraft Characteristics - as a successor to AO-0306 improving Wave 1 performance using advanced techniques such as big data / machine learning but also more digital information from MET providers or directly downloaded from the aircraft. -
- AO-0315 - (to be refined) : More efficient departure separation delivery between departing aircraft using better prediction of rolling distance and departure speed profile and trajectory also using big data / machine learning techniques as a successor to AO-0323.
- AUO-0504: Dynamic Adjustment of Spacing Depending on Wake Turbulences
- AUO-0505: Improved air safety using data exchange via e.g. ADS-B for Wake Turbulence prediction. Or the following ones that could be proposed as potential candidates for future datasets:
- AUO-0507: Airborne Spacing Monitoring under IFR (ATSAW - Spacing monitoring)
- AO-0315 - (to be refined) : More efficient departure separation delivery between departing aircraft using better prediction of rolling distance and departure speed profile and trajectory also using big data / machine learning techniques as a successor to AO-0323.
- AO-0324: Dynamic Pair Wise Separations for Departures - as a successor to AO-0323 improving Wave 1 performance using advanced techniques such as big data / machine learning but also more digital information from MET providers or directly downloaded from the aircraft.
- TS-0312: Increased runway capacity through allowance of simultaneous runway occupancy (only for departures).
- New proposed AO-XXXX - (to be further refined): More efficient arrival separation delivery between arriving aircraft using better prediction of runway occupancy time and final speed profile also using big data / machine learning techniques as a successor to AO-0328 and AO-0310 and considering the effect of separation delegation (AUO-0507) on separation delivery tool and the effect of traffic mix using Increased glide slope or displaced thresholds.

New proposed AO-XXXX - (to be further refined): Improved runway safety by predicting runway exit overshoot and developing advance lighting system also supporting safe use of increased glide slope or displaced threshold.

Solution PJ.02-W2-04 Advanced geometric GNSS based procedures in the TMA

Relevant links to the ATM Master Plan Level 2 (DS18a): This SESAR Solution follows-up the work performed by SESAR Solutions in Wave 1 and covers the following OI steps:

- AOM-0607 — Enhanced Terminal Area for Efficient Curved Operation. It may also include aspects of solution #02-02 not fully validated in the Wave 1 timeframe:
- AO-0322 - Enhanced Arrival procedures using double slope approach

In case parallel arrivals are addressed, this SESAR Solution would cover the following OI steps:

- AOM-0606 — Enhanced Parallel Approach Operations using PBN/RNP transitions to xLS.
- AOM-0608 — Enhanced Parallel Approach Operations using PBN/RNP transitions to RNP (for at least one runway)

In addition, the following OIs are potential candidates for inclusion in future integrated roadmap datasets that may be addressed by this SESAR solution:

- AO-0312- Increased Runway Throughput by reducing separation distance defining CSPR
- AO-0313 - More flexible SID design for advanced GNSS based procedures in TMA

Solution PJ.02-W2-17 Improved access to secondary airports

Relevant links to the ATM Master Plan Level 2 (DS18a): This SESAR Solution follows-up the work performed by SESAR Solution #02-06 in Wave 1 and covers the following OI steps:

- SDM-0301 — Improved Capacity and Safety of Runway Operations at Secondary Airports in Low Visibility Conditions.
- AO-0333 — Improved Approach procedures into Secondary Airports in Low Visibility Conditions. However, due to the enlarged scope, the SESAR Solution may require additional OI step in future integrated roadmap datasets.
- AO-0108 - Airport Safety Nets for Controllers at Secondary Airports
- AUO-0410 Equivalent Visual Approach and Landing operations providing improved resilience to LVC

Expected Benefits

- **Capacity** – Significant capacity enhancements are expected from the application of pair wise wake turbulence separation rules for final approach and for departures based on finer granularity of aircraft characteristics.
 - Time Based Separation (TBS) increase runway throughput as a function of traffic mix and weather conditions, directly proportional to the proportion of wake-constrained pairs in peak hours.
 - Low noise procedure allows modulating the curfew and enables additional slots.
 - Optimising runway-aiming point optimises runway occupancy. Runway second aiming point can serve as an enabler for reducing wake turbulence separations as well as reducing the wake vortex encounter risk.
 - Minimum-Pair separation at constrained airports increases RWY throughput, mainly for arrival (depending on traffic mix) during peak periods.
 - Improved access into small/medium airports in low visibility conditions increases the major airport capacity by allowing the use of small/medium airports in LVP.
 - Traffic optimisation on single and multiple runway airports increases runway capacity and in mixed mode minimises the interference between the two flows.
- **Predictability** enhancements are expected from TBS tools and procedures, which help to maintain arrival throughput in adverse headwind conditions thus reducing arrival delay that would otherwise occur.

Traffic optimisation on single and multiple runway airports contribute to a more realistic and precise accuracy of landing and departure times, thereby increasing the predictability, allowing airspace users to improve planning, avoiding passengers delays and cancelations during the execution phase.

Environment The various proposals for increased glide slopes reduce the noise footprint in the immediate area of the airport, as proposals for displaced touchdown zone(s)

Efficiency Automation and system support integrating various information sources and functions to support Tower Controller decision making.

- **Safety** is increased as the rotorcraft traffic is removed from the active runway by separating fixed-wing traffic from rotorcraft traffic by enabling rotorcraft approach and departure procedures to and from segregated landing and take-off areas.

The provision of a support tool for implementing the complex options for tactical and planning of runway traffic management improves the safety and capacity outcome of controller decisions.

Wave2

Performance expectations ; solutions are expected to have a positive impact on:

- **Airport and Airspace TMA Capacity** - Increase runway and airspace throughput (e.g. reducing runway occupancy time, or arrivals and departures wake turbulence separation) and resilience;
- **Improved access** to secondary airports, with expected benefits in resilience, cost-efficiency and increased flexibility under non-nominal conditions at the airport;
- **Predictability** - increased predictability and airport efficiency (e.g. runway occupancy time, runway exit and departure rotation);
- **Environmental Sustainability** - reduced fuel consumption and noise near the airports.

2.4.2 Integrated Surface Management

Surface movement management solutions are developed into an integrated environment dedicated to maintaining or improving current safety levels together with increased reliability and improved predictability of the planning milestones associated with aircraft taxi times. This includes the integration of RPAS into surface operations.

The new processes and technologies focus on the surface movement management in the two ground segments of the trajectory of the aircraft - taxi-in and taxi-out.

To ensure safe and seamless surface movement a common picture of the existing situation with a clear understanding of the intentions of the various ATM stakeholders is available to all involved. Thus, the ATC surveillance data is integrated into a system providing both positional information and intent between ATCOs, vehicle drivers and Flight Crews.

Global Navigation Satellite System (GNSS) information on position, speed and time displayed in the cockpit may be used in low surface visibility conditions to increase Flight Crew awareness of their position and heading.

The integrated surface management system generates a complete ground traffic situation display including routing information and guidance instructions that is presented in the tower, aircraft and vehicles operating on the manoeuvring area.

Enhanced Guidance Assistance to Aircraft and Vehicles on the Airport Surface Combined with Routing provides enhanced guidance to ATCOs, Flight Crews and vehicle drivers based on holistic surface traffic planning.

The holistic plan is broken down to individual, optimised route instructions reducing conflicting situations and being transmitted to Flight Crews or vehicle drivers via data link or radio telecommunication.

The ATCO HMI integrates all A-SMGCS functions and exchange relevant information with support systems such as AMAN, DMAN and the AOP.

Additionally, automated exchanges between Tower Controllers and Vehicle Drivers via datalink for ground-related clearances and information improve the efficiency of surface operations.

The guidance environment may be supplemented by a re-configurable virtual stop bar system flexibly adding virtual stop bars on the airport moving map in mobiles and on the ATCO's HMI.

Enhanced Visual Operations enables more efficient landing, taxi and take-off in low visibility conditions through Enhanced flight vision system operation enhancing vision or providing synthetic information of the surrounding terrain obstacles including airport surface information on Head Up Display, Head Down Display and/or a handheld device.

Surface Operations by Remotely Piloted Air Systems (RPAS) are integrated into the surface traffic management with procedures and regulations developed for manned mobiles.

On the aerodrome surface, RPAS interact with ATC much the same as manned operations: remote pilots request taxi clearance, report aerodrome hazards, and accept instructions concerning surface movement. Upon request, remote pilots will receive ATC clearance. The clearance may be loaded to RPAS automation or carried out manually by the remote Pilot-In-Command (PIC) or ground support personnel overseen by the remote PIC.

When taxiing, RPA needs to be capable of identifying and avoiding surface hazards (e.g. vehicles), adhering to ATC movement clearances and abiding by all aerodrome signage and markings unless alternate methods are developed and agreed.

Wave 2:

Solution PJ.02-W2-21 **Digital evolution of integrated surface management** is to increase pilots' and vehicle drivers' situational awareness by providing them with supplementary guidance means in all weather conditions which includes the extension of the A-SMGCS routing functions to support tactical conflict management in the taxi phase. Routing and planning are optimised to deliver more accurate taxi times, and minimise delays and controllers' workload. The input from airport DCB processes (that takes into consideration information provided by airport sequencing tools e.g. AMAN/DMAN) is integrated in order to determine potential conflicting situations when planning taxi routes.

The use of new algorithms, artificial intelligence/expert systems and AI/machine learning to increase automation for ATC in tower ground positions.

The advanced use of datalink and the extension of datalink below FL285 and on the airport surface will open new communication possibilities, it is expected that the use of datalink between controllers and pilots will increase and the use of voice decrease.

Enhanced guidance assistance to aircraft and vehicles on the airport surface combined with routing(PJ03a01)

- Guidance Assistance to Aircraft on the Airport Surface Combined with Routing for trajectory based operations (AUO-0603-B)
- Enhanced Guidance Assistance to Airport Vehicle Driver Combined with Routing (AO-0206)
- Airport ATC provision of ground-related clearances and information to vehicle drivers via datalink (AO-0215)
- Full Guidance Assistance to mobiles using "Follow the Greens" procedures based on Airfield Ground Lighting (aprons/taxiways/runways) (AO-0222-B)
- Enhanced Safety in LVP through use of Dynamic Virtual Block Control (AO-0223-B)
- Advanced Automated Assistance to Controller for Surface Movement Planning and Routing (AO-0224)
- Datalink Services used for Provision of Ground-related Clearances and Information for trajectory based operations (AUO-0308-B)

Enhanced Visual Operations(PJ03a-04)

- Equivalent visual take-off, landing and taxi operations performed through Enhanced Visual Systems and Synthetic Visual Systems for Operations in LVC. (AUO-0405, AUO-0406)

Surface Operations by RPAS(PJ03a-09)

- Enabling Integrated RPAS Surface Operations (AUO-0617)
- RPAS are integrated adequately into Surface Operations.

Wave 2

Solution PJ.02-W2-21 Digital evolution of integrated surface management

Relevant links to the ATM Master Plan Level 2 (DS18a): This SESAR Solution follows-up the work performed by SESAR Solution #03a-01 and #03b-01 in Wave 1 and covers the following OI steps:

- AO-0206 — Enhanced Guidance Assistance to Airport Vehicle Driver Combined with Routing.
- AO-0215 — Airport ATC provision of ground-related clearances and information to vehicle drivers via datalink.
- AO-0222-B — Full Guidance Assistance to mobiles using 'Follow the Greens' procedures based on Airfield Ground Lighting (aprons/taxiways/runways).
- AO-0223-B — Enhanced Safety in LVP through use of Dynamic Virtual Block Control.
- AO-0224 — Advanced Automated Assistance to Controller for Surface Movement Planning and Routing.
- AUO-0308-B — Datalink Services used for Provision of Ground-related Clearances and Information for trajectory based operations.
- AUO-0603-B — Enhanced Guidance Assistance to Aircraft on the Airport Surface Combined with Routing for trajectory based operations.
- AO-0104-B — Extended Airport Safety Nets for Controllers at A-SMGCS Airports

Expected benefits

Safety and performance is increased through Integrated Surface Management. It also increases **punctuality, predictability and flight efficiency** through better Trajectory Management and improved data exchange with partnering systems.

Reduced Flight Crew and ATCO workload through common situational awareness, improves low visibility procedures including airport accessibility minima, and reduces the probability of occurrence of hazardous situations in low visibility conditions by digital vision systems for Flight Crews.

Fuel Efficiency System support for the taxi route generation process, including planning information creates more time and fuel efficient taxi Routings resulting in less fuel burn and corresponding emissions in both the taxi-in and the taxi out phase, provided the Routing and timing is integrated with departure management information. The surface routing algorithms employed within the systems can consider more constraints than a human considers and thereby decrease overall arrival and departure delays and smooth overall flow.

Airspace and Airport Capacity through improved coordination for pre-departure management and arrival metering the workload is reduced in the integration of arriving/departing traffic.

Predictability benefits are expected in the Airport environment from taxi clearances with more predictable taxi times through system support and integration of planning information.

Wave 2

- **Safety:** maintain or increase runway, taxiway and apron safety levels, increase situational awareness, ATC Workload maintained or reduced;

2.4.3 Airport Safety Nets

Safety Nets at the airport are enhanced in order to contribute to the safety of operations.

The Tower Ground Controller uses the A-SMGCS to monitor the aircraft movement and track its progress against the issued taxi-out route and its position with respect to other surface traffic. He is provided with an automatic alert if the system detects potential and actual conflicting situations, incursions and non-conformance to procedures or ATC clearances.

For the Flight Crew, this means a system that alerts the crew in case of risk of collision with an aircraft or a ground vehicle on runways or taxiways. Furthermore, conformance-monitoring alerts (non-compliance with ATC clearances, procedures or airport configuration) are provided by an on-board system to the Flight Crew.

Prevention of runway excursions is improved through two new concepts. Firstly, an on-board support tool alerts Flight Crews in case a runway overrun is detected before or during landing. Secondly, advanced systems (e.g. built-in runway sensors, runway condition models, weather forecast) allow developing a more accurate measure of the actual and predicted runway condition, which can be shared with all airport actors.

Solution PJ.02-W2-25 Safety support tools for avoiding runway excursions

Safety support tools for avoiding runway excursions will improve runway condition awareness to increase safety in order to prevent runway excursions (take-off and landing), Runway status information on secondary airports is enhanced to increase safety (e.g. airport with short runways).

In future operations, additional information will be provided by the aircraft, which can be potentially used for improving both the current and predicted Runway Condition Code (RWYCC). ATM systems will relay the current and predicted RWYCC to the flight crew via any suitable means and it will be disseminated towards the other interested stakeholders i.e. APOC and AOC to enhance their situational awareness.

Enhanced Airport Safety Nets for Controllers (PJ03b-01)

- AO-0104-B Extended Airport Safety Nets for Controllers at A-SMGCS Airports
- AO-0108 Airport Safety Nets for Controllers at Secondary Airports
- AO-0109 Enhanced Airport Safety thanks to Time Critical Weather Alerts
- AO-0110 Airport Safety Enhanced by Prediction and by Detection of Adverse Traffic Patterns based on Ground Surveillance

Conformance monitoring safety net for Pilots (PJ03b-03)

- AUO-0614 Conformance Monitoring Safety Nets for Pilots
- The Flight Crew is provided with appropriate alerts, generated by on-board systems, to alert from non-compliance with airport configuration (e.g. closed taxiway/runway, restricted areas) and non-conformance to ATC clearances and procedures

Traffic alerts for pilots for airport operations (PJ03b-05)

- AUO-0605 Traffic Alerts for Pilots during Runway Operations
- AUO-0615 Traffic Alerts for Pilots during Taxiway Operations

- Traffic Alerts for pilots for airports operations refer to enhancing on-board systems in order to detect potential and actual risks of collision with other traffic during runway and taxiway operations). In all cases, the flight crew are provided with appropriate alerts.

Safety support tools for avoiding runway excursions (PJ03b-06)

- AO-0107 Improved Airport Safety with Better Prevention of Runway Excursions for Tower Controllers
- AO-0216 Enhanced Runway Condition Awareness
- AUO-0616 Improved Safety with Better Prevention of Runway Excursions for Pilots
-
- Support tools are provided to Flight Crew and Tower Controllers for assessing the risk of runway excursion (AUO-0616, AO-0107).
- Enhanced runway condition awareness for pilots and Tower Controllers (AO-0216).

Wave 2

Solution PJ.02-W2-25 Safety support tools for avoiding runway excursions

Relevant links to the ATM Master Plan Level 2 (DS18a): This SESAR Solution follows-up the work performed by SESAR Solution #03b-06 in Wave 1 and covers the following OI steps:

- AO-0107 — Improved Airport Safety with Better Prevention of Runway Excursions for Tower Controllers.
- AO-0216 — Enhanced Runway Condition Awareness.
- AUO-0616 — Improved Safety with Better Prevention of Runway Excursions for Pilots.

Expected benefits

Safety, Significant safety benefits are expected from new support tools and safety nets for surface operations. They will improve the situational awareness of all actors, including Flight Crews, ATCOs, vehicle drivers and the airport operator and alert them in case of a risk of collision, a risk of runway excursion or non-conformance to procedures or ATC clearances. This results in a reduction in the number of runway incursions, excursions, collisions and incidents on aprons, taxiways and gate areas.

Enhanced Visual Operations has a high impact on Safety thanks to the ability of Flight Crew to maintain an awareness of runway centreline.

Situation awareness is improved with the introduction of safety support tools for pilots. The number of conflicting situations during airport surface operations is also expected to be reduced.

Safety of runway operations is improved thanks to the provision of support to the pilots and ATC for assessing the risk of runway excursion during take-off and landing operations.

2.4.4 Total Airport Management

Airport DCB is part of the ATM network DCB processes incorporating key parameters of the Airport Operations Plan (AOP), which are used for planning/managing capacity and are shared with stakeholders through the Network Operations Plan (NOP). The airspace Demand and Capacity processes are optimised by more updated, accurate and coordinated information, coming from the Airport Operations Plan and reflected in the Network Operations Plan. This makes it easier to predict and act strategically to delays, thereby increasing airport capacity and overall network capacity.

This is possible due to enhanced information exchange capabilities (through SWIM). The synchronised Arrival and Departure planning at each airfield is linked to the departure times from (upstream) origin airfields and the arrival times for the (downstream) destination airfields.

Turnaround operation is an Airspace User controlled process. However, by sharing some key aspects with the APOC all actors can monitor the operation from a wider perspective and receive early warnings of inefficiencies/issues/failures resulting in potential delays and act pro-actively for coping with the disrupting event.

The solution does not alter the ownership of the turnaround, its activities or the decision making involved in achieving a turnaround. It however provides all impacted stakeholders with an alert so that they can pro-actively manage the situation, thereby minimising the delay to the departure flight itself and the knock-on impact to the Airport and the Network.

A set of milestones in the turn-round process are established at the airport and flight progress is monitored against those milestones. The information is shared by all relevant actors such as the Network Manager and APOC at destination airport. The completion of a milestone triggers decision making processes for downstream events. Shared information on the progress of turn-round is used to estimate departure demand and enable arrival/departure balancing.

The DCB process feeds the AOP with refined information needed in each of the different phases. UDPP is expected to be mostly utilised during the Medium to Short Term phase to cope with sudden DCB imbalances.

The Enhanced Collaborative Airport Performance Management is achieved through incorporation of Total Airport Management processes such as passenger, baggage, aircraft flows plus environmental and meteorological impacts showing current performance. Inclusion of outputs from landside processes (passenger & baggage) allows for improved accuracy and predictability of the airside operations.

Wave2: Data and video analytics, big data and machine learning techniques will improve the management of airport performance and support what-if decision support capabilities and using the power of data to understand how to incorporate historical operational knowledge. The purpose is to improve the management of airport performance and to anticipate special/crisis situations by correlation with historical data to mitigate as much as possible any airport performance decrease, better allocation of airport resources and thus improve resilience, predictability and cost efficiency.

The solution shall coordinate with solution 21 “Digital evolution of integrated surface management”, solution 1 “Next generation AMAN for 4D environment” and solution 49 “Collaborative Network Performance Management”.

A sudden capacity shortfall in the airport (i.e. any event not planned or predicted) affects each trajectory differently; requiring different actions for each individual flight in order to restore the balance of demand and capacity at the airport.

If flights are in an early pre departure status with sufficient time to react, the possible imbalances can be addressed through the normal DCB processes (CDM negotiation and UDPP) as a part of the normal maturity process of the trajectory with no necessity for arbitration by the Network Manager.

As the trajectory maturity progresses to its completeness towards flight execution, other processes are applied taking into account that there may be less time for reaching an agreement among the affected stakeholders involved in the negotiation.

Flights in execution are the ones with less flexibility for negotiating alternate solutions in a real time environment therefore DCB actions vary from dDCB (dynamic DCB) in the tactical area of intervention, deployment of previously agreed Contingency Plans where solutions are pre-described and finally UDPP if time permits¹⁸.

The Airport Operations Plan linked with the Network Operations Plan

The AOP connects the relevant stakeholders, notably the FOC in a Collaborative Decision-Making Process (CDM) framework [3]. It contains information relating to the different planning phases and is a rolling plan, which evolves over time.

In each flight phase, every stakeholder is responsible for producing and negotiating information needed to feed the AOP to get the full operational benefit of the CDM processes, as the optimum outcome of this relies on the quality and accuracy of the information exchanged between the processes.

Unscheduled flights are integrated in the AOP/NOP, at a very close time to flight execution as soon as trajectory data is delivered by the Airspace User.

Every single airport must have an AOP¹⁹, adapted to local airport characteristics but with a minimum content of key ATM elements that must be shared with the Network Operations Plan (NOP).

The rolling AOP/NOP is continuously updated by sharing of information between all stakeholders. This results in a single (rolling) plan for the Airport and the Network for all stakeholders involved.

The synchronised Arrival and Departure planning at each airport are linked to the departure times from (upstream) origin airports and the arrival times for the (downstream) destination airports. Any

¹⁸ UDPP is further described in the “Optimised Network Management” section of this document

¹⁹ The AOP content must be linked to a single airport. What could be done is common management of several AOPs from a single airport. There is a defined minimum core set of airport & traffic data that all airports need to publish to the network

delay in an ATOT (Actual Take-Off Time) should not only update the ELDT (Estimated Landing Time) at destination but possibly all future time estimations in planned trajectories of the same aircraft.

During the execution phase AOP monitor and detect any deviation from the planned operational expectations. The deviations are resolved in the Airport Operations Centre (APOC), involving the appropriate stakeholders through A-CDM.

Meteorological impacts on the AOP are pro-actively managed by decision support tools that can assess the impact of key meteorological conditions and proposes pre-defined solution scenarios.

Collaborative recovery procedures and decision support tools are put in place to support airport, network, AU and ANSP to anticipate and collaboratively manage large scale disruptive events to reduce impact and knock-on effects and optimising solutions.

Environmental sustainability of operations need to take into account the (typically local) environmental restrictions and considerations.

Post-Operations Analysis

All data recorded from the AOP/NOP throughout the planning and execution phases are used as the main source to build post-operations reports at airport level, those which affect to network performance shall be standardised at Network level.

The Post Operations Analysis (POAS) collects a coherent set of support tools for the timely analysis of specific events and post operations reporting to support a continuous airport learning environment.

Enhanced collaborative airport performance planning and monitoring (PJ04-01)

- Collaborative Airport Planning Interface (AOP fully integrated with NOP & local business rules) (AO-0801-B)
- A-CDM process enhanced through integration of landside (passenger and baggage) process outputs (AO-0802-B)
- Extended Turn-round monitoring (AO-0818)
- Post-Operations Analysis support solutions and reporting capabilities (AO-0821)
- Network-connected Regional Airports (AO-0824)
- Airside process is enhanced with the inclusion of landside (passenger and baggage flow) process.
- Delays are monitored through the extended Turn-round monitoring within the APOC for early warnings.
- Post-Operations Analysis support solutions for the timely analysis of specific events and reporting capabilities to support a continuous airport learning environment
- Airport Operational Plan is fully integrated with the Network Operational Plan via a Collaborative Airport Planning Interface
- Network-connected Regional Airports

Enhanced collaborative performance management (PJ04-02)

- Enhancement of Airside/Landside Performance Management (AO-0814)
- Pro-active management of meteorological impacts on the AOP (AO-0819)
- Pro-active Collaborative Management of Predicted Performance Deterioration via UDPP (AO-0820)
- Enhancement of Environmental Performance Management (AO-0822)
- Enhancement of Airside/Landside Performance Management fed with all landside and airside leading key performance indicators:
- Passenger, baggage, aircraft flows showing current performance.

- Meteorological impacts are pro-actively managed
- Enhancement of Environmental Performance Management
- All OI steps above include the possible creation of an airport learning environment and intelligent optimisation linked to Airport Performance Management for:
 - (a) improving what-if computation parameters and forecasts,
 - (b) adjusting Airport Performance Framework parameters (e.g. warning, alerts thresholds),
 - (c) supporting the retrieval from past experiences of most suitable solution(s) for an on-going event with observed impact and
 - (d) supporting the detection of airport performance issues with possible causes (performance drivers indicators).

Wave 2²⁰

PJ04 W2 Total airport management

Solution PJ.04-W2-28 Enhanced Collaborative Airport Performance Planning and Monitoring

Relevant links to the ATM Master Plan Level 2 (DS18a): This SESAR Solution follows-up the work performed by SESAR Solution #04-01 in Wave 1 and covers the following OI steps:

- AO-0801-B — Collaborative Airport Planning Interface (AOP fully integrated with NOP & local business rules)
- AO-0802-B — A-CDM process enhanced through integration of landside (passenger and baggage) process outputs.
- AO-0818 — Extended Turn-round monitoring.
- AO-0821 — Post-Operations Analysis support solutions and reporting capabilities.
- AO-0824 — Network-connected Regional Airports (note: this OI Step could reach V3 in Wave1)

Solution PJ.04-W2-29 Digital Collaborative Airport Performance Management

Relevant links to the ATM Master Plan Level 2 (DS18a): This SESAR Solution follows-up the work performed by SESAR Solution #04-02 in Wave 1 and covers the following OI steps:

- AO-0805 Improved airport collaboration in “Service to Schedule principle”
- AO-0807 Dynamic target profiles for airport performance
- AO-0808 — AOP for a group of airports
- AO-0809 Setting performance targets for a group of airports
- AO-0810 — Monitoring Performance for a group of airports
- AO-0811 Digital smart airport
- AO-0812 Improved Airport Performance via its Integration in a Multimodal Transport Network
- AO-0813 Enhanced Collaborative Airport Performance Management
- AO-0814 Enhancement of Airside/Landside Performance Management
- AO-0815 (Lite) APOC for regional airports
- AO-0816 Connecting regional APOC
- AO-0819 Pro-active management of meteorological impacts
- AO-0820 Pro-active Collaborative Management of Predicted Performance Deterioration via UDPP.
- AO-0822 Enhancement of Environmental Performance Management

²⁰ Several New OI-Steps for wave 2 solutions will be created in DS20

- AO-0825: Pro-active Collaboration Hub Airport / Network Management of Predicted Performance Deterioration
- AO-0826 Pro-active Collaborative Regional Airport / Network Management of Predicted Performance Deterioration
- New OI steps:
- What-If analysis
- Airport Performance Prediction Techniques
- APOC dashboard integrated in NM
- APOC Business re-engineering

Expected benefits

Capacity is increased by Enhanced Collaborative Airport Performance Management

The TMA and En Route Demand and Capacity processes are optimised by more updated, accurate and coordinated information, coming from the Airport Operations Plan and reflected in the Network Operations Plan.

Reaction to unexpected events is addressed more strategically acting over a rich information environment thereby increasing airport capacity and overall network capacity.

Predictability Inclusion of outputs from landside processes (passenger & baggage) allows for improved accuracy and predictability of the airside operations.

The creation of an airport learning environment and intelligent optimisation linked to Airport Performance Management, which would make an extensive use of post operations data for better planning of airport and network operations.

High Performance benefits from Advanced DCB Increases **predictability** in planning airport operations during adverse weather conditions, contributing to more predictable business trajectories.

By sharing some key aspects with the APOC all actors can monitor the operation from a wider perspective and receive early warnings of inefficiencies / issues / failures resulting in potential delays, and act pro-actively for coping with the disrupting event.

Fuel efficiency is increased on the ground through increased predictability of operations before trajectory execution, which contributes to optimisation of the pre-departure sequence

Environment Better estimation of the planned off block time, may lead to improved accuracy of the actual off block time (less difference between the planned OBT and the actual OBT), enabling smaller taxi queues. Less queuing results in less fuel burn. Additionally, should enable improved stand management and lead to a reduction in taxi or apron waiting for incoming aircraft. Reduction in waiting leads to a reduction in fuel burn.

Wave2 Performance expectations

The delivered solutions are expected to have a positive impact on:

- Better use of existing **airport capacity** and its optimisation to comply with traffic demand;
- A better situational awareness through SWIM information sharing, enabling provision and reception of Airport CDM data including MET and AIM;
- A significant increase in the **predictability, efficiency, environmental sustainability and flexibility** of airport operations thanks to a performance-driven airport through KPIs monitoring and early detection of deviations, collaborative decision making using support tools and what-if functions, post-operations analysis used as learning process, etc.;

- Proactive management of predicted impacts to normal operations, quicker reactions on deviations;
- Increased **resilience** through shorter and effective recovery to normal operations and collaboration with the Network, from predicted or unpredicted adverse operating conditions; Increased **safety** in the airport environment due to reduced uncertainty of operations and reduced congestion through better planning.

2.4.5 Remote Tower for multiple airports /Remote Tower centre

Remote Control Towers allow ANSPs to provide Aerodrome Control Service²¹ or Aerodrome Flight Information Service for two or more aerodromes by a single ATCO/ Aerodrome Flight Information Service Officer (AFISO) from a remote location, i.e. not from a conventional control tower at the aerodrome. This not only has a major impact on the service providers cost by allowing a single ATCO (or AFISO) to service multiple aerodromes, but there is also a positive impact on the Airspace Users that would benefit from such service at the small or rural aerodromes. A Remote Tower Services (RTS) solution also gives these airports an opportunity to expand technology related to safety in a cost efficient way.

Remote Tower Centers (RTC) provide Remote Tower Services to a large number of airports with a highly flexible and dynamic allocation of airports connected to different Remote Tower Modules (RTM). The ATCOs will have planning tools to manage aerodromes with more traffic. RTC supervisor will have planning tools for planning the allocation of airports managed by the RTC. The planning tools ensure a possibility to use the available Remote Tower Module in an efficient way, both for multiple and single airport operations.

The provision of remote ATS service to the aerodromes can be dynamically assigned (over time) to any Remote Tower Module within a Remote Tower Centre.

The Multiple Remote Tower Service needs to be supported by automated controller support tools in order to allow a high number of airports and traffic to be controlled by RTM.

Remote Tower solution may also offer the potential for contingency measure at a single major airport.

Wave 2: PJ05 W2 Digital technologies for Tower

Advanced automation functions to maintain ATCO situational awareness in operations involving multiple remote towers e.g. voice recognition, alerting and warnings for conflict detection and conflict resolution advisories, etc.; as well the integration of artificial intelligence (AI) and machine learning algorithms for the intelligent data provision in order to minimise the load and mental strain on the Tower controllers (especially under high traffic density situations, low visibility conditions, etc.). The objective is to improve ATCO productivity supporting the ATCO to get the benefits of a more automation level, while maintaining / increasing the level of safety.

The solution shall take into consideration the inputs from exploratory projects such as MALORCA and RETINA.

²¹ The main target for the Multiple Remote Aerodrome ATS concept is Small to Medium Aerodromes

PJ.05-02 Remotely Provided Air Traffic Service for Multiple Aerodromes

- **Remotely Provided Air Traffic Service for Multiple Aerodromes (up to 3 aerodromes) (SDM-0207)**

PJ.05-03 Remotely Provided Air Traffic Services from a Remote Tower Centre with a flexible allocation of aerodromes to Remote Tower Modules

- **Highly Flexible Allocation of Aerodromes to Remote Tower Modules (SDM-0210)**

Wave 2

PJ05 W2 Digital technologies for Tower

- **Solution PJ.05-W2-35 Multiple Remote Tower and Remote Tower Centre**

Relevant links to the ATM Master Plan Level 2 (DS19): This SESAR Solution follows-up the work performed by SESAR Solution #05-03 in Wave 1 and covers the following OI steps:

- SDM-0210 — Highly Flexible Allocation of Aerodromes to Remote Tower Modules.

- **Solution PJ.05-W2-97 HMI Interaction modes for Airport Tower**

Relevant links to the ATM Master Plan Level 2 (DS19): This SESAR Solution follows-up the work performed by SESAR Solution #16-04 and in ER project RETINA (but it may include additional features). Relevant OI steps and enabler(s) need to be created in the ATM MP Level 2 to cover the scope of the solution, on top of those already available:

- AO-0212 — Equivalent Visual Operations for Tower Control in Low Visibility

Expected benefits

Cost efficiency Better cost efficiency can be achieved through large-scale operation with co-location of staff, management, training, etc.

Safety and accessibility is improved by making advanced technology for surveillance available to smaller airports without high costs

Wave2

The solutions to be developed under this topic are expected to provide:

- Improved cost-efficiency with the optimisation of the use of available resources;
- Increased safety with an increased awareness of the traffic situation thanks to the use of digital display information available to the tower controller;
- Increased ATCO efficiency with the use of technologies allowing ATCOs to focus on key tasks.

3 High Level Operational Requirements

3.1 Wave 1 High Level Operational Requirements

Introduction

This section provides High Level Operational Requirements applicable to the SESAR 2020. For the sake of clarity, these requirements are structured by Operational Projects and Operational Solutions, considered as concept parts (parts of Key Areas).

This section does not anticipate the allocation of these requirements to H2020 Projects. This distinction may be easier to understand by considering that:

1. an Operational Project here is a sub-level of one Key Area ;
2. for a given part of the concept (i.e. an Operational Project or Operational Solution), there might be a coordination between various H2020 projects, or between their Solutions, to develop and validate the parts of the concept addressed by the Requirements. This is applicable to Waves 1 and 2.

The Operational Requirements, presented in a tabular format, include two levels:

- the Top Level Operational Requirements that address a given operational project (from PJ 01 to PJ 11)
- (at a lower level) the High Level Operational Requirements that address a given operation Solution within PJs 01 to 11; in this 2019 edition, the High Level Operational Requirements also address the Operational solution 18-02a of PJ 18.

Links between both levels (Top Level Operational Requirements and High Level ones) are provided in order to ensure the top/down traceability of the cascading in between. It is to be noted that a High Level Operational Requirement (addressing a given operational solution) may be originated by Top Level Requirements belonging to different PJs.

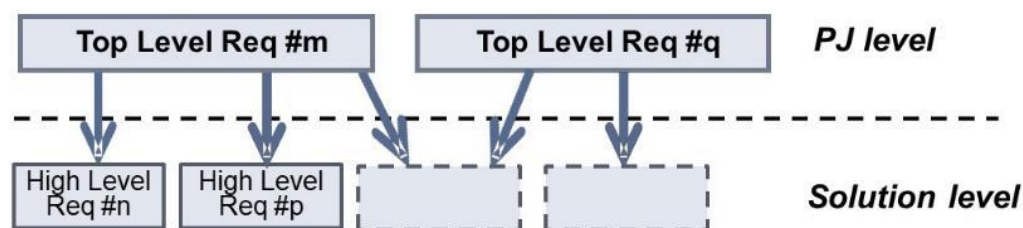


Figure 3: Requirement traceability matrix

Any Top Level Operational Requirements applicable to an operational PJ xx is structured this way:

- The identification of the requirement (numbered Pxx-TLOR-mmm)
- The semantic core of the requirement (see below)
- The perimeter, i.e. the operational scope [when needed for comprehensiveness]
- The main characteristics of the environment [whenever useful]
- The successor High Level requirements.

Every Top Operational Requirement is consistent with the concept described in chapter 2.

The semantic core mentions:

- The addressee ; mostly a global ATM process ; sometimes an operational actor
- The main target ATM Benefit(s)
- The (high level) conceptual way to achieve the ATM benefits²²
- Additional considerations if any

Every High Level Operational Requirement is structured this way:

- The identification of the requirement (numbered Sxx-yy-HLOR-mmm)
- The semantic core of the requirement
- The predecessor Top Level Requirement(s)
- Identified OI Step(s) addressed by the given High Level Requirement
- (Integrated Roadmap reference is Data Set 19 [4])

²² Detailed solutions are in hands of Operational Projects

3.1.1 Operational Requirements PJ 01

▪ Top Level Requirements PJ 01

Top Level Requirements applicable to PJ 01 Enhanced Arrivals and Departures		Applicable Solutions Requirements
P01-TLOR-01	Management of Arrivals and Departures shall be improved through the delivery of better queue management techniques and improved use of routes and airspace to ensure optimisation of the Extended TMA and consistent traffic flows through: • Arrival sequencing towards multiple TMAs • Coordination of dependent departure traffic flows from multiple airports within the TMA. • Use of predicted arrival and departure demand information to identify and resolve complex interacting traffic flows. • Use of CTA for sequencing in a High density/complexity environment • Dynamic use of routes to facilitate optimised climbs and descents • Enhanced use of Parallel Approach Operations based on PBN/RNP capabilities • Optimised use of airspace and controller capacity While: • managing potentially conflicting sequencing constraints in En Route sectors • ensuring integration with airport and network-wide DCB • ensuring a consistent and manageable delivery through areas of constraint (hotspots) within the TMA and into the En Route phase of flight • enabling aircraft to fly as efficiently as possible	S01-01-HLOR-01 S01-02-HLOR-01 S01-03-HLOR-01 S01-03-HLOR-02 S01-03-HLOR-03
	<u>Perimeter – Operational scope</u> • Low to Medium to High Density/Complexity Airspace • TMA • Extended TMA (Systemised Airspace encompassing the TMA and the climb/descent phases below FRA) • En Route (including FRA)	
	<u>Environmental contexts</u> • Mixed Mode Traffic regarding RNP and i4D capabilities	
	P01-TLOR-02 Management of Arrivals and Departures shall support and provide opportunities for improved sequencing and merging in the TMA and on approach, and better integration of	S01-05-HLOR-01 S01-06-HLOR-01

	rotorcraft operations through: • enabling Flight Crew to achieve and maintain a given spacing with another aircraft • Improved flight efficiency facilitating Visual approach combined with Visual Separation²³ in marginal VMC • In high density reducing the number of Go-arounds by supporting the visual acquisition when separation with preceding aircraft cannot be maintained due to compression effect • Designing rotorcraft procedures to allow improved access (under IFR) to VFR FATOs, based on advanced (e.g. curved) SBAS guided Point-in-Space (PinS) approaches and departures • Improved connections to a low level IFR route network for rotorcraft operations While: • Target aircraft may not be flying direct to the merge point • Operating in adverse weather conditions <u>Perimeter – Operational scope</u> • TMA • Extended TMA (Systemised Airspace encompassing the TMA and the climb/descent phases below FRA) <u>Environmental contexts</u> • Scheduled Aviation • Rotorcraft • Mixed Equipage Traffic	S01-07-HLOR-01
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²³ This operation is also named generically « CDTI Assisted Visual separation »

■ High Level Requirements PJ 01

High level Requirements applicable to PJ.01-01	Predecessor operational requirements
S01-01-HLOR-01 Extended Arrival Management with overlapping AMAN operations and interaction with DCB shall deliver improved queue management techniques through: • The simultaneous contribution to arrival sequencing towards multiple TMAs • Integrating Arrival Management constraints with constraints applied for DCB purposes • Providing an AMAN Allocated Time for aircraft departing from airports within the AMAN horizon • Use of CTA for sequencing in a High density/complexity environment	P01-TLOR-01 P09-TLOR-02
Identified, addressed OI Steps TS-0109, TS-0305-B, TS-0315	
High level Requirements applicable to PJ.01-02	Predecessor operational requirements
S01-02-HLOR-01 Use of Arrival and Departure Management Information for Traffic Optimisation within the TMA shall deliver improved optimisation of the Extended TMA through: • Support to departure metering and coordination of dependent traffic flows from multiple airports • Use of predicted demand information from local Arrival and Departure Management systems to identify and resolve complex interacting traffic flows • Balancing the demand within the TMA, optimising the airspace and controller workload. A tailored modification of demand will only take place if no further capacity can be added.	P01-TLOR-01
Identified, addressed OI Steps TS-0302, TS-0307	

High level Requirements applicable to PJ.01-03A	Predecessor operational requirements
S01-03-HLOR-01 Improved parallel approach operations shall be enhanced through the application of PBN capabilities/RNP navigation specifications, with a transition to final approach segments guided with xLS.	P01-TLOR-01
Identified, addressed OI Steps AOM-0606, AOM-0608	

High level Requirements applicable to PJ.01-03B	Predecessor operational requirements
S01-03-HLOR-02 Dynamic E-TMA for A-CCO and A-CDO shall deliver improved optimisation of the E-TMA through: The ability to support the transition into and out of airports and En Route sectors with both fixed route and free route airspace	P01-TLOR-01
<u>Identified, addressed OI Steps</u> AOM-0806	
S01-03-HLOR-03 Dynamic E-TMA for A-CCO and A-CDO shall deliver improved facilitation of optimum climb and descent profiles even in high density operations through: - Use of new controller tools and enhanced airborne functionalities - Use of a dynamic PBN route structure and dynamically assigned traffic flows	P01-TLOR-01
<u>Identified, addressed OI Steps</u> AOM-0702-B, AOM-0705-B	

High Level Requirements applicable to PJ.01-05	Predecessor operational requirements
S01-05-HLOR-01 Airborne Spacing Flight Deck Interval Management shall deliver opportunities for improved operations in the TMA through: The application of Interval Management clearance by ATC, enabling Flight Crew to achieve, capture and/or maintain a given spacing (the spacing shall be in time) with a designated aircraft which may not be flying direct to a merge point	P01-TLOR-02
<u>Identified, addressed OI Steps</u> TS-0108	

High Level Requirements applicable to PJ.01-06	Predecessor operational requirements
S01-06-HLOR-01 Enhanced Rotorcraft operations in the TMA shall deliver improved integration of rotorcraft into the RNP route structure through: - Design of rotorcraft procedures to allow increased access (under IFR) to/from VFR FATOs, based on new procedure design requirements (e.g. curved segments, lower RNP) and SBAS guided Point-in-Space (PinS) approaches and departures - Support the crew to improve performance and safety when flying the advanced Point-in-Space RNP approach and departure procedures	P01-TLOR-02 P02-TLOR-01
<u>Identified, addressed OI Steps</u> AOM-0104-B	

High Level Requirements applicable to PJ.01-07	Predecessor operational requirements
S01-07-HLOR-01 Approach Improvements through Assisted Visual Separation shall: - Facilitate Visual approach combined with Visual Separation to increase opportunities to shorten track and then improve flight efficiency - Facilitate delegation to the Flight Crew of the separation with respect to preceding traffic, in order to : - bring flexibility to manage speed optimally (instead of ATC vectoring) - continue the approach during a temporary loss of visual contact (resulting for instance from sun or city light dazzling) - In high density reduce the number of Go-arounds when separation with preceding aircraft cannot be maintained due to compression effect by supporting the visual acquisition and the maintaining of separation thanks to traffic information of the preceding aircraft provided to the crew, based on ADS-B data.	P01-TLOR-02
<u>Identified, addressed OI Steps</u> AUO-0507	

3.1.2 Operational Requirements PJ 02

▪ Top Level Requirements PJ 02

Top Level Requirements applicable to PJ.02 Increased Runway and Airport Throughput		Applicable Solutions Requirements
P02-TLOR-01	Airport operations shall increase throughput by the integration and combination of several aspects, inter alia, separation distances derived from time-based wake-vortex separation minima, Minimum Radar Separations, ROT predictions and integration of arrivals and departures management processes, through: • Application of S-PWS in arrivals and departures. • Separation minima adjustment to airport's local traffic mix patterns. • Application of Weather Dependant Separations (WDS) for departures and arrivals. • The improvement in the use of MET information in planning and execution flight phases for impact analysis and optimised procedures around adverse weather. • The integration of rotorcraft instrumental arrival and take-off operations into the traffic mix of the airport using FATOs. • Improved ROT techniques as differentiated glide paths, displaced thresholds and multiple runway aiming points. <u>While:</u> • Development of adequate system support for ATCOs and Flight Crews. • Ensuring avoiding Wake Vortex encounters. • Development of GBAS and SBAS enablers to a full maturity. • Reducing Minimum Radar Separation taking into account the new Time-Based Wake-Vortex distance separations, maintaining the same operational safety level. • Advancing in the maturity of using GNSS procedures for enriching approach procedures based on RNP. • Incrementing the operational resilience in adverse conditions. • Using aircraft broadcasted MET info and ground based sensors for feeding a Runway Usage Management Sub-system. • Assessing the planning conditions of the aforementioned new techniques during the transition periods between good and adverse weather/degraded operational conditions. Developing ground based ROT prediction tools.	S02-01-HLOR-01 S02-02-HLOR-01 S02-03-HLOR-01 S02-05-HLOR-01 S02-06-HLOR-01 S02-08-HLOR-01 S02-11-HLOR-01 S03b-06-HLOR
	<u>Perimeter - Operational Scope</u> • Medium and Large Airports • Medium and Small Airport with a significant number of	

	rotorcraft operations - Initial Departure Common Path. - Final Approach Segment. - Intermediate Approach Segment. - Initial Approach.	
	<u>Environmental contexts</u> - All Weather - Adverse Weather including strong head winds, LVC, etc. - All runway configurations and operations - Degraded operational conditions - Recovery Transitional Periods - All Traffic mixes	

■ High Level Requirements PJ02

High Level Requirements applicable to PJ.02-01	Predecessor operational requirements
S02-01-HLOR-01 The Optimisation of Wake Turbulence Separation shall: - increase runway throughput - ensure more refined and efficient wake separation than current ICAO rules - ensure consistent and efficient management of spacing on final approach and the initial departure phase of flight - by automatically managing the complexity of applying the required wake separation between each aircraft pair through: - the implementation of S-PWS and WDS - the use of the associated ATC support tools - while: - supporting passive wake vortex decay devices	P02-TLOR-01
<u>Identified, addressed OI Steps</u> AO-0304, AO-0306, AO-0310, AO-0323, AO-0325, AO-0327, AO-0328, AO-0329	

High Level Requirements applicable to PJ.02-02	Predecessor operational requirements
S02-02-HLOR-01 The Enhanced Arrival Procedures shall: ○ Increase runway throughput ○ Mitigate noise • by helping controllers integrate aircraft flying the following new procedures on the final approach : ○ Dual Thresholds (DT) ○ multiple Runway Aiming Points (MRAP) ○ Increased Glide Slope (IGS) ○ Adaptive Increased Glide Slope (A-IGS) ○ double slope approach (DS)	P02-TLOR-01
Identified, addressed OI Steps AO-0308, AO-0319, AO-0320, AO-0321, AO-0322, AO-331	

High Level Requirements applicable to PJ.02-03	Predecessor operational requirements
S02-03-HLOR-01 The Reduced Minimal-pair Separation shall: ○ Increase runway throughput • by: ○ reducing the arrival separation of non-wake turbulence constrained aircraft pairs to 2.0 NM • through: ○ the implementation of appropriate Required Surveillance Performance standards	P02-TLOR-01
Identified, addressed OI Steps <u>AO-0309</u>	

High Level Requirements applicable to PJ.02-05	Predecessor operational requirements
S02-05-HLOR-01 Independent Rotorcraft operations shall: ○ permit independent IFR rotorcraft operations to/from FATO located at airport: • by using, for rotorcraft: ○ point in space procedures ○ SVS / EVS / CVS capabilities ○ DGNSS navigation • while ○ offering simultaneous fixed wing operation in a non-interfering environment.	P02-TLOR-01

Identified, addressed OI Steps

AO-0316

High Level Requirements applicable to PJ.02-06	Predecessor operational requirements
S02-06-HLOR-01 Improved access into secondary (small/medium) airports shall: - o facilitate LVC operations - o increase throughput • by combining - o low cost visual sensors (cameras) for tower operations - o (SBAS/GBAS) or Localiser Performance with Vertical Guidance (LPV 100) equipment and related procedures (CVS) - o Robust EFVS supporting functions • while - o limiting major capital investment	P02-TLOR-01
<u>Identified, addressed OI Steps</u> SDM-0301, AO-0333	

High Level Requirements applicable to PJ.02-08	Predecessor operational requirements
S02-08-HLOR-01 The Advanced Coupled AMAN/DMAN shall: - o Optimise runway throughput - o increase predictability and improve strategic optimisation of runway arrival and departure demand - o improve accuracy of TLDT, TTOT and TSAT • by all or parts of : - o providing improved landing and departure times integrated with routing functions - o DCB tools for runways configuration in planning phase - o improved ROT calculation capability - o giving ATC more precise information about aircraft behaviour during first or last part of their flight - o linking the capability to new Routing and Planning systems that compute a more dynamic taxi-out time • through - o re-tactical use of the improved predictability - o improved prediction of variable taxi times	P02-TLOR-01
<u>Identified, addressed OI Steps</u> TS-0301, TS-0313, AUO-0704	

High Level Requirements applicable to PJ.02-11	Predecessor operational requirements
S02-11-HLOR-01 The Enhanced Terminal Area shall: ○ reduce the impact of transition between arrival route and final approach ○ reduce Flight Crew workload ○ reduce controller workload • by ○ supporting precise and efficient GNSS curved route navigation in the TMA	P02-TLOR-01
<u>Identified, addressed OI Steps</u> AOM-0607	

3.1.3 Operational Requirements PJ 03a

■ Top Level Requirements PJ 03a

Top Level Requirements applicable to PJ 03a Integrated Surface Management		Applicable Solutions Requirements
P03a-TLOR-01	Integrated surface management shall take into account the following aspects: • Arrival and Departure Managers (AMAN/DMAN) fully integrated with the advanced surface management and airport operations management functions. • Predictable/variable taxi times as essential inputs for turnaround process integration in the surface management operations. • Integrated and efficient planning for optimisation of the underutilised available resources. • Integrated DMAN/AOP/A-SMGCS/AMAN systems for better planning taxi-In and taxi-Out phases and to enable routing changes reaction available to the aircraft. • Increase of situational awareness of ATCOs, Flight Crews and Vehicle Drivers during operations and especially during LVC. • Integration of RPAs, GA and Rotorcraft into the airport operations.	S03a-01-HLOR-01 S03a-03-HLOR-01 S03a-04-HLOR-01 S03a-09-HLOR-01
	While: • Increasing the reliability and predictability level of the taxi times. • Increasing information sharing among AMAN/DMAN/Surface Management Systems in a secure mode with proper cyber security architecture. • Increasing predictability and efficiency of the turnaround process. • Airport management ability to adapt to unknown or unplanned constraints. • Release heavy using of radio communications introducing taxi clearances.	
	<u>Perimeter - Operational Scope</u> • Small, medium and large airports • Recovery Transitional Periods	
	<u>Environmental contexts</u> • Mixed mode runway or dependent runways operations. • All Weather • Low Visibility Conditions.	

▪ **High Level Requirements PJ03a**

High Level Requirements applicable to PJ.03a-01	Predecessor operational requirements
S03a-01-HLOR-01 The Enhanced Guidance Assistance to Aircraft and Vehicles shall: ○ deliver more accurate taxi times ○ minimise delays ○ minimise controllers' workload ○ minimisation of waiting time for mobiles at intersections • by ○ Optimising routing and planning ○ providing enhanced guidance assistance to Flight Crews, vehicle drivers and controllers to handle airport operations ○ considering potential conflicting situations when planning taxi routes ○ using AGL as speed control ○ consolidating the set of instructions that produces relevant benefits when exchanged via data link ○ defining an Aerodrome Mapping Database (AMDB) to be shared, via SWIM, between ground and on-board systems • while ensuring, through datalink: ○ the exchange of VSB positions and statuses ○ the exchange of clearances / instructions between vehicle drivers and controllers	P3a-TLOR-01
Identified, addressed OI Steps AO-0206, AO-0215, AUO-0603-B, AO-0222-B, AO-0223-B, AO-0224, AO-0308-B	

High Level Requirements applicable to PJ.03a-04	Predecessor operational requirements
S03a-04-HLOR-01 Enhanced Visual Operations shall increase access in LVC: - by augmenting Flight Crew vision during landing (EFVS) , taxi, and take-off through the use of: - legacy, and new higher performance active sensors - Heads up and Heads down displays - while - defining requirement for sensors, displays and human factors. - Studying the opportunity for new MET observations and MET needs for the safe operation of these systems.	P3a-TLOR-01
Identified, addressed OI Steps AUO-0405, AUO-0406	

High Level Requirements applicable to PJ.03a-09	Predecessor operational requirements
S03a-09-HLOR-01 The integration of RPAS, capable to fly IFR rules and capable of flying SIDs and STARs as designed for manned operations on the airport surface shall provide useful input to the integration of RPAS in surface operations - by identifying the particular requirements of remotely piloted surface operations in relation to: - Detect & Avoid (D&A); - Automated landing and take-off; - Platform operations; - Ground movements; - Contingency; - while - Identifying technologies that could support unmanned surface operations - Investigating interoperability between ATC and RPAS safety nets - Investigating procedural issues to support unmanned surface operations	P3a-TLOR-01
Identified, addressed OI Steps AUO-0617	

3.1.4 Operational Requirements PJ 03b

■ Top Level Requirements PJ 03b

Top Level Requirements applicable to PJ.03b Airport Safety Nets		Applicable Solutions Requirements
P3b-TLOR-01	Airport Safety Nets shall enhance the Safety of airport operations by : • Providing Flight Crews with airborne-generated Traffic alerts in case of a risk of collision with an aircraft or ground vehicle on runway and taxiways. • Providing Flight Crews with airborne-generated Conformance monitoring alerts related to ATC clearances and procedures. • Providing Safety tools for assessing risk of runway excursions in both take-off and landing While: • Integrating GA, Rotorcraft and RPAS operations • Complying with regulation (EU) No 716/2014. • Supported by new standards (related to the new airport Safety nets) which will be required for appropriate interoperability, complying with other, non-airport related Safety Net requirements for applications and core surveillance performance.	S03b-01-HLOR-01 S03b-03-HLOR-01 S03b-05-HLOR-01 S03b-06-HLOR-01
	<u>Perimeter - Operational Scope</u> • All airports, especially those with complex layouts, high traffic density, or small airports without safety nets for controllers.	
	<u>Environmental contexts</u> • All Weather • Low Visibility Conditions. • All traffic mixes	

High Level Requirements PJ03b

High Level Requirements applicable to PJ.03b-01	Predecessor operational requirements
S03b-01-HLOR-01 The Enhanced airport safety support tools for controllers shall increase airport safety : - by developing Airport Safety Nets that: - Detect conflicting situations, excursions, incursions and non-conformance, involving mobiles (and stationary traffic) on runways, taxiways and in the apron/stand/gate area as well as unauthorised / unidentified traffic - Provide solutions to secondary (small/medium) airports with limited surveillance capabilities - Provide solutions to controllers to better address weather hazard situations and provide Time Critical Weather Alerts - while - minimising the level of false and nuisance alerts - Considering of other types of Airspace Users (GA, Rotorcraft and Civil RPAS).	P3b-TLOR-01
Identified, addressed OI Steps AO-0104-B, AO-0108, AO-0109, AO-0110	

High Level Requirements applicable to PJ.03b-03	Predecessor operational requirements
S03b-03-HLOR-01 The Conformance monitoring alerts for pilots shall increase Flight Crew situation awareness - by - integrating alert indications into the cockpit display system for both the regional and the business platforms - while - providing sufficient acknowledgement - considering low visibility conditions	P3b-TLOR-01
Identified, addressed OI Steps AUO-0614	

High Level Requirements applicable to PJ.03b-05	Predecessor operational requirements
S03b-05-HLOR-01 The Traffic alerts for pilots for airport operations shall Increase airport safety - by - detecting potential and actual risk of collision with other traffic during taxiway and runway operations - providing the Flight Crew with the appropriate alert - while insuring a compatibility between all traffics (e.g. based on worldwide shared standards)	P3b-TLOR-01
<u>Identified, addressed OI Steps</u> AUO-0605, AUO-0615	
High Level Requirements applicable to PJ.03b-06	Predecessor operational requirements
S03b-06-HLOR-01 The Safety support tools for avoiding runway excursions shall: - Increase airport safety - reduce the risk of runway excursions - by using data from ground-based and airborne sources to: - provide a Runway Overrun Awareness and Avoidance System (ROAAS) for business aircraft - provide an objective evaluation of the runway condition - improve the collaborative use of the runway capacities through a shared view of the current situation - while - share this information among the main airport stakeholders - satisfying the new regulatory requirements in the ICAO Annex 15	P02-TLOR-01 P3b-TLOR-01
<u>Identified, addressed OI Steps</u> AO-0107, AO-0216, AUO-0616	

3.1.5 Operational Requirements PJ 04

▪ Top Level Requirements PJ 04

Top Level Requirements applicable to PJ 04 Total Airport Management		Applicable Solutions Requirements
P04-TLOR-01	The evolutions towards a performance-based airport management shall be achieved, through: • Full integration of AOP/NOP inter-related functions. • Full integration of landside/airside performance monitoring and management. • Refining the turnaround monitoring with the APOC in coordination with Airspace Users. • Integrating all aspects from De-Icing operations into the AOP to allow refinement of airport planning. • Development of the Total Airport Demand and Capacity processes and tools built upon the work achieved in SESAR 1. • Consolidation of the KPIs monitored and managed through the APOC, focusing on leading indicators of future performance. • Develop an airport learning environment (Airport Business Intelligence). <u>While:</u> • Integrating environmental impacts into the planning and execution timeframes of the AOP. • Improving the impact assessment tools available to the APOC in coordination with the Airspace Users. • Further development of the collaborative recovery procedures and support tools in coordination with all relevant stakeholders (Network Manager, Airport Operators, ANSPs, Airspace Users). • Fully integrating execution tools (AMAN, DMAN, A-SMGCS...) and resource allocation planning tools (Stand/gate Allocation Planner) into the Airport Demand and Capacity processes. • Refining the “What if” decision suite to support decision making between airport stakeholders. • Facilitating the pro-active management of predicted performance deteriorations. • Addressing Post-Operations Analysis processes, support tools and reporting capabilities, using post operations data • Using a combination of relevant MET data (satellite, radar, models) and advancing forecasting methodologies developed in SESAR 1.	S04-01-HLOR-01 S04-02-HLOR-01

	<u>Perimeter - Operational Scope</u> • All Airports • Network Operations • Recovery Transitional Periods	
	<u>Environmental contexts</u> • All Weather • GA and Rotorcraft • Civil RPAS • All traffic mixes plus a significant number of rotorcraft operations	

■ High Level Requirements PJ04

High Level Requirements applicable to PJ.04-01	Predecessor operational requirements
S04-01-HLOR-01 The Enhanced Collaborative Airport Performance Planning and Monitoring shall: ○ increase operational resilience ○ increase departure time predictability ○ increase operational resilience of the pre-departure sequence • by ○ extending airport performance process monitoring to the airport landside processes ○ developing an airport dashboard to support pro-active management of ATM related disruptions ○ Providing tools to support post-operations analysis to support the recording and timely analysis of specific events ○ Provide the ability to quickly find, merge, analyse and replay airport data with linkage to other stakeholder tools ○ refining, in association with Airspace Users, a suitable level of turnaround process planning and monitoring within the APOC • while ○ complementing the Airport's legacy systems ○ defining a suitable level of Turnaround Process Monitoring within the APOC	P04-TLOR-01
<u>Identified, addressed OI Steps</u> AO-0801-B, AO-0802-B, AO-0818, AO-0821, AO-0824	

High Level Requirements applicable to PJ.04-02	Predecessor operational requirements
S04-02-HLOR-01 The Enhanced Collaborative Airport Performance Management shall: ○ enhance collaborative management of airport performance ○ increased stakeholder common situational awareness • by ○ sharing relevant information in the AOP with the Network Manager and other NOP users ○ developing collaborative procedures between the airport and Network Manager ○ developing decision support functionalities (what if scenario capabilities, trend prediction) ○ orchestrating tactical execution support tools (arrival, departure and surface management systems) via the AOP • while ○ insuring security compliancy is in line with required standards for on line critical systems	P04-TLOR-01
<u>Identified, addressed OI Steps</u> AO-0814, AO-0819, AO-0820, AO-0822	

3.1.6 Operational Requirements PJ 05

▪ Top Level Requirements PJ 05

Top Level Requirements applicable to PJ.05 Remote Tower for Multiple Airports		Applicable Solutions Requirements
P05-TLOR-01	The provision of ATS from remote tower for multiple airports shall take into account the following aspects: • To support simultaneous operations in different aerodromes • To apply SWIM-enabled functions. • To maintain or even increase operational situation awareness. • To address human performance and roles needed for supporting ATS. While: • Maintaining safe working methods as safe as current service provision. • Addressing different kind of operational environments considering the mix of complexity, controller workload and type of traffic. • Maintaining an acceptable network quality of service and operational resilience. • Further developing Controller working position (CWP). • Considering additional automation functionalities. • Gradually increasing the operating range of the concept. • Addressing fall back and standardisation needs.	S05-02-HLOR-01 S05-03-HLOR-01
	<u>Perimeter - Operational Scope</u> • Small to medium airports • Recovery Transitional Periods	
	<u>Environmental contexts</u> • Several airports connected to a site with several modules for remote air traffic service. • Various types of traffic mix, including RPAS, GA, BA and Rotorcraft. • Adverse Weather	

■ High Level Requirements PJ05

High Level Requirements applicable to PJ.05-02	Predecessor operational requirements
S05-02-HLOR-01 Remotely Provided Air Traffic Service for Multiple Aerodromes shall increase RTM cost efficiency : - by incorporating into the RTM: - features for automation and support for operators in an environment with several connected aerodromes - representations of MET information for multiple airports and airports with more than one runway - enhanced PTZ functionality for a CWP suitable for several aerodromes - enhanced VCS - overlays to cope with several aerodromes at the same time - ATCO planning tools in the Multiple RTM (MRTM) - while - addressing cyber security - addressing SWIM related service models - handling planning issues such as the daily operative work	P05-TLOR-01
Identified, addressed OI Steps SDM-0207	

High Level Requirements applicable to PJ.05-03	Predecessor operational requirements
S05-03-HLOR-01 The Remotely Provided Air Traffic Services from a Remote Tower Centre with flexible allocation of aerodromes to Remote Tower Modules shall: - optimise the balance between traffic demand and the number of ATCOs/AFISOs required - increase cost-efficiency - by - flexible allocation of airports to the different RTMs within the RTC - supporting the RTC supervisor with appropriate functionalities - supporting the ATCO/AFISO with appropriate functionalities - integrating MET information in the RTM - supporting RTC supervisor planning tasks with appropriate tools - connexion with ATCO planning tools - while : - finding the balance between providing all information required at a certain moment while avoiding clutter of information - applying handover procedures for a flexible transfer of aerodromes between MRTMs	P05-TLOR-01

Identified, addressed OI Steps

SDM-0210

3.1.7 Operational Requirements PJ 06

▪ Top Level Requirements PJ 06

Top Level Requirements applicable to PJ 06 Trajectory Based Free Route		Applicable Solutions Requirements
P06-TLOR-01	Free Route concept shall enable Airspace Users to plan and fly trajectories without reference to a fixed route network or published directs within high complexity environments in order to provide them with significant opportunities to optimise their associated flights in line with their individual operator business needs or mission requirements while ensuring: • Efficient FOC/WOC operations to define and maintain trajectories in planning and execution phases of the flight • ATC workload and complexity to remain at acceptable levels • Optimised use of maintained capacity and high performing service thanks to DCB operations both at local, sub-regional and regional levels • Airspace Management and civil/military flexible Use of Airspace • To respect the associated Safety and Human Performance Objectives	S06-01-HLOR-01 S06-01-HLOR-02 S06-01-HLOR-03 S06-02-HLOR-01 S06-02-HLOR-02 S06-02-HLOR-03 S06-02-HLOR-04 S06-02-HLOR-05
	<u>Perimeter – Operational scope</u> • Mixed Mode Traffic • RPAS • General Aviation and rotorcraft • Scheduled Air Transport • Compatibility with Flight Centric operations • Interaction with TMA • Transition aspects with conventional ATS Route network	
	<u>Environmental contexts</u> • En Route • Low to High Density/Complexity Airspace	

■ High level Requirements PJ 06

High level Requirements applicable to PJ 06-01	Predecessor operational requirements
S06-01-HLOR-02 Free Route operations shall allow ATC services to be performed across boundaries, with maintained or increased capacity compared to current operations, even without reference to published directs or fixed route network, while ensuring: • smooth entry and exit to/from Free Route Airspace and adapted procedures for transition to/from conventional ATS route Network, in both vertical and horizontal plans, • full situation awareness and manageable workload for ATC thanks to advanced Controller toolset for coordination and management of trajectories interactions.	P06-TLOR-01
Identified, addressed OI Steps AOM-0505	

High level Requirements applicable to PJ 06-02	Predecessor operational requirements
AOM-0506	
S06-02-HL-02 Free Route operations shall allow ATC services to be performed across boundaries, with maintained or increased capacity compared to current operations, even without reference to published directs or fixed route network, while ensuring; • smooth entry and exit in/from Free Route Airspace and adapted procedures for transition to/from conventional ATS route Network, in both vertical and horizontal plans, • complexity resolution, traffic synchronisation and separation provision in execution phase thanks to revision process using non published waypoints defined in lat/long or bearing/range if needed, • full situation awareness and manageable workload for ATC thanks to advanced Controller toolset for coordination and management of trajectories interactions.	P06-TLOR-01
Identified, addressed OI Steps AOM-0506	

S06-02-HLOR-05 Free Route operations shall ensure a seamless transition from/to ETMA/TMA with FRA. A flexible structure must provide the optimal and efficient link between FRA airspace (in which CCO may be continued and CDO may be initiated) and SID/ STAR to be used in ETMA/TMA. The overall structure must be manageable by ground and airborne systems so that pilots and ATCOs understand the shared operational context.	P06-TLOR-01
<u>Identified, addressed OI Steps</u> AOM-0506	

3.1.8 Operational Requirements PJ 07

■ Top Requirements PJ 07

Top Level Requirements applicable to PJ.07 Optimised Airspace User Operations		Applicable Solutions Requirements
P07-TLOR-01	Optimised Airspace User Operations shall • Increase the Airspace Users operations and flight cost efficiency • Increase the Airspace Users flexibility to plan and re-plan a flight; • Contribute to an optimised demand and capacity management; • Enable Airspace Users to optimise their flights throughout the whole life cycle of the flight by defining and implementing Trajectory Management processes on AU side that feed into ATM DCB processes that : • Integrate the trajectories calculated/ planned by the Airspace User throughout the whole lifecycle of the flight; • Integrate the Airspace User defined flight priorities and preferences throughout the whole lifecycle of the flight; • Allow the civil Airspace User to swap slots between eligible flights, • Allow the Airspace User to revise the RBT/RMT While: • Support an integrated briefing throughout all flight phases; • Supporting prioritisation and re-prioritisation throughout all flight phases; • Supporting fleet delay assignment and flight protection; • Supporting AU driven RBT revisions throughout the whole flight execution • Integrating civil and military Airspace Users into the Trajectory Management, DCB and UDPP when required.	S07-01-HLOR-01 S07-01-HLOR-02 S07-02-HLOR-01 S07-02-HLOR-02 S07-04-HLOR-01 S07-04-HLOR-01
	<u>Perimeter - Operational scope</u> • Manned and unmanned commercial civil Airspace Users (scheduled) operations • Manned and unmanned commercial civil Airspace Users (non-scheduled) operations • Fixed-route and Free Route airspace • Manned and unmanned general aviation operations	
	<u>Environmental contexts</u> • All ATM environments	

P07-TLOR-02	Optimised Airspace User Operations shall: • Increase the effectiveness of mission trajectories effectiveness and flight efficiency ; • Increase the military Airspace Users' mission flight efficiency and safety • Facilitate optimisation of ATM network operations by defining and implementing ATM solutions that : • integrates the mission trajectory into the ATM network operating environment throughout the whole ATM lifecycle integrates SMTs into CDM on the Target Time negotiation when required enable a Pan-European mission and training planning and enables planning and executing of military training at Pan-European scale facilitates capacity optimisation : • Providing enhanced data consistency through the use of a harmonised improved OAT FPL format and content for military IFR operations • Increasing predictability and awareness on state AU demand • Supporting the integration of military operational requirements related to meteorological conditions; • Supporting the integration of military preferences and mission trajectories for DCB processes; • Developing a Pan-European OAT-IFR Transit Service (OATTS); • Allowing military Airspace Users to protect RMTs. <u>Perimeter – Operational scope</u> • Manned and unmanned military Airspace Users operations <u>Environmental contexts</u> • All ATM environments	S07-02-HLOR-01 S07-02-HLOR-02 S07-03-HLOR-01 S07-03-HLOR-02
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■ High Level Requirements PJ07

High Level Requirements applicable to PJ.07-01	Predecessor operational requirements
S07-01-HLOR-01 Trajectory definition processes shall allow civil Airspace User to plan optimised trajectories that best consider their own operational requirements while fulfilling the requirements of the other ATM stakeholders expressed with ATM constraints through • involvement of the civil Airspace User throughout all appropriate ATM planning activities; • implementing a CDM process assessing what-if scenarios that allows civil Airspace Users to join the DCB optimisation process; • Allowing civil Airspace Users to re-plan their flight trajectories whenever seen necessary or required by any ATM constraint; • Integration of the 4D trajectories provided by the civil AUs in the SBTs; • Allowing civil Airspace User to trigger the RBT when the trajectory is mature enough to safely and efficiently operate the flight; and • Provision detailed constraint information to civil AUs that is adequate to understand the reason behind the constraint and plan an alternative trajectory, which can be provided as adjusted/updated SBT.	P07-TLOR-01
<u>Identified, addressed OI Steps</u> AUO-0207, AUO-0208, AUO-0219	
S07-01-HLOR-02 Trajectory definition processes shall be as efficient and effective as possible to reach: • An optimum trade-off between planning effort and flight operations benefit; especially in the long-term and early medium-term planning phases; • High degree of process automation on AU side to reduce human workload and systems scale as much as possible; and • High degree of flight-cost efficiency during the conduct of the flight by allowing the planning of optimised trajectories.	P07-TLOR-01
<u>Identified, addressed OI Steps</u> AUO-0207, AUO-0208, AUO-0219	

High Level Requirements applicable to PJ.07-02	Predecessor operational requirements
S07-02-HLOR-01 All types of civil Airspace User shall be able to participate to DCB processes throughout the whole lifecycle of a flight by influencing the allocation of ATM constraint (to a flight) by - Protecting individual flights for capacity constraint situations; - Flexibly assign fleet delays for capacity constraint situations; - Flexibly re-prioritise flights in case of changing conditions and changing capacity constraint situations; and - Allowing proposing a flight priority order to NM in case of deteriorated operations.	P07-TLOR-01 P07-TLOR-02
Identified, addressed OI Steps AUO-0109, AUO-0110, AUO-0106, AUO-0107	
S07-02-HLOR-02 UDPP processes shall be as efficient and effective as possible to reach: - An optimum trade-off between planning effort and flight operations benefit; - High degree of process automation on AU side to reduce human workload & intervention and systems scale as much as possible; and - High degree of flight-cost efficiency during the flight conduction by allowing the planning of optimised trajectories.	P07-TLOR-01 P07-TLOR-02
Identified, addressed OI Steps AUO-0106, AUO-0107	

High Level Requirements applicable to PJ.07-03	Predecessor operational requirements
S07-03-HLOR-01 Trajectory management shall allow the planning and executing of the most efficient (from military AU perspective) military missions by - Involving military and civil Airspace Users into the DCB processes through iSMT Target Time negotiation when deemed necessary; - Involving WOC in refinement and iterative update of MT via CDM with due regard to AUs priorities. - Exploiting the capability of a Pan-European OAT-IFR Transit Service for cross-border operations.	P07-TLOR-02
Identified, addressed OI Steps AUO-0210, AUO-0211, AOM-0303, AOM-0304-A, AUO-0215, AUO-0228	

High Level Requirements applicable to PJ.07-03	Predecessor operational requirements
S07-03-HLOR-02 Trajectory definition processes shall be as efficient and effective as possible to reach: • An optimum trade-off between planning effort and subsequent flight operations benefit in execution; • High degree of process automation on state AU side to reduce human workload and systems scale as much as possible; and • High degree of flight efficiency during the flight operations by allowing the optimisation of trajectories	P07-TLOR-02
<u>Identified, addressed OI Steps</u> AUO-0210, AOM-0303, AOM-0304-A, AUO-0215, AUO-0228	

3.1.9 Operational Requirements PJ 08

▪ Top Requirements PJ 08

Top Level Requirements applicable to PJ.08 Advanced Airspace Management		Applicable Solution Requirements
P08-TLOR-01	Dynamic Configuration of Airspace shall enhance support to sector design and configurations allowing airspace to be managed as a continuum (i.e. coordinated single sky) in order to make optimum use of available airspace resource, staffing versus Airspace Users demand through: • Optimisation of sector design, configurations and opening schemes by using integrated predicted workload function and confidence index; • Definition of new airspace design principles (such as Dynamic Mobile Areas (DMAs)²⁴), satisfying users' need whilst minimising impacts on the network; • Full integration into DCB process. While: • Recognising national prerogative on airspace allocation • Supporting civil and military preferred trajectories; • Developing flexible airspace design for Long Term planning predicting needed resources to meet the demand; • Initiating Dynamic Airspace Configurations (DAC) based on traffic flow complexity and historical data; • Defining airspace structure and interfaces of DMAs; • Contributing to the refinement of coordination and interfaces with other processes (e.g. DCB, flight planning); • Ensuring adequate coverage of GA, Rotorcraft and civil RPAS operations.	S08-01-HLOR-01 S08-01-HLOR-02 S08-02-HLOR-01
	<u>Perimeter - Operational scope</u> • Mixed Equipage Traffic • Fixed route and Free Route airspaces	
	<u>Environmental contexts</u> • Scheduled Air Transport • Low to High Density/complexity Airspace • En Route • TMA	

²⁴ Dynamic Mobile Area (DMA) are defined and detailed in § 1.9.

■ High Level Requirements PJ 08

High Level Requirements applicable to PJ.08-01	Predecessor operational requirements
S08-01-HLOR-01 Dynamic configuration of Airspace shall allow optimum use of available airspace and staffing on a performance based approach by: - Utilising fully integrated collaborative decision making at local/sub-regional/regional levels for airspace management and airspace reservation; - Employing sector design and configurations unconstrained by predetermined boundaries to allow a seamless and coordinated approach for airspace configurations from planning to execution phases; - Utilising increased flexibility from “modular sectorisation adapted to traffic flows” concept in defining sectors/airspace building blocks, including cross-border; - Assessing and conciliating local level airspace configurations to build optimal configuration at network level; - Using ARESDMA Type and 2 sharing concepts extended at European level (including Cross Border Operations/Cross Border Areas).	P08-TLOR-01
Identified, addressed OI Steps AOM-0208-B, AOM-0805, AOM-0809-A	
S08-01-HLOR-02 <u>Dynamic configuration of Airspace</u> shall support rapid/automated assesement of dynamic sectorisation and constraint management allowing management/optimisation of workload and complexity at local level in order to: - Manage the airspace as a continuum to meet users’ expectations in ENR/TMA at all complexity levels; - Fully integrate ASM/DCB/ATS layered processes, optimising airspace configuration based on workload and complexity; - Make best use of available airspace and staffing at any given time.	P08-TLOR-01 P09-TLOR-02
Identified, addressed OI Steps CM-0102-B	

High Level Requirements applicable to PJ.08-02	Predecessor operational requirements
S08-02-HLOR-01 Dynamic configuration of Airspace shall allow enhanced use of available airspace by: - Using of Dynamic Mobile Areas of type 3; - Using Hazard zone in order to mitigate the hazardous impact of for instance the meteorological phenomena on ATM network operations.	P08-TLOR-01

Identified, addressed OI Steps

AOM-0208-C, AOM-0209

3.1.10 Operational Requirements PJ 09

▪ Top Requirements PJ 09

Top Level Requirements applicable to PJ.09 Advanced Demand and Capacity Balancing		Applicable Solutions Requirements
P09-TLOR-01	Advanced Demand and Capacity balancing shall expand the dynamic integration of operational data for improving tailored real time performance assessment and allow continuous monitoring of Network Performance through: • Provision of probabilistic traffic demand; • Sharing of common awareness on capacity constraints; • Agreement of performance criteria. While: • Developing confidences indexes allowing evaluation of accuracy of the trajectory data; • Defining metrics and algorithms for predicting, detecting or assessing traffic complexity; • Integrating the NOP in ATM stakeholders' planning processes and working method; • Providing prognosis of network performance for assessing impact of intentions and/or actions versus agreed overall optimum; • Developing advanced planning tools; • Supporting enhanced CDM process by providing What-if capabilities.	S09-01-HLOR-01 S09-03-HLOR-01 S09-03-HLOR-02
	<u>Perimeter – Operational scope</u> • Mixed Mode Traffic • En Route • Low to High Density/Complexity Airspace • TMA • Extended TMA • Free Route Airspace	
	<u>Environmental contexts</u> • Scheduled Aviation	

P09-TLOR-02	Advanced Demand and Capacity balancing shall ensure from planning up to execution at local, sub-regional and regional levels seamless and fully collaborative Flow & Capacity Management operations through: • Integration of Airspace Users (including UDPP), Airport (including A-CDM), ATC and Network planning processes in a holistic network planning management; • Integration of DAC and DCB; • Supervision of the network. While: • Developing collaborative constraint management processes (reconciliation mechanisms of multiple constraints); • Enhancing complexity assessment and resolution techniques; • Improving DCB interfaces with stakeholders; • Enhancing INAP capabilities. <u>Perimeter – Operational scope</u> • Mixed Mode Traffic • Low to High Density/Complexity Airspace • TMA • Extended TMA • Free Route Airspace <u>Environmental contexts</u> • Scheduled Aviation	S01-01-HLOR-01 S08-01-HLOR-02 S09-01-HLOR-01 S09-02-HLOR-01 S09-03-HLOR-01 S09-03-HLOR-02
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■ High Level Requirements PJ 09

High Level Requirements applicable to PJ.09-01	Predecessor operational requirements
S09-01-HLOR-01 The Network prediction and performance processes within a continuum from planning through to execution, shall provide increased efficiency by: - Assessment of performance of network operations performed at local/sub-regional/regional levels are continuously monitored through network performance KPA/KPI, with stakeholders able to evaluate the impact of their intentions and decisions on capacity and Quality of Service performance; - Automated tools supporting traffic complexity assessment to predict congestion and to support resolution; - Exploitation of the shared iterative SB/MT development providing continuous traffic demand forecast from long term planning to execution phases in order to allow enhanced business trajectory forecast methodology (probabilistic demand) to provide a standardised confidence indicator and define common operational usage of 4D trajectory forecast.	P09-TLOR-01 P09-TLOR-02
Identified, addressed OI Steps CM-0103-B, DCB-0211, DCB-0212	

High Level Requirements applicable to PJ.09-02	Predecessor operational requirements
S09-02-HLOR-01 Integrated local DCB processes shall provide to all actors in the INAP environment access to all flow, trajectory and capacity options by: - Automated tool supporting the local DCB and Extended ATC Planning actors in traffic complexity/ATC workload assessment at the planning and execution phases in order to increase capacity by providing timely and accurate prediction on upcoming congestions and providing appropriate input to tools handling hotspot/complexity resolution. - Seamless integration of airspace management functions and dynamic airspace configurations; - Support of the identification, assessment (including the use of 'what if' modelling) resolution of local complexity in a dynamic airspace management environment; - Collaborative flight planning and constraints management; - Improvement of situational awareness.	P09-TLOR-02
Identified, addressed OI Steps CM-0104-B, CM-0302, DCB-0210, DCB-0213	

High Level Requirements applicable to PJ.09-03	Predecessor operational requirements
S09-03-HLOR-01 The Collaborative <u>Network Management Functions</u> shall provide, in a continuum from planning up to execution, network stability, increased capacity and increased efficiency by: • Conciliating multiple constraint resolution strategies from all the stakeholders in order to identify the best measure given nature, scope and time horizon; • Monitoring Network performance and provide network impact assessment; • Drawing latest information shared via SWIM; • Providing stakeholders with access to enhanced “What-if” functionalities.	P09-TLOR-01 P09-TLOR-02
Identified, addressed OI Steps DCB-0103-B, DCB-0214, DCB-0215, DCB-0217	
S09-03-HLOR-02 Collaborative inter-airport decision-making, as a result of multi-airport interaction, shall deliver a more stable AOP for all airports, and hence a more stable network operation by specific monitoring/management of connected APOCs, at airports with a significant proportion of inter-dependent traffic.	P09-TLOR-01 P09-TLOR-02
Identified, addressed OI Steps AUO-0108	

3.1.11 Operational Requirements PJ 10

▪ Top Requirements PJ 10

Top Level Requirements applicable to PJ.10 Separation Management En Route and TMA		Applicable Solutions Requirements
P10-TLOR-01	Enhanced Separation Management in En Route and TMA shall manage separation in a safe, orderly and expeditious way through sensible combinations of : • Improved performance of enhanced ATC assistance tools; • New functionalities for detection, resolution and coordination tasks; • Shared information provision between all concerned stakeholders; • New separation tools for ATC assistance and Automated support; • More efficient sector team organisations and responsibilities; • Flight-centred ATC in En Route environment;	
	while: • respecting the associated Safety and performance objectives of capacity, productivity, cost efficiency; • Achieving more efficient flight profiles; • Minimising ATFCM constraints.	S10-01a-HLOR-01 S10-01b-HLOR-01 S10-01c-HLOR-01 S10-02a-HLOR-01 S10-02b-HLOR-01 S10-05-HLOR-01 S10-06-HLOR-01
	<u>Perimeter - Operational scope</u> • Outbound from single or multiple airports ; from single or multiple sectors (TMA, ACC) ; from individual runway(s) • Delivery to single or multiple sectors (TMA, ACC) ; to single or multiple airports ; to individual runway(s) • Mixed Mode Traffic • Scheduled Air Transport • Free Route Airspace • General Aviation and rotorcraft • RPAS	
	<u>Environmental contexts</u> • Low to High Density/Complexity Airspace • TMA • Extended TMA • En Route	

■ High level Requirements PJ 10

High level Requirements applicable to PJ 10-01a	Predecessor operational requirements
S10-01a-HLOR-01 The High Productivity Controller Team Organisation, based on one Planner Controller (Multi-Sector Planner) to several Executive Controllers, shall provide efficient and safe distribution of responsibility for traffic and separation management across the team, in order to provide efficient flight profiles through the airspace sectors under control and ensure an efficient distribution of tactical workload between the Executive Controllers.	P10-TLOR-01
<u>Identified, addressed OI Steps</u> CM-0303 ²⁵ , CM-0304	

High level Requirements applicable to PJ 10-01b	Predecessor operational requirements
S10-01b-HLOR-01 Flight Centric ATC in low, medium, high and very high complexity environments shall improve in flexibility and cost effectiveness by assigning ATCOs the responsibility for a specific number of aircraft throughout their flight segment within a given airspace in order to dissolve sector boundaries and to have one controller in charge of a single flight to guide it through a large airspace. • Need to ensure the transition with sector-based ATC (in the vertical or horizontal plane) • Check compatibility with DCB processes (Flow management and INAP...) • Support for Coordination between ATCOs • Support for unambiguous and synchronised sharing of : ○ Conflict detection status ○ assignment of responsibilities for conflict solving ○ validation of the resolution including with respect to 3rd party aircraft • Compatibility with safety nets and adapted procedures in case of multiple/ complex conflicts configuration)	P10-TLOR-01
<u>Identified, addressed OI Steps</u> CM-0200-B, CM-0200-C	

²⁵ Reference MSP - see footnote section 2.3.2 regarding impending change of scope for MSP

High level Requirements applicable to PJ 10-01c	Predecessor operational requirements
S10-01c-HLOR-01 Collaborative Control, based on new responsibilities adapted to sector team organisations, shall remove the systematic need for coordination between different ATS units in En Route and TMA in order to: • reduce ATC workload, • remove strategic constraints at AoR boundaries (leading to optimised trajectories and improving the performance of conflict detection and resolution tools),	P10-TLOR-01
<u>Identified, addressed OI Steps</u> CM-0306, CM-0309, CM-0310	

High level Requirements applicable to PJ 10-02a	Predecessor operational requirements
S10-02a-HLOR-01 Improved conflict detection and resolution tools with more accurate trajectory prediction and aircraft derived data shall enhance the performance in the provision of separation by : • supporting ATC decision making during execution of the flight • providing ATCOs with increased opportunity to optimise the execution of trajectories (thanks to longer look-ahead time, integration of both flow management and Trajectory Management options, including across ATSUs/ borders).	P10-TLOR-01
<u>Identified, addressed OI Steps</u> CM-0209, CM-0209-B, CM-0210, CM-0210-B, 0206, CM-0208-A, CM-0605, CM-0606	

High level Requirements applicable to PJ 10-02b	Predecessor operational requirements
S10-02b-HLOR-01 Advanced Separation Provision shall increase the quality of services of separation management by introducing automation mechanisms in order to reduce the ATCO workload, reduce separation buffers and facilitate new controller team organisation by • Improved ground trajectory prediction; • Resolution support providing effects on flight efficiency to allow prioritisation; • Improved conformance monitoring taking into account the calculated ground system trajectories.	P10-TLOR-01
<u>Identified, addressed OI Steps</u> CM-0207-B, CM-0208-B, CM-0407, CM-0408, CM-0607, 0608	

High level Requirements applicable to PJ 10-05	Predecessor operational requirements
S10-05-HLOR-01 RPAS capable to fly IFR rules and capable of flying SIDs and STARs as designed for manned operations shall be able to use technical capabilities or procedural means to be safely integrated in ATM including compliance with ATC instructions in order to be integrated in non-segregated airspace by: • understanding and determining whether RPAS fit into the current manned aircraft classification criteria, or whether there is a need to establish a specific RPAS operation classification; • addressing flight preparation, requiring information management for flight planning, where all intended flight-trajectories are planned in a manner compatible with the ATM Network; • assessing the impact of the behaviour of RPAS with regards to other aircraft, due to the potential latency and a different flight awareness of the crew, and its possible influence on separation provision; • understanding of RPAS-specific trajectories that are not easy to describe in the existing B/MT format • understanding the sensitivity of RPAS to severe weather conditions and their ability to anticipate them; • addressing ATC that will need awareness of RPAS activities in their area of responsibility, as well as the other Airspace Users; • addressing whether RPAS is able to fly IFR in managed IFR airspace where VFR flight is permitted • analysing the effects of loss-of-C2 link procedures, which will be developed for RPAS contingencies conditions.	P10-TLOR-01
Identified, addressed OI Steps AUO-0618	

High level Requirements applicable to PJ 10-06	Predecessor operational requirements
S10-06-HL-01 Generic (non-geographical) Controller Validations shall identify the needs that enable a more flexible ATCO regime, allowing a Controller to operate in any airspace classified as a particular type. These types shall define the additional aids (information, support in emergencies, fallback modes of operation, etc.) that are needed in addition to the pure sector and separation management ones, allowing a safe operation with more flexibility of Controller deployment.	P10-TLOR-01
Identified, addressed OI Steps SDM-0203	

3.1.12 Operational Requirements PJ 11

▪ Top Requirements PJ 11

Top Level Requirements applicable to PJ.11 Enhanced Air & Ground Safety Nets		Applicable Solutions Requirements
P11-TLOR-01	Enhanced air and ground based safety nets shall comply with future operations management, as defined in ICAO ASBU Block 2, and new separation modes (with lower separation minima) through: - the use of enhanced surveillance means (taking advantage of ADS-B information) - minimising negative operational interactions between Ground & Airborne Collision Avoidance processes (Safety nets) ; - optimising the rules for triggering Resolution Advisories; - Including compatibility with passive sources; while: - respecting the associated Safety and Human Performance Objectives; - ensuring their continuous role of last ATC safety layer against the risk of collision (and other hazards); - timely identifying hazardous situations for AUs with disparate levels of equipment; - avoiding undesired/unnecessary alerts; - maintaining independent Ground and Airborne Safety Nets systems; <u>Perimeter – Operational scope</u> - Mixed Mode Traffic - Scheduled Air Transport - General Aviation and rotorcraft - Civil RPAS <u>Environmental contexts</u> - Low to High Density/Complexity Airspace - TMA - Extended TMA - En Route including Free Route Airspace	S11-G1-HLOR-01 S11-A1-HLOR-01 S11-A2-HLOR-01 S11-A3-HLOR-01 S11-A4-HLOR-01

■ **High level Requirements PJ 11**

High level Requirements G11	Predecessor operational requirements
S11-G1-HLOR-01 Enhanced ground-based safety nets adapted to future operations, as defined in ICAO ASBU Block 2, shall provide: • An alerting system, which is robust against information error, or missing information. • Improvements on the Ground-based safety net performance, e.g. to detect that the separation mode has been compromised and to provide/propose resolution action. By: • Using enhanced information sharing ADS-B information • Keeping ACAS and STCA independent at functional level • Ensuring compatibility between ACAS and STCA • Adapting STCA to new separation modes (i.e. lower separation minima)	P11-TLOR-01
Identified, addressed OI Steps CM-0812, CM-0813, CM-0814 CM-0815	

High level Requirements applicable to PJ 11-A1	Predecessor operational requirements
S11-A1-HLOR-01 Airborne Collision Avoidance for Commercial Air Transport (ACAS Xa) shall be improved in terms of safety and operational compatibility by: • Using information from the more accurate surveillance source available; • Optimising the rules for triggering a Resolution Advisory by applying state-of-the-art mathematical processes and modelling. • Without significantly changing the cockpit procedures.	P11-TLOR-01
Identified, addressed OI Steps CM-0808-a	

High level Requirements applicable to PJ 11-A2	Predecessor operational requirements
S11-A2-HL-01 Airborne Collision Avoidance for Remotely Piloted Aircraft Systems (ACAS Xu), capable to fly IFR rules and capable of flying SIDs and STARs as designed for manned operations, shall be improved in terms of safety and operational compatibility by: • Using information from the more accurate surveillance source available (i.e. ADS-B and other sensors installed on the RPAS); • Optimising the rules for triggering a Resolution Advisory by applying state-of-the-art mathematical processes and modelling adapted to the RPAS capabilities.	P11-TLOR-01
Identified, addressed OI Steps CM-0808-u	

High level Requirements applicable to PJ 11-A3	Predecessor operational requirements
S11-A3-HL-01 ACAS for Commercial Air Transport specific operations (ACAS Xo) shall be improved in terms of safety and operational compatibility by: • Using information from the more accurate surveillance source available; • Optimising the rules for triggering a Resolution Advisory by applying state-of-the-art mathematical processes and modelling. While avoiding unnecessary triggering of Resolution Advisories in new separation modes (i.e. ASAS), in particular if lower separation minima are considered.	P11-TLOR-01
Identified, addressed OI Steps CM-0808-o	

High level Requirements applicable to PJ 11-A4	Predecessor operational requirements
S11-A4-HL-01 Airborne Collision Avoidance for General Aviation and Rotorcraft (ACAS Xp) shall be provided by making GA/RC aircraft capable of responsive coordination (i.e. after receiving an ADS-B squitter that identifies them as an intruder in an ongoing Resolution Advisory on-board of another aircraft, the GA/RC aircraft would be aware of the Resolution Advisory direction of the other aircraft, and would use it to determine its own manoeuvre with GA/RC adapted advisories).	P11-TLOR-01
Identified, addressed OI Steps CM-0808-p	

3.1.13 - Operational Requirements PJ 18

■ High level Requirements applicable to Solution 18-02a

S18-02a-HLOR-01 Enhanced Air/Ground operational procedures for TBO using complex clearances supported by data link and increased automation shall improve the following KPIs:

- Predictability,
- Sector Capacity
- Fuel Efficiency
- Safety

through :

- 2D trajectory modifications in the current sector and in advance, via complex clearances as needed, safely and efficiently implementable on board the capable aircraft
- An initial set of 3D trajectory modifications
- The accommodation by the ATCOs of a Flight crew requested trajectory
- 2D consistency check using ADS-C EPP

by :

- Reducing air/ground communication workload
- Reducing ATCO workload to issue 2D and an initial set of 3D clearances through system support, including the concatenation of CPDLC messages, and therefore allowing optimal distribution of tasks and improved usage of ATCO's time
- Allowing ground-ground exchange of proposed 2D trajectories between sectors, which may include the proposals originating from the flight crew
- Allowing a consistency check between ground and airborne trajectories

Identified, addressed OI Steps

SDM-0215, SDM-0214, SDM-0213

Perimeter - Operational scope

- Inbound and outbound flights within or out of ECAC area
- Scheduled Air Transport
- Free Route Airspace

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route
- IFR operations
- Wide availability of ADS-C / ATN-B2, Enhanced Surveillance, ADS-B
- Basic IOP

3.2 High Level Operational Requirements for Wave 2 Solutions

Introduction

This section provides High Level Operational Requirements applicable to the Operational solutions of SESAR 2020 Wave 2. For the sake of clarity, these requirements are structured by Operational Projects and Operational Solutions, considered as concept parts (parts of Key Areas).

This section does not anticipate the allocation of these requirements to H2020 Projects. This distinction may be easier to understand by considering that:

1. an Operational Project here is a sub-level of one Key Area ;
2. for a given part of the concept (i.e. an Operational Project or Operational Solution), there might be a coordination between various H2020 projects, or between their Solutions, to develop and validate the parts of the concept addressed by the Requirements. This is applicable to Waves 1 and 2.

The Operational Requirements are presented in a tabular format.

Every High Level Operational Requirement is structured this way:

- The identification of the requirement (numbered Sxx-yy-HLOR-mmm)
- The semantic core of the requirement
- Identified OI Step(s) addressed by the given High Level Requirement

Integrated Roadmap reference is Data Set 19 [9].

3.2.1 Operational Requirements PJ 01

Important note: The Requirements allocated to PJ 01 were defined based on the closest available view of its future content. The High Level Requirements consider:

- A solution 06 unchanged with respect to the DOW
- A solution 8A taking charge of the OI steps TS-0302, TS-0307 targeted to V3, allocated to Solution 08 in the DOW, as well as some conceptual elements allocated to solution 01 in the DOW
- A solution 08B taking charge of the OI Steps AOM-0702-B, AOM-0705-B, AOM-0806 targeted to V2, allocated to Solution 08 in the DOW

- **High level Requirements applicable to Solution PJ.01-06**

S01-06-HLOR-01 Advanced Rotorcraft Operations in the TMA shall integrate different AUs like rotorcraft, GA and RPAS in the same operational environment and airspace under the same rules through:

- Suitable ATC procedures
- Suitable HMI for ATCO and pilot
- The use of new technologies not previously evaluated for rotorcraft operation

Identified, addressed OI Steps

CM-0808-p

(The Solution covers additional aspects that will require the creation of a new Operational Improvement step (OIs) for future Datasets)

Perimeter - Operational scope

- Rotorcraft
- GA
- RPAS

Environmental contexts

- Low to High Density/Complexity Airspace
- TMA
- CTR
- G-class airspace

- **High level Requirements applicable to Solution PJ.01-08A**

S01-08A-HLOR-01 Use of Arrival and Departure Management Information for Traffic Optimisation within the TMA shall deliver improved optimisation of the Extended TMA through:

- Departure metering and coordination of dependent traffic flows from multiple airports
- Use of predicted demand information from local Arrival and Departure Management systems to identify and resolve complex interacting traffic flows
- Balancing the demand within the TMA, optimising the airspace and controller workload.
- Use of support tools to aid decision making and improve situational awareness'

Identified, addressed OI Steps

TS-0302, TS-0307

Perimeter - Operational scope

- Traffic with destination regulated by AMAN ,
- Traffic flying through congested airspace regulated by AMAN
- Departing traffic from airports inside the E-AMAN horizon

Environmental contexts

- Low to Very High Density/Complexity Airspace
- En Route
- TMA,
- Airspace between ToD/ToC and TMA border (so called Extended TMA, E-TMA)
- E-AMAN In-horizon Departure Airports

- **High level Requirements applicable to Solution PJ.01-08B**

S01-08B-HLOR-01 Dynamic E-TMA for A-CCO and A-CDO shall deliver improved facilitation of optimum climb and descent profiles even in high density operations through:

- Use of new controller tools and enhanced airborne functionalities
- Descent speed support
- Use of a dynamic PBN route structure and dynamically assigned traffic flows
- air-ground sharing of AMAN and EPP planned times

Identified, addressed OI Steps

AOM-0702-B, AOM-0705B

Perimeter - Operational scope

- Arriving traffic descending and entering TMA

Environmental contexts

- Low to Very High Density/Complexity Airspace
- En Route, TMA

S01-08B-HLOR-02 Dynamic E-TMA for A-CCO and A-CDO shall enable a response to traffic demand variations shifting the trade-off thresholds between KPAs, e.g. emphasis on environmental aspects versus focus on capacity through:

- Dynamic route structures supporting arrival traffic synchronisation
- In certain environments strategic de-confliction of arrival and departure traffic flows.

Identified, addressed OI Steps

AOM-0806 – Dynamic Management of Terminal Airspace Routes and Transition

Perimeter - Operational scope

- Departing traffic climbing and exiting TMA
- Arriving traffic descending and entering TMA

Environmental contexts

- Low to High Density/Complexity Airspace
- TMA

S01-08B-HLOR-02 Dynamic E-TMA for A-CCO and A-CDO shall enable a response to traffic demand variations shifting the trade-off thresholds between KPAs, e.g. emphasis on environmental aspects versus focus on capacity through:

- Dynamic route structures supporting arrival traffic synchronisation
- In certain environments strategic de-confliction of arrival and departure traffic flows.

Identified, addressed OI Steps

AOM-0806 – Dynamic Management of Terminal Airspace Routes and Transition

Perimeter - Operational scope

- Departing traffic climbing and exiting TMA
- Arriving traffic descending and entering TMA

Environmental contexts

- Low to Very High Density/Complexity Airspace
- TMA
- Airspace between ToD/ToC and TMA border (so called Extended TMA, E-TMA)

3.2.2 Operational Requirements PJ 02

- High level Requirements applicable to Solution 4

S4-HLOR-01 The Enhanced Traffic Sequencing shall enhance the segregation of arriving traffic flows, based on the wake categories and approach speeds in order to obtain :

- Increased flight efficiency
- Easy transition between arrival route and final approach
- Reduced Flight Crew workload
- Reduce controller workload

by combination of :

- Precise and efficient GNSS curved route navigation in the TMA for capable aircraft
- Use of geometric altitudes for capable aircraft
- Parallel arrival slopes on same runway
- Lateral displacements
- Minimum PBN requirements
- Safe Minimum vertical separations

Identified, addressed OI Steps

AOM-0607, AO-0322

Perimeter - Operational scope

- Medium and Large Airports
- Single runways
- Final Approach Segment.
- Intermediate Approach Segment.
- Initial Approach
- Fixed-wing aircraft

Environmental contexts

- Mixed traffic (ex. GNSS geometric guidance or not)
- All Weather including high temperatures
- Adverse Weather including strong head winds, LVC, etc.
- Degraded operational conditions

S4-HLOR-02 The Enhanced Traffic Sequencing shall enhance the variety and flexibility of departures thanks to :

- Precise and efficient GNSS curved route navigation in the TMA for capable aircraft
- Use of geometric altitudes for capable aircraft
- Minimum PBN requirements
- Safe Minimum vertical separations

In order to obtain :

- Safer and more efficient separation between departures and approaches
- Efficient avoidance of noise-sensitive areas
- Reduced Flight Crew workload
- Reduce controller workload

Identified, addressed OI Steps

AOM-0607, AO-0313

Perimeter - Operational scope

- Medium and Large Airports
- Single runways
- Fixed-wing aircraft

Environmental contexts

- All Weather
- Adverse Weather including strong head winds, LVC, etc.
- Degraded operational conditions
- Noise-sensitive areas

S4-HLOR-03 The Enhanced Traffic Sequencing shall enhance the operations from and towards parallel runways by :

- Precise and efficient GNSS curved route navigation in the TMA for capable aircraft
- Use of geometric altitudes for capable aircraft
- Minimum PBN requirements
- Safe Minimum vertical separations

In order to obtain :

- Safer and more efficient separation between departures and approaches
- Increased runway throughput
- Efficient avoidance of noise-sensitive areas

• Reduced Flight Crew workload • Reduce controller workload
<u>Identified, addressed OI Steps</u> AOM-0606
<u>Perimeter - Operational scope</u> • Medium and Large Airports • Approach and departure to and from Parallel runways • Fixed-wing aircraft
<u>Environmental contexts</u> • All Weather • Adverse Weather including strong head winds, LVC, etc. • Degraded operational conditions

▪ **High level Requirements applicable to Solution 14**

S14-HLOR-01 The Enhanced Separation for Departures and Arrivals shall:

- increase runway throughput
 - ensure more refined and efficient wake separation than current ICAO rules
 - ensure consistent and efficient management of spacing on final approaches and on the initial departure phase of flight
- through combinations of :
- static pairwise separations for successive arrivals
 - static pairwise separations for successive departures
 - static pairwise separations between arrivals and departures
 - weather dependent separation minima for successive arrivals
 - dynamic Pairwise Separation minima (ex. based on flight-specific aircraft characteristics taken from the eFPL, downlinked from the aircraft ...)
 - Differentiated rotation positions between departing aircraft
 - steeper climb profiles between departing aircraft
 - Accurate vertical flights information (from the Aircraft and/or taken from the eFPL)
 - Accurate estimations of Runway Occupancy Times
 - the use of enhanced ATC support tools
 - Advanced wind now-casting
 - Adapted runway lighting
- while:

- supporting passive wake vortex decay devices

Identified, addressed OI Steps

AO-0307, AO-0315, AO-0324, AUO-0505, TS-0312, TS-0316, AUO-0504, AUO-0507, (+ AUO-0332, AUO-0334, AUO-0335, AUO-0336 created in DS20) possibly additional OIs.

Perimeter - Operational scope

- Medium and Large Airports
- Medium and Small Airport with a significant number of rotorcraft operations
- Final Approach Segment.
- Single runways
- Closely Spaced Parallel Runways
- Runways more than 760m apart
- mix-mode operations

Environmental contexts

- All Weather
- Adverse Weather including strong head winds, LVC, etc.
- All runway configurations and operations
- Degraded operational conditions
- Recovery Transitional Periods
- All Traffic mixes
- Large availability of ADS-C / ATN-B2 and of IOP
- Efficient TWR-APP coordination
- Local availability of compatible SDM services including FF-ICE.

▪ **High level Requirements applicable to Solution 17**

S17-HLOR-01 Improved Traffic Sequencing into secondary (small/medium) airports shall:

- facilitate LVC operations
- increase airport throughput

- by combining :

- low cost visual sensors (cameras) for tower operations
- (SBAS/GBAS) or Localiser Performance with Vertical Guidance (LPV 100) equipment and related procedures (CVS – Combined Vision Systems)
- Robust EFVS/SVS/EVS supporting functions
- markings and lightings (CVS)

• low-cost surveillance system (ex. based on ADS-B and/or multilateration)
<u>Identified, addressed OI Steps</u> SDM-0301, AO-0333, AO-0108, AUO-0410 + TBD new OI steps
<u>Perimeter - Operational scope</u> • Medium and Small Airports • Final Approach and landing • Take-off • Taxiing
<u>Environmental contexts</u> • All Weather • Adverse Weather including LVC • All runway configurations and operations • Degraded operational conditions • All Traffic mixes

▪ **High level Requirements applicable to Solution 21**

S21-HLOR-01 The Enhanced Guidance Assistance to Aircraft and Vehicles shall: • deliver more accurate taxi times • minimise delays • minimise controllers' workload • minimisation of waiting time for mobiles at intersections - by : • Optimising routing and planning • providing enhanced guidance assistance to Flight Crews, vehicle drivers and controllers to handle airport operations • considering potential conflicting situations when planning taxi routes • using AGL as speed control • consolidating the set of instructions that produces relevant benefits when exchanged via data link • defining an Aerodrome Mapping Database (AMDB) to be shared, via SWIM, between ground and on-board systems • while ensuring, through datalink: ○ the exchange of VSB positions and statuses

the exchange of clearances / instructions between vehicle drivers and controllers
<u>Identified, addressed OI Steps</u> AO-0206, AO-0215, AO-0222-B, AO-0224, AUO-0308-B , AUO-0603-B
<u>Perimeter - Operational Scope</u> - Small, medium and large airports - Recovery Transitional Periods
<u>Environmental contexts</u> - Mixed mode runway or dependent runways operations. - All Weather - Low Visibility Conditions.

S21-HLOR-02 The Enhanced Guidance Assistance to Aircraft and Vehicles shall detect conflicting situations and non-conformance, involving mobiles (and stationary traffic) on runways, taxiways and in the apron/stand/gate area as well as unauthorised / unidentified traffic - While : - Minimising the level of false and nuisance alerts - Considering all types of Airspace Users (including GA, Rotorcraft and Civil RPAS).
<u>Identified, addressed OI Steps</u> AO-0223-B, AO-0104-B
<u>Perimeter - Operational Scope</u> - Small, medium and large airports - Mixed mode runway or dependent runways operation - Recovery Transitional Periods
<u>Environmental contexts</u> - All Weather - Low Visibility Conditions - SWIM availability

▪ **High level Requirements applicable to Solution 25**

S25-HLOR-01 The future Approach and landing Operations shall :

- Increase airport safety
- reduce the risk of runway excursions

- by using data from ground-based and airborne sources to:

- provide a Runway Overrun Awareness and Avoidance System (ROAAS) for business aircraft
- provide an objective evaluation of the runway condition
- improve the collaborative use of the runway capacities through a shared view of the current situation

- while :

- share this information among the main airport stakeholders
- satisfying the new regulatory requirements in the ICAO Annex 15

Identified, addressed OI Steps

AO-0107, AO-0216, AUO-0616

Perimeter - Operational Scope

- Small, medium and large airports
- Mixed mode runway or dependent runways operation
- Recovery Transitional Periods

Environmental contexts

- All Weather
- Low Visibility Conditions
- SWIM availability

3.2.3 Operational Requirements PJ 04

▪ High level Requirements applicable to Solution 28

S04-28-HLOR-01 The Enhanced Collaborative Airport Performance Planning and Monitoring shall: • increase operational resilience • increase departure time predictability • increase operational resilience of the pre-departure sequence - by • extending airport performance process monitoring to the airport landside processes • developing an airport dashboard to support pro-active management of ATM related disruptions • Providing tools to support post-operations analysis to support the recording and timely analysis of specific events • Provide the ability to quickly find, merge, analyse and replay airport data with linkage to other stakeholder tools • refining, in association with Airspace Users, a suitable level of turnaround process planning and monitoring within the APOC - while • complementing the Airport's legacy systems • defining a suitable level of Turnaround Process Monitoring within the APOC
<u>Perimeter - Operational Scope</u> • All Airports • Network Operations • Recovery Transitional Periods
<u>Environmental contexts</u> • All Weather • GA • Civil RPAS • All traffic mixes plus a significant number of rotorcraft operations
<u>Identified, addressed OI Steps</u> AO-0801-B, AO-0802-B, AO-0818, AO-0821, AO-0824

3.2.4 Operational Requirements PJ 05

▪ High level Requirements applicable to Solution 35

S05-35-HLOR-01 The Remotely Provided Air Traffic Services from a Remote Tower Centre with flexible allocation of aerodromes to Remote Tower Modules shall:

- optimise the balance between traffic demand and the number of ATCOs/AFISOs required
- increase cost-efficiency

- by :

- flexible allocation of airports to the different RTMs within the RTC
- supporting the RTC supervisor with appropriate functionalities
- supporting the ATCO/AFISO with appropriate functionalities
- integrating MET information in the RTM
- supporting RTC supervisor planning tasks with appropriate tools
- connexion with ATCO planning tools
- while :
- finding the balance between providing all information required at a certain moment while avoiding clutter of information
- applying handover procedures for a flexible transfer of aerodromes between MRTMs

Perimeter - Operational Scope

- Small to medium airports
- Recovery Transitional Periods

Environmental contexts

- Several airports connected to a site with several modules for remote air traffic service
- Various types of traffic mix, including RPAS, GA, BA and Rotorcraft
- Adverse Weather

Identified, addressed OI Steps

SDM-0210

3.2.5 Operational Requirements PJ 07

▪ High level Requirements applicable to Solution 38

S38-HLOR-01 Enhanced integration of AU trajectory definition and network management processes shall reduce the ATM planning impact on AU costs of operations through:

- provision of enhanced access for AUs to DCB information and ATM resource management through improved FOC participation in network collaborative processes (with equal provision for non-FOC operators). This is to allow AUs to better manage the impact of ATM constraints (including STAM) and to take advantage of opportunities in planning;
- automation of AU flight planning/NM interaction, including the inclusion of AU priority and preference;
- provision and exploitation of AU Preliminary FPL;
- enhanced CDM between AUs and ANSPs

Identified, addressed OI Steps (referenced to W2 Call tech specs)

AUO-0207, AUO-0208, AUO-0219, AUO-0206, DCB-0217

▪ High level requirements applicable to solution 39

S39-HLOR-01 Application of the collaborative framework shall allow the integration and coordination of 4D constraints resulting from capacity issues on arrival from stakeholders at airport, ANSP, AU and NM level through:

- management of such constraints via the NOP using new rules and mechanisms, synchronisation and reconciliation of multiple measures per flight/flow – replacing the current first planned/first served slot allocation principle;
- improved integration to streamline all prioritisation processes from all stakeholders concerned with arrivals, linking to pre-departure regulation at origin.

Identified, addressed OI Steps (referenced to W2 Call tech specs)

AUO-0109, AUO-0110, DCB-0213

▪ High level requirements applicable to Solution 40

S40-HLOR-01 The Integrated Management of the Military AU demand shall combine request for ARES allocated via ASM process and mission trajectory profile connected to the ARES delivering to NM the iSMT data pack expressed in the form of the harmonised iOAT FPL.
Identified, addressed OI Steps AUO-0210, AUO-0214, AOM-0208-B, AUO-0228, AUO-0211
Perimeter – Operational scope • Low to Medium to High Density/Complexity Airspace • TMA • Extended TMA (Systemised Airspace encompassing the TMA and the climb/descent phases below FRA) • En Route (including FRA)
Environmental contexts • Mixed Mode Traffic regarding RNP and i4D capabilities

S40-HLOR-02 The integrated management of the Military demand shall be enabled by enhanced automation and participation in CDM providing opportunities to : • Define the most efficient geographical location of ARES (DMA type1/2) • Reduce human interaction in the processing of the iOAT FPL with iSMT related data pack • Increase credibility of the ISMT data integrated in to airspace configuration • Negotiate the military AU preferences • To mitigate adverse impact on the ATM network performance through target time TTO negotiation
Identified, addressed OI Steps AUO-0210, AUO-0214, AOM-0208-B, AUO-0228, AUO-0211
Perimeter – Operational scope • Low to Medium to High Density/Complexity Airspace • TMA • Extended TMA (Systemised Airspace encompassing the TMA and the climb/descent phases below FRA) • En Route (including FRA)
Environmental contexts

- Mixed Mode Traffic regarding RNP and 4D capabilities

S040-HLOR-03 DMA type 1 and 2 shall be defined and allocated to be fully compatible with MT while enabling optimum configuration of airspace based on:

- Digitalisation and automation of DMA data set definition, sharing and integration into MT and airspace configuration
- Integrated civil-military collaborative decision making based ASM-ATFM processes for airspace management and airspace reservation at local/sub-regional and regional levels
- Preservation of national prerogative on airspace allocation and on prioritisation of MTs

Identified, addressed OI Steps

AUO-0210, AUO-0214, AOM-0208,-B AUO-0228, AUO-0211

Perimeter – Operational scope

- Low to Medium to High Density/Complexity Airspace
- TMA
- Extended TMA (Systemised Airspace encompassing the TMA and the climb/descent phases below FRA)
- En Route (including FRA)

Environmental contexts

- Mixed Mode Traffic regarding RNP and 4D capabilities

3.2.6 Operational Requirements PJ 09

▪ High Level Requirements applicable to solution 44

S44-HLOR-01 Airspace capacity for both civil and military users shall be increased within/across ANSPs' AoRs through increased granularity and flexibility in airspace configuration and management by;

- integration of concepts and procedures providing flexible sectorisation boundaries to be dynamically modified based on demand (Free Routing trajectories and AFUA needs), so that the ATM environment is able to match resources to hotspots;
- employing CDM processes relating to DAC, and with coordination between DAC and DCB to define optimal solutions;
- sharing DAC information with stakeholders via the NOP, with new capability to allow AUs to contribute to the DAC process before its completion;
- management of the RBT in the execution as the DAC process evolves from the planning phase;
- utilising automated tool functionalities for airspace management processes.

Identified, addressed OI Steps

AOM-0805, AOM-0809-A, AOM-0809-B, CM-0102-B, DCB-0210

▪ High level requirements applicable to solution 45

S45-HLOR-01 The accuracy of the NM traffic prediction from D-2 to execution shall be improved through:

- New trajectory management features, such as the preliminary FPL from FF-ICE;
- Adaptation of existing methodologies and algorithms for demand prediction and regional complexity assessment;
- Provision of accurate and timely indication of imbalance to relevant actors involved in DCB;
- Integration of the airborne trajectory via EPP

Identified, addressed OI Steps

CM-0103-B, DCB-0211, DCB-0103-B

▪ High level requirements applicable to solution 49

S49-HLOR-01 Network Performance shall be transparent and shared with stakeholders in Network Management pre-tactical and tactical phases through:

- a shared situational awareness and agreed set of network performance indicators, allowing real-time performance monitoring in support of network operations via the NOP;

- trade-off indication and what-if impact assessment capability for stakeholders to assess the impact of their action against the overall objective of optimising the network performance.
- development of network resilience and network states, using advanced performance monitoring tools;
- utilisation of data analytics, which include AI, deep-learning techniques etc. to continuously improve the underlying algorithms related to what-is and what-else functionalities.

Identified, addressed OI Steps

DCB-0212, DCB-0214

3.2.7 Operational Requirements PJ 10

▪ High level Requirements applicable to Solution 70

S70-HLOR-01 The Multi-sector planner (MSP) role and Single Person Operations (SPO) in En Route shall improve the following KPIs:

- ATCO cost efficiency

- Through:

- The role of Multi-Sector Planner (MSP) where the Planning Controller is responsible for the airspace that is under the control of two or more independent Executive Controllers²⁶
- Single Person Operations where the role of planning and execution for an En Route sector can be undertaken by a unique controller

- By:

- Defining suitable coordination agreements between sectors and the MSP assisting in managing the workload of the Tactical Controllers, thus ensuring that potentially critical traffic situations and the associated workload are manageable
- The development of appropriate procedures, roles and responsibilities
- The development of appropriate system support to the ATCOs
- Applying collaborative control working method where possible

Identified, addressed OI Steps

CM-0303, CM-0307

Perimeter – Operational scope

- Low to Medium to High Density/Complexity Airspace
- En Route (including FRA)

Environmental contexts

- Mixed Mode Traffic regarding RNP and i4D capabilities

S70-HLOR-02 Collaborative control shall improve the following KPIs:

- ATCO Cost efficiency
- Fuel efficiency

²⁶ Reference MSP - see footnote section 2.3.2 regarding impending change of scope for MSP

- By:

- Allowing controllers to work in a collaborative way, and to issue clearances into or operate traffic within all of the sectors in the area without the requirement to pre-coordinate.
- Alleviating the traditional requirement to coordinate traffic at all sector boundaries for an area covering two or more sectors, and applying constraints only when it is necessary to fulfil a specific ATM function, e.g. separation, or traffic synchronisation

- With the support of :

- The development of advanced controller tools that support reduced coordination and display of relevant traffic and conflicts to the controllers
- The development of appropriate procedures, roles and responsibilities

Identified, addressed OI Steps

CM-0306, CM-0309

Perimeter – Operational scope

- Low to Medium to High Density/Complexity Airspace
- TMA
- Extended TMA (Systemised Airspace encompassing the TMA and the climb/descent phases below FRA)
- En Route (including FRA)

Environmental contexts

- Mixed Mode Traffic regarding RNP and i4D capabilities

▪ High level Requirements applicable to Solution 73

S73-HLOR-01 Flight Centric ATC and improved distribution of separation responsibility between ATCOs in medium and high complexity environments shall improve the following KPIs:

- cost efficiency

- Through:

- Optimised workload distribution between ATCOs within a given airspace by assigning them the responsibility for a specific number of aircraft without reference to sector boundaries.
- Reduced coordination workload between ATCOs within a Flight Centric Area or between ATCOs in adjacent areas

- By:

- Allocating aircraft to responsible ATCOs based on predicted workload within the flight-

centric airspace • Designing appropriate procedures and tool support for coordination and conflict resolution, which allow reducing the need for coordination between ATCOs within the Flight Centric Airspace, or between adjacent airspaces with a Flight Centric or Conventional ATC set up • Designing appropriate procedures and cost-effective tool support for Air/Ground coordination
<u>Identified, addressed OI Steps</u> CM-0200-B, CM-0200-C, CM-0200-D, CM-0306, CM-0309, CM-0310
<u>Perimeter – Operational scope</u> • Low to Medium to High Density/Complexity Airspace • En Route (including FRA)
<u>Environmental contexts</u> • Mixed Mode Traffic regarding RNP and i4D capabilities

3.2.8 Operational Requirements PJ 13

▪ High level Requirements applicable to Solution 111

S111-HLOR-01 The RPAS shall apply the ICAO-defined Remain Well Clear (RWC) procedure²⁷ in order to insure the general safety of separation with any other aircraft/rotorcraft in the vicinity, through two mandatory processes :

- A sufficient and efficient anticipation of possible losses of separation or risks in the actual separation, based on available information on board and/or provided to the Remote Pilot
- Coordination with ATC, time permitting, to confirm, reject or solve the anticipated conflicting situation, whenever needed based on additional clearances provided to the RPA or to the related aircraft/rotorcraft

- by local application to RPAS RWC function of :

- appropriate time thresholds
- appropriate airspace volumes for alerts
- where appropriate, appropriate advisories for anticipation (prior to alerts)

- while:

- being fully compatible with the Accommodation procedures²⁸

Identified, addressed OI Steps

CM-0808-u + TBD OI steps

Perimeter - Operational scope

- Civil and Military RPAS in non-segregated airspaces classes A, B or C
- ECAC-wide applicable Specific regulations and procedures for such accommodation
- Outbound from single or multiple sectors (TMA, ACC)

²⁷ Working note : The RPAS shall apply the ICAO-defined traffic avoidance procedures to ensure safe separation with any other aircraft/rotorcraft in the vicinity, by :

- Exploiting Remain Well Clear function complementary to collision avoidance that provide quantified parameters of the volume and its corresponding threshold in order to ensure safe execution of the flight depending on the airspace classification.
- Coordination with ATC, time permitting, to confirm, reject or solve the anticipated conflicting situation, whenever needed based on additional clearances provided to the RPA or to the related aircraft/rotorcraft

²⁸ Cf Requirements applicable to Solution 115

- Delivery to single or multiple sectors (TMA, ACC)
- Cross-border operations
- Free Route Airspace
- Transit from/to High Level Altitude Operations

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route, TMA
- IFR operations between dedicated airfields
- Partial availability of ADS-C / ATN-B2 and of IOP
- Local availability of compatible SDM services including FF-ICE
- Adapted regulation²⁹

S111-HLOR-02 RPAS shall safely and efficiently Detect and Avoid any non-cooperative aircraft/rotorcraft in the vicinity³⁰, as soon as the Time to Conflict exceeds a defined threshold that does not allow a formal coordination with ATC, and reducing to the minimum the adverse impacts on flight efficiency of other aircraft/rotorcraft.

Identified, addressed OI Steps

TBD

Perimeter - Operational scope

- Civil and Military RPAS in non-segregated airspaces classes A, B or C
- ECAC-wide applicable Specific regulations and procedures for such accommodation
- Outbound from single or multiple sectors (TMA, ACC)
- Delivery to single or multiple sectors (TMA, ACC)
- Cross-border operations
- Free Route Airspace
- Transit from/to High Level Altitude Operations

Environmental contexts

- Low to High Density/Complexity Airspace

²⁹ Ex. Certified, Class VI RPAS

³⁰ As a result of an external degraded situation, such a failed TCAS in a close A/C.

- En Route, TMA
- Adapted regulation³¹

S111-HLOR-03 RPAS shall safely and efficiently Detect and Avoid any cooperative aircraft/rotorcraft in the vicinity, by exploiting of ACAS II and/or ACAS X systems functionalities.

Identified, addressed OI Steps

TBD

Perimeter - Operational scope

- Civil and Military RPAS in non-segregated airspaces classes A, B or C
- ECAC-wide applicable Specific regulations and procedures for such accommodation
- Outbound from single or multiple sectors (TMA, ACC)
- Delivery to single or multiple sectors (TMA, ACC)
- Cross-border operations
- Free Route Airspace
- Transit from/to High Level Altitude Operations

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route, TMA
- Adapted regulation³²

S111-HLOR-04 After termination of a Collision Avoidance the RPAS and ATC authority shall ensure recovery of the trajectory in a safe and efficient manner as regards:

- Preservation of the global safety of operations in the vicinity
- Reduction to the minimum needed of the negative impacts (time, patch stretching) on every aircraft/rotorcraft/RPA as regards their intended Flight / Mission

Identified, addressed OI Steps

TBD

³¹ Ex. Certified, Class VI RPAS

³² Ex. Certified, Class VI RPAS

Perimeter - Operational scope

- Civil and Military RPAS in non-segregated airspace classes A, B or C
- ECAC-wide applicable Specific regulations and procedures for such accommodation
- Outbound from single or multiple sectors (TMA, ACC)
- Delivery to single or multiple sectors (TMA, ACC)
- Cross-border operations
- Free Route Airspace
- Transit from/to High Level Altitude Operations

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route, TMA
- Adapted regulation³³

▪ High level Requirements applicable to Solution 115

S115-HLOR-01 Accommodation of RPAS in airspace classes A, B, C shall provide a ECAC-wide generic, operational concept for individual RPAS types and types of operations, through the following ATM Components:

- Airspace Management
- Demand and Capacity Balancing
- Conflict Management (Separation Management)
- Traffic Synchronisation

- by application to RPAS of:

- Adapted separation minima between RPAS them and with other Aircraft, providing required minima for safe separation reduced to the needed ones
- Performance Based Navigation requirements, or equivalent
- Adapted Performance Based Communications (possible consideration of expected latencies in C2 & C3 communications) including availability and continuity of service
- Adapted Flow management and/or declared capacities in sectors expected to accommodate one or several RPAs

³³ Ex. Certified, Class VI RPAS

- Adapted I non nominal and/or emergency procedures, particularly in the case of loss of C2 or C3 links - while: - Reducing to the minimum the adverse impacts on flight efficiency of other aircraft/rotorcraft
<u>Identified, addressed OI Steps</u> TBD
<u>Perimeter - Operational scope</u> - Civil and Military RPAS in non-segregated airspaces classes A, B or C - ECAC-wide applicable Specific regulations and procedures for such accommodation - Outbound from single or multiple sectors (TMA, ACC) - Delivery to single or multiple sectors (TMA, ACC) - Cross-border operations - Free Route Airspace - Transit from/to High Level Altitude Operations
<u>Environmental contexts</u> - Low to High Density/Complexity Airspace - En Route, TMA - IFR operations between dedicated airfields - Partial availability of ADS-C / ATN-B2 and of basic IOP - Partial availability of compatible SDM services including FF-ICE

S115-HLOR-02 The ATM Components mentioned in S115-HLOR-01, to support RPAS accommodation, shall be compatible of anticipated Collision Avoidance procedures and enablers, in particular - Remain well Clear - Situation awareness³⁴ - Non-cooperative Detect and Avoid - Collision Avoidance
<u>Identified, addressed OI Steps</u>

³⁴ Ex. Remote Pilot in the loop

CM-0808-u + TBD

Perimeter - Operational scope

- Civil and Military RPAS in non-segregated airspaces classes A, B or C
- ECAC-wide applicable Specific regulations and procedures for such accommodation
- Outbound from single or multiple sectors (TMA, ACC)
- Delivery to single or multiple sectors (TMA, ACC)
- Inbound and outbound flights within or out of ECAC area
- Free Route Airspace
- Transit from/to High Level Altitude Operations

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route, TMA
- IFR operations between a selection of aerodromes / airports
- Wide availability of ADS-C / ATN-B2
- Local availability of compatible (adapted) SDM services including FF-ICE
- Adapted regulation³⁵
- ACAS II, ACAS X

S115-HLOR-03 The accommodation of RPAS, in order to favour the party-line as possible wherever the R/T link is the primary one, and reduce to the minimum the impacts on ATC systems and procedures, shall rely on:

- Legacy communication protocols, enablers and links, especially between the RPA and the controlling sectors, this link serving as a relay between the RPS and the controlling sector
- As seldom as feasible, on additional link between the RPS and the controlling sector
- Communications for in-flight re-planning
- ECAC -generic fall-back procedures in case of loss of C2 or C3 links

Identified, addressed OI Steps

TBD

Perimeter - Operational scope

- Civil and Military RPAS in non-segregated airspaces classes A, B or C

³⁵ Ex. Certified, Class VI RPAS

- ECAC-wide applicable Specific regulations and procedures for such accommodation
- Outbound from single or multiple sectors (TMA, ACC)
- Delivery to single or multiple sectors (TMA, ACC)
- Inbound and outbound flights within or out of ECAC area
- Free Route Airspace
- Transit from/to High Level Altitude Operations

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route, TMA
- IFR operations between a selection of aerodromes / airports
- Wide availability of ADS-C / ATN-B2
- Local availability of compatible SDM services including FF-ICE
- Adapted regulation

S115-HLOR-04 In the operational context of accommodation of RPAS, the contingency and emergency procedures shall be generic at ECAC wide area and ensure:

- Avoiding to the minimum any additional negative impacts on legacy operations compared to the current emergency ones
- Reduce to the minimum possible the local procedures at Network Operation, ATC and Airport level
- Maintaining the global safety of operations, especially during the possible rerouting of the RPA to any anticipated or defined area / airfield (other aircraft, overflown areas ...)

Identified, addressed OI Steps

TBD

Perimeter - Operational scope

- Civil and Military RPAS in non-segregated airspaces classes A, B or C
- ECAC-wide applicable Specific regulations and procedures for such accommodation
- Outbound from single or multiple sectors (TMA, ACC)
- Delivery to single or multiple sectors (TMA, ACC)
- Inbound and outbound flights within or out of ECAC area
- Free Route Airspace
- Transit from/to High Level Altitude Operations

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route, TMA

- IFR operations between a selection of aerodromes / airports
- Wide availability of ADS-C / ATN-B2
- Local availability of compatible SDM services including FF-ICE
- Adapted regulation

▪ **High level Requirements applicable to Solution 117³⁶**

S117-HLOR-01 IFR RPAS in airspace classes A-C shall be managed similarly (like other airspace users) by Air Traffic Control by :

- Treating RPAS in a similar manner to manned aircraft
- Considering RPAS technical capabilities
- Considering ATC planning and ATC tools
- Considering ATC procedural means

- While:

- Taking into account ATC workload
- Maintaining or increasing operational aviation safety, capacity and efficiency

Identified, addressed OI Steps

AUO – 0618

Perimeter – Operational scope

- Outbound from single or multiple airports; from single or multiple sectors (TMA, ACC)
- Inbound to single or multiple sectors (TMA, ACC); to single or multiple airports; to individual runway(s)
- Mixed Mode Traffic
- Civil/military RPAS
- Scheduled Air Transport
- Free Route Airspace
- All weather

Environmental contexts

- Low to Medium to High Density/Complexity Airspace
- TMA
- Extended TMA

³⁶ Note : IFR RPAS integration in airspace classes A-C shall address RPAS of EASA's 'Certified' category that are capable to carry out IFR operations (Network included), TMA operations with capability of flying SIDs and STARs as designed for manned operations through the consideration of:

- MALE RPAS,
- HALE RPAS,
- Rotary wing RPAS,
- Fixed wing RPAS,

for civil and military operations, including BRLOS (Beyond Radio Line of Sight) operations.

- En Route

S117-HLOR-02 IFR RPAS integration in airspace classes A-C shall consider the Remain Well Clear and Collision Avoidance functions and the automatic management separation in compliance with manned aviation and ATC instructions through:

- Definition of Mid-Air Collisions hazards
- Identification of separation hazards

Identified, addressed OI Steps

AUO – 0618

Perimeter – Operational scope

- Outbound from single or multiple airports; from single or multiple sectors (TMA, ACC)
- Inbound to single or multiple sectors (TMA, ACC); to single or multiple airports; to individual runway(s)
- Mixed Mode Traffic
- Civil/military RPAS
- Scheduled Air Transport
- Free Route Airspace
- All weather

Environmental contexts

- Low to Medium to High Density/Complexity Airspace
- TMA
- Extended TMA
- En Route

S117-HLOR-03 IFR RPAS integration in airspace classes A-C shall consider nominal and emergency conditions taking into account future ATM infrastructures through:

- Systems and procedures compatibility.
- Systems and procedures interoperability

- By :

- implementing harmonised standards and recommended practice and procedures

Identified, addressed OI Steps

AUO – 0618

Perimeter – Operational scope

Founding Members



- Outbound from single or multiple airports; from single or multiple sectors (TMA, ACC)
- Inbound to single or multiple sectors (TMA, ACC); to single or multiple airports; to individual runway(s)
- Mixed Mode Traffic
- Civil/military RPAS
- Scheduled Air Transport
- Free Route Airspace
- All weather

Environmental contexts

- Low to Medium to High Density/Complexity Airspace
- TMA
- Extended TMA
- En Route

3.2.9 Operational Requirements PJ 18

Important note: The Requirements allocated to PJ 18 were defined based on the closest available view of its future content. As regards PJ 18, the High Level Requirements consider:

- A solution 53 taking charge of Improvements on the ground TP, of Separation and Monitoring tools (CD&R 2D/3D Conformance and Discrepancies Tools), so mainly close to the DOW³⁷
- A solution 56 taking benefit of Solution 53 results as regards CD&R, in order to mature Air-Ground Trajectory Management exchanges (vertical and horizontal change clearances ; clearances in advance)
- A solution 57 taking charge of the overall Trajectory Management: Ground-Ground and Air-Ground data exchanges associated to Trajectory Revision for the ATM actors. This including new higher automation of A/G and G/G Trajectory exchanges based on very innovative enablers.

▪ High level Requirements applicable to Solution 53

S53-HLOR-01 Enhanced Conflict Management shall enhance the following KPIs in the provision of separation in En Route :

- Sector (Traffic) Capacity
- ATCO Cost Efficiency
- Predictability
- Fuel Efficiency
- Safety

- through :

- Advanced conflict detection and resolution tools fed by most accurate trajectory predictions (including ADS-C), accurate aircraft position and other available aircraft derived data, and fully compatible with 2D RNP environments
- Resolutions primarily supported by CPDLC closed-loop instructions

by :

- Reducing to the minimum the spurious detection of conflicts
- Reducing to the minimum the need for the ATCOs to actively detect separation infringements
- Increasing situational awareness through highlighting relevant traffic

³⁷ TBC for management of merging traffic.

- Providing resolution support to the ATCOs for some conflicts and therefore reducing related cognitive workload
- Providing ATCOs with an increased opportunity to optimise trajectories with a longer look-ahead time

Identified, addressed OI Steps

CM-0209, CM-0209-B, CM-0211, CM-0605, CM-0407, CM-0607

Perimeter - Operational scope

- Outbound from single or multiple sectors (TMA, ACC)
- Delivery to single or multiple sectors (TMA, ACC)
- Inbound and outbound flights within or out of ECAC area
- Scheduled Air Transport
- Free Route Airspace
- General Aviation and rotorcraft
- RPAS in airspaces classified A, B or C

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route
- IFR operations
- Wide availability of ADS-C / ATN-B2

S53-HLOR-02 Enhanced Conflict Management shall enhance the following performances in the provision of separation in TMA :

- Sector (Traffic) Capacity
- ATCO Cost Efficiency
- Predictability
- Fuel Efficiency
- Safety

- through :

- Advanced conflict detection and resolution tools fed by most accurate trajectory predictions (including ADS-C), accurate aircraft position and other available aircraft derived data, and fully compatible with 2D RNP environments
- Resolutions supported (time permitting) by CPDLC closed-loop instructions

- by :

- Reducing to the minimum the spurious detection of conflicts
- Reducing to the minimum the need for the ATCOs to actively detect separation

Founding Members



infringements - Increasing situational awareness through highlighting relevant traffic - Providing resolution support to the ATCOs for some conflicts and therefore reducing related cognitive workload - providing ATCOs with an increased opportunity to optimise the execution of trajectories with a longer look-ahead time
<u>Identified, addressed OI Steps</u> CM-0206, CM-0606, CM-0408, CM-0608
<u>Perimeter - Operational scope</u> - Outbound from single or multiple sectors (TMA, ACC) ; from individual runway(s) - Delivery to single or multiple sectors (TMA, ACC) ; to single or multiple airports ; to individual runway(s) - Out of RNP AR operations - Mixed Mode Traffic - Scheduled Air Transport - General Aviation and rotorcraft - RPAS in airspaces classified A, B or C
<u>Environmental contexts</u> - Low to High Density/Complexity Airspace - TMA - IFR & VFR operations - All weather - Wide availability of ADS-C / ATN-B2, Enhanced Surveillance, ADS-B - Basic IOP

S53-HLOR-03 Enhanced Conflict Management shall enhance the following performances in the provision of separation in En Route and in TMA :

- ATCO Cost Efficiency
- Safety

- through :

- Advanced Conformance Monitoring tools fed by most accurate trajectory predictions and intent (including up-to-date RBT/electronic Flight-Plan), accurate aircraft position and other available aircraft derived data, and fully compatible with 2D RNP environments
- Detection of all actual non-conformances in the related Area of Interest
- Providing early notification of any predicted non-conformances along the trajectory in the related AoI

- and :
- Reducing to the minimum the spurious detection of non-conformances - Reducing to the minimum the need for the ATCOs to actively detect non-conformances
<u>Identified, addressed OI Steps</u>
CM-0208-A, CM-0210, CM-0210-b, CM-0207-B, CM-0208-B
<u>Perimeter - Operational scope</u>
- Outbound from single or multiple sectors (TMA, ACC) - Delivery to single or multiple sectors (TMA, ACC) ; to single or multiple airports - Inbound and outbound flights within or out of ECAC area - Mixed Mode Traffic - Scheduled Air Transport - Free Route Airspace - General Aviation and rotorcraft - RPAS in airspaces classified A, B or C
<u>Environmental contexts</u>
- Low to High Density/Complexity Airspace - En Route - IFR operations - Wide availability of ADS-C / ATN-B2, Enhanced Surveillance, ADS-B - Basic IOP

▪ **High level Requirements applicable to Solution 56**

S56-HLOR-01 Enhanced Air/Ground operational procedures for TBO using complex clearances supported by data link and increased automation shall improve the following KPIs: - Predictability - Sector Capacity - Fuel Efficiency - Safety - through : 2D and 3D trajectory modifications in the tactical and strategic timeframe, via complex clearances as needed, safely and efficiently implementable on board the capable aircraft - The accommodation by the ATCOs of a Flight crew requested trajectory - Systematic 2D/3D consistency check using ADS-C EPP - by:

- Reducing air/ground communication workload
- Reducing ATCO workload to issue 2D/3D clearances through system support, including concatenation of CPDLC messages, and therefore allowing optimal distribution of tasks and improved usage of ATCO's time
- Reducing ATCO workload to issue clearances through allowing the system, or other roles, such as the EAP, to take over these tasks when possible
- Allowing ground-ground exchange of proposed trajectories between sectors, including the proposals originating from the flight crew
- Allowing a 2D/3D consistency check between ground and airborne trajectories

Identified, addressed OI Steps

SDM-0213, SDM-0214, SDM-0215

Perimeter - Operational scope

- Inbound and outbound flights within or out of ECAC area
- Scheduled Air Transport
- Free Route Airspace

Environmental contexts

- Low to High Density/Complexity Airspace
- En Route
- IFR operations
- Wide availability of ADS-C / ATN-B2, Enhanced Surveillance, ADS-B
- Basic IOP

▪ **High level Requirements applicable to Solution 57**

S57-HLOR-01 Advanced ground/ground operational procedures for TBO shall improve, among others, the following KPIs:

- Predictability,
- Sector Capacity
- Fuel Efficiency
- Safety

- Through:

- Enhanced RBT revision initiated and executed by relevant roles.
- Enhanced CDM between an extended set of ground roles within the appropriate timeframe, considering AU preferences when possible.

- By:

- Allowing the initiation of a revision by any ground role within the appropriate timeframe and subject to rules and procedures. Ground roles may include Planning and Executive

Founding Members



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ATCOs, EAP, NMF. AU requested trajectories – through FOC or Flight Crew if equipped – might be used to trigger the revision process. • Allowing both static and/or dynamic AU preferences to be taken into account in the decision making process, while keeping workload of the operators within acceptable limits. • Enabling the ground roles to perform the CDM process, through the support of “what-if” trajectories, and resulting in a Ground Agreed Trajectory when needed
<u>Identified, addressed OI Steps</u> SDM - 0214 + TBD revisions or additional OI step as identified further
<u>Perimeter - Operational scope</u> • All or part of end-to-end of RBTs, after the transition SBT → RBT • Consideration of Inbound and outbound flights within or out of ECAC area • Scheduled Air Transport • Free Route Airspace • General Aviation and rotorcraft • RPAS in airspaces classified A, B or C
<u>Environmental contexts</u> • All categories of Density/Complexity Airspaces, both TMA & En Route • All categories of airports • IFR operations • Wide availability of ADS-C / ATN-B2, Enhanced Surveillance, ADS-B • Full IOP, SWIM, FF-ICE

S57-HLOR-02 Advanced air/ground operational procedures for TBO shall improve, among others, the following KPIs: • Predictability, • Sector Capacity • Fuel Efficiency • Safety - Through: • RBT revision initiated by relevant ground roles and/or the aircrew • CDM between all relevant air and ground roles within the appropriate timeframe • Enhanced air/ground synchronisation of trajectories • Reduced air/ground communication workload - By : • Designing advanced procedures and system support for easy generation of revised RBTs by
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ground actors and the Flight Crew, through e.g. a graphical interface. • Taking maximum benefit of expected and/or new Air-Ground communication technologies. • Allowing easy implementation on-board of any revised trajectory, as agreed with ground actors, while maintaining situational awareness of the flight crew and reducing any latency issues. • Enabling RPAS integration in such RBT revisions. • Enabling air/ground exchange of AU preferences.
<u>Identified, addressed OI Steps</u> SDM - 0213 + TBD revisions or additional OI step as identified further
<u>Perimeter - Operational scope</u> • All or part of end-to-end of RBTs, after the transition SBT → RBT • Consideration of Inbound and outbound flights within or out of ECAC area • Scheduled Air Transport • Free Route Airspace • General Aviation and rotorcraft • RPAS in airspaces classified A, B or C
<u>Environmental contexts</u> • All categories of Density/Complexity Airspaces, both TMA & En Route • All categories of airports • IFR operations • Wide availability of ADS-C / ATN-B2, Enhanced Surveillance, ADS-B • Possible extensions of CPDLC (areas, altitudes and semantic) beyond ATN – B2 standard • Full IOP, SWIM, FF-ICE

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Appendix A Glossary

A.1 List of Acronyms

Term	Explanation
AAT	AMAN Allocated Time
ACC	Area Control Centre
ACAS	Airborne Collision and Avoidance Systems
A-CCO	Advanced –Continuous Climb operations
A-CDM	Airport Collaborative Decision Making
A-CDO	Advanced-Continuous Climb Operations
ADEP	Aerodrome of Departure
ADS-B	Automatic Dependant Surveillance - Broadcast
ADS-C	Automatic Dependant Surveillance - Contract
AFISO	Aerodrome Flight Information Service Officer
AFUA	Advanced Flexible Use of Airspace
A/G	Air/Ground
AGL	Above Ground Level
AIM	Aeronautical Information Management
AI	Artificial Intelligence
AIS	Aeronautical Information Service
AMAN	Arrival Manager
AMDB	Aerodrome Mapping Database
ANSP	Air Navigation Services Provider
AOP	Airport Operations Plan
AoR	Area of Responsibility
APOC	Airport Operations Centre
APP	Approach Control

Term	Explanation
AR	Augmented Reality
ARES	Airspace Reservation/Restriction
ASAS	Airborne Separation Assistance Systems
ASBU	Aviation System Block Upgrade (ICAO)
ASM	Airspace Management
A-SMGCS	Advanced Surface Movement Guidance and Control System
ATC	Air Traffic Control
ATCO	Air Traffic Control Officer
ATFCM	Air Traffic Flow and Capacity Management
ATM	Air Traffic Management
ATN	Aeronautical Telecommunication(s) Network
ATOT	Actual Take-Off Time
ATS	Air Traffic Services
ATSAW	Airborne Traffic Situational Awareness
ATSU	Air Traffic Services Unit
ATV	Airport Transit View
AU	Airspace User
BVLOS	Beyond Visual Line Of Sight
C2	Command & Control link
CAT	Category
CCO	Continuous Climb Operations
CCS	Capacity Constrained Situations
CD&R	Conflict Detection & Resolution
CDM	Collaborative Decision Making
CDO	Continuous Descent Operations
CNS	Communication, Navigation, Surveillance

Term	Explanation
CONOPS	Concept of Operations
CPDLC	Controller Pilot Data Link Communication
CTA	Controlled Time of Arrival
CVS	Combined Vision System
CWP	Controller Working Position
D&A	Detect & Avoid
DAC	Dynamic Airspace Configurations
DCB	Demand and Capacity Balancing
dDCB	Dynamic Demand and Capacity Balancing
DCL	Departure Clearance
DMA	Dynamic Mobile Area
DMAN	Departure Manager
E-AMAN	Extended Arrival Manager
EAD	European AIS Database
EAP	Extended ATC Planner
EC	Executive Controller
ECAC	European Civil Aviation Conference
eFPL	extended Flight Plan
EFVS	Enhanced Flight Vision System
EGNOS	European Geostationary Navigation Overlay Service
ELDT	Estimated Landing Time
ENR	En Route
EPP	Extended Projected Profile
ETA	Estimated Time of Arrival
E-TMA	Extended Terminal Control Area
ETO	Estimated Time Over

Term	Explanation
EVS	Enhanced Vision System
FAA	Federal Aviation Administration
FAB	Functional Airspace Block
FATO	Final Approach and Take-Off area
FF-ICE	Flight and Flow Information for a Collaborative Environment
FIR	Flight Information Region
FL	Flight Level
FMS	Flight Management System
FPL	Flight Plan
FOC	Flight Operations Centre
FPL	Filed Flight Plan
FRA	Free Route Airspace
GA	General Aviation
GBAS	Ground-Based Augmentation System
G/G	Ground / Ground
GNSS	Global Network Satellite System
H24	Continuous day and night service
HMI	Human Machine Interface
IAF	Initial Approach Fix
ICAO	International Civil Aviation Organisation
IFR	Instrument Flight Rules
IGS	Increased Glide Slope
IM	Interval Management
INAP	Integrated Network Management and (Extended) ATC Planning
IOP	Interoperability (between ground systems)
i-OAT	Initial Operational Air Traffic

Term	Explanation
IoT	Internet of Things
iRMT	Initial Reference Military Trajectory
iSMT	Initial Shared Military Trajectory
KPA	Key Performance Area
KPI	Key Performance Indicator
LCS	Low-Cost Surveillance System
LVC	Low Visibility Condition
LVP	Low Visibility Procedures
MET	Meteorological
MRAP	Multiple Runway Aiming Points
MRS	Monitoring and Ranging Stations
MSP	Multi Sector Planner
MT	Mission Trajectory
NM	Network Manager
NMF	Network Management Function
NOP	Network Operations Plan
NTZ	Non Transgression Zone
OAT	Operational Air Traffic
OATTS	Pan-European OAT-IFR Transit Service
OBT	Off Block Time
OCD	Operational Concept Document
OI	Operational Improvements
PBAOM	Performance Based Aerodrome Operational Minima
PBN	Performance Based Navigation
PBN IR	PBN Implementing Regulation
PC	Planning Controller

Term	Explanation
PFD	Predicted Flight Data
PFP	Preliminary Flight Plan
PIC	Pilot-In-Command
POAS	Post Operations Analysis
PTZ	Pan Tilt Zoom (Camera)
PWS	Pair Wise Separation
RA	Resolution Advisory
RC	Rotorcraft
R&D	Research & Development
RMT	Reference Military Trajectory
RNP	Required Navigation Performance
RNP AR APP	RNP Authorisation Required Approach
ROAAS	Runway Overrun Awareness and Avoidance System
ROT	Runway Occupancy Time
RPAS	Remotely Piloted Aircraft System
RPS	Remote Pilot Station
RSP	Required Surveillance Performance
RTC	Remote Tower Centre
RTM	Remote Tower Modules
RTS	Remote Tower Services
RWC	Remain Well Clear
RWY	Runway
RWYCC	Runway Condition Code
SBAS	Satellite-Based Augmentation System
SBT	Shared Business Trajectory
SDM	Service Delivery Management

Term	Explanation
SESAR	Single European Sky ATM Research
SFL	Selected Flight Level
SID	Standard Instrument Departure (Route)
SMT	Shared Mission Trajectory
SNI	Simultaneous Non-Interfering
SPO	Single Person Opera
SSR	Secondary Surveillance Radar
STCA	Enhanced Short Term Conflict Alert
STAM	Short Term ATFCM Measure
STAR	Standard Terminal Arrival Route
SV	Synthetic Vision
SVS	Synthetic Vision System
SVGS	Synthetic Vision Guidance System
SWIM	System Wide Information Management
TBO	Trajectory Based Operations
TBS	Time Based Separation (Wake Turbulence)
TLDT	Target Landing Time
TM	Trajectory Management
TMA	Terminal Manoeuvring Area
TMR	Trajectory Management Requirement
ToC	Top of Climb
ToD	Top of Descent
TPO	Tactical Parallel Offset
TSAT	Target Start-up Approval Time
TTO	Target Time Over
TTOT	Target Take-Off Time

Term	Explanation
TWR	Tower
UAS	Unmanned Aerial System
UDPP	User Driven Prioritisation Process
VCS	Voice Communication System
VDL	VHF Data Link
VFR	Visual Flight Rules
VLOS	Visual Line Of Sight
V-NAV	Vertical Navigation
VPA	Variable Profile Area
VTS	Vehicle Tracking Systems
WOC	Wing Operation Centre
WDS	Weather Dependent Separation
Wx	Weather
xLS	x Landing System

A.2 Definition of Terms

When the following terms are used in this document, they will have the meaning as specified below.

Term	Definition	Source
4D trajectory	A set of consecutive segments linking waypoints and/or points computed by airborne or by ground systems to build the vertical profile and the lateral profile where each point is defined by a longitude, latitude, a level and a time.	SESAR1
ADS-C ETA min/max report	ADS-C ETA min/max report is the ADS-C report containing the earliest and latest values of ETA computed by the aircraft system on the point specified by ATC (e.g. IAF).	SESAR1
Advanced Continuous Descent Operation (A-CDO)	A-CDO is a Continuous Descent Operation giving optimal uninterrupted descent from cruise ToD to Final Approach and in high density operations with use of minimum required thrust setting, normally idle thrust.	SESAR 1

Term	Definition	Source
Advanced Continuous Climb Operation (A-CCO)	A-CCO is a Continuous Climb Operation giving uninterrupted climb from point of departure to ToC (at RFL) and in high density operations at operator's preferred thrust and speed setting.	SESAR 1
Aircraft Derived Data	A generic term that refers to avionics data transmitted from the aircraft to the ground and possibly to other aircraft for surveillance application. These data may be displayed to the Air Traffic Controller (ATCO) or used in ground processing functions. These data can be transmitted by several technical means such as SSR, ADS-B and ADS-C. Note: Airborne position, airborne velocity, heading, airspeed, aircraft category, aircraft identification, aircraft airborne status, selected vertical intention, status data (selected altitude, barometric pressure setting, selected heading, Status of FCU mode), ACAS active resolution advisory, current gross mass, when received directly from the aircraft through legacy means, are part of Aircraft Derived Data. The same data transmitted by non-legacy means (e.g. by FOC via SWIM) or trajectory prediction coming from aircraft (e.g. ETO, predicted speed, predicted gross mass ...) are not considered as Aircraft Derived Data.	SESAR1
AMAN Allocated Time	This is a time at a point of constraint, allocated by the Extended AMAN system. It is facilitated by ATC and may be communicated to a flight to be incorporated into the trajectory.	SESAR 2020
ASAS Operations	A set of operational procedures for controllers and flight crews, supported in particular by specific capabilities on groundside and airborne side, which enables the flight crew to respect either a specified spacing or a specified separation of their aircraft from one or more aircraft, and provides flight information concerning surrounding traffic.	SESAR1
ASAS Spacing	ASAS spacing is an ASAS-based separation provision mode in which the separation responsibility remains with the controller, who instructs the flight crew of the own ship to maintain a specified time or distance from a designated aircraft, who is referred to as the target aircraft	SJU
Building Block	Elementary sections of modularised airspace (as defined by the appropriate airspace provider) that are too small individually for controlling purposes, but instead form the basic constituent parts of a Controlling Block as part of an optimising process. This block may also be referred to as a Pixel.	SESAR1

Term	Definition	Source
Dynamic Mobile Area	<u>DMA Type 1</u> is a volume of airspace of defined dimensions described as integral part of MT at flexible geographical locations agreed upon a CDM process, satisfying Airspace Users requirements in terms of a time and/or distance constraint parameters from a reference point as specified by AU (e.g. Aerodrome of Departure).	SESAR1
	<u>DMA Type2</u> is a volume of airspace of defined dimensions described as integral part of Mission Trajectory and agreed upon a CDM process, satisfying the Airspace Users requirements. This volume of airspace may be planned and used at any geographical location along the trajectory.	
	<u>DMA Type 3</u> is a volume of airspace of defined dimensions described as an integral part of a MT and agreed upon in a CDM process satisfying the Airspace Users requirements. This volume of airspace is designed around moving aircraft (an aircraft centric bubble) requiring specific separation criteria based on aircraft configuration (e.g. fighter, tanker, formation).	
Enhanced Flight Vision System operation	EFVS operation means an operation in which visibility conditions require an EFVS to be used in lieu of natural vision to perform an approach or landing, identify required visual references or conduct a rollout. The term EFVS is introduced to describe a system that can be used for operational credits as opposed to 'EVS', which refers to a system to be used only for improved situational awareness	SESAR 2020
Enhanced flight vision system	Enhanced flight vision system (EFVS)' is an electronic means to provide the flight crew with a real-time sensor-derived or enhanced display of the external scene topography (the natural or manmade features of a place or region especially in a way to show their relative positions and elevation) through the use of imaging sensors; an EFVS is integrated with a flight guidance system and is implemented on a head-up display or an equivalent display system	SESAR 2020
ETA min/max	ETA min/max is the earliest/latest ETA at a waypoint, provided the aircraft flies the 4D trajectory at its max/min allowable speed, wind/temp error is also taken into account, in order to guarantee that any CTA defined within associated ETA min/max interval will be satisfied with high probability.	SESAR1
Extended TMA	Describes the airspace encompassing both the TMA and the airspace beyond, which has been designed to service the need for a seamless transition between the TMA and FRA (and vice versa).	SESAR 2020
Free Routing	The ability of an Airspace User to plan/re-plan a route according to	Free Route

Term	Definition	Source
	the User defined segments.	Task Force Conclusions V00.01.00
Intent	Ground Automation Intent – The behaviour of the aircraft that is expected by the ground automation (e.g. flight plan and amendments provide long-term intent, interim altitudes provide medium-term intent) Controller Intent – Desired behaviour by the controller that will have an impact on the aircraft trajectory. The controller intent may not be entirely observable or quantifiable. For example, a controller may vector an aircraft by assigning a heading, the controller intent includes a turn-back, but the precise magnitude and timing of that manoeuvre may not be known a priori. Aircraft Intent – Behaviour by the aircraft that will have an impact on the aircraft trajectory. Pilot Intent – Desired behaviour by the pilot that will have an impact on the aircraft trajectory. An example includes the placement of the top-of-descent during descents at the pilot's discretion.	FAA / EUROCONTROL COOPERATIVE R&D (White Paper "Common Trajectory Prediction-Related Terminology")
Pair Wise Separation	Procedure designed for use between arrivals on final approach or between departures from the runway, based on an optimum adjusting of the needed separation between two aircraft taking into account their respective characteristics and giving Controllers more dynamic separation options.	SESAR1
SWIM	SWIM (System Wide information Management) consists of standards, infrastructure and governance enabling the management of ATM information and its exchange between qualified parties via interoperable services. It is a distributed processing environment which replaces data level interoperability and closely coupled interfaces with an open, flexible, modular and secure data architecture totally transparent to users and their applications	SESAR Integrated Dictionary, http://www.eurocontrol.int/lexicon/lexicon/en/index.php/SESAR
Trajectory Management (4D)	Trajectory management is the process by which the Business or Mission Trajectory of the aircraft is planned, agreed, updated and revised. It is achieved through Collaborative Decision Making (CDM) processes between Airspace users (Airspace Users) and ATM Service Providers (ANSP, Airports, Network Manager) or directly between Flight Crew and Controller during the execution phase when time does not permit CDM.	SESAR1

END OF DOCUMENT